

Modal Estimation of Autoregressive Distributed Lag Model Applied to COVID-19 Pandemic Data

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Abstract

The current study applies modal regression to the Autoregressive Distributed Lag (ARDL) model. Using the proposed Modal Autoregressive Distributed Lag (MARDL) model, we not only provide asymptotic theory on the estimators of the unknown parameters in the model but also analyze short-run dynamics and long-run cointegration relationships. We also validate our theoretical results through Monte Carlo simulations by verifying that the Wald tests using the short- and long-run parameter estimators follow the chi-square distribution under the null. We further apply the MARDL method to analyze COVID-19 cases globally and in two sets of European countries (Belgium and Germany, Denmark and Norway) to assess the dynamic changes in confirmed cases and deaths pre- and post-vaccination periods and the impact of government response policies on the case fatality rate. Our empirical findings demonstrate that **the in-sample fit delivered by MARDL outperforms those delivered by ARDL and QARDL** in terms of **Coverage** Quantile Function (CQF), confirming that vaccination and government response policies have a positive impact on Covid-19.

Key Words: MARDL; Cointegration; Modal Regression; Conditional Mode; Covid-19; Vaccination; Government Response Policy.

JEL Classifications: C22, C53, I18, G35.

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1 Introduction

Over the past four decades, the statistical properties of cointegrated time series have been widely applied and studied in the field of economics. Research by [Phillips and Kheifets \(2024\)](#) suggests that the cointegrated time series often exhibit associations between variables and economic identities within constituent variables and economic data. For example, common relationships exist between variables such as capital stock and fixed investment, inventory investment and inventories, completed housing construction, and units under construction. These variables often exhibit non-stationary characteristics, and cointegration models are commonly used in empirical work studying such time series. The Fully Modified Ordinary Least Squares (OLS) estimation method proposed by [Phillips and Hansen \(1990\)](#) finds widespread application in linear time series modeling. However, this semi-parametric approach is static, only displaying long-run relationships between integrated series, leaving the associated short-run dynamics to be recovered through potentially inefficient two-step procedures (e.g., [Engle and Granger \(1987\)](#)). To overcome this limitation, [Pesaran and Shin \(1998\)](#) introduce the Autoregressive Distributed Lag Error Correction (ARDL-EC) model, which is applied to analyze cointegration relationships within the first-order integrated processes enabling simultaneous study of long-run relationships and short-run dynamics in a fully parameterized manner. Notably, they not only extend the cointegration ARDL methodology but also develop asymptotic theory for estimation and inference. Furthermore, [Pesaran et al. \(2001\)](#) further extend this by introducing a bounds-testing framework, allowing for mixed orders of integration among variables entering the ARDL model. The comprehensive application of this series of methods provides powerful tools for in-depth analysis of cointegrated time series, offering a framework for a more comprehensive understanding of the dynamic relationships between economic variables. These two papers provide a method for modeling the long-run relationships between variables of unknown integration orders. It is worth mentioning that the ARDL model offers an intuitive and clear error correction interpretation, which can be estimated via OLS, and can handle serial correlation by selecting appropriate lag orders. Even in the presence of weak endogeneity in the explanatory variables, the model can provide consistent estimates of long-run parameters. In recent years the autoregressive distributed lag approach has been extended into the realm of quantile regression. [Cho et al. \(2015\)](#) propose the Quantile Autoregressive Distributed Lag (QARDL) model, which jointly analyzes short-run dynamics and long-run cointegrating relationships across a range of quantiles.

In recent years, the literature on modal regression (MR) has grown rapidly. Research on modes traces back to [Parzen \(1962\)](#), whose Parzen window method lays the foundation for estimating modes alongside kernel density functions. Subsequently, research on modes, both theoretical and applied, gradually emerges

including works by [Chernoff \(1964\)](#), [Comaniciu and Meer \(2002\)](#), and others. The study of conditional mode gives rise to the concept of MR. [Sager and Thisted \(1982\)](#) first proposes the idea of estimating conditional mode in isotonic regression. This is followed by more detailed studies on MR under the assumption of independent observations by [Collomb et al. \(1986\)](#), who theoretically prove the consistent convergence of non-parametric mode estimators to conditional mode functions under certain conditions. Subsequently, [Lee \(1989\)](#) pioneers MR from the perspective of supervised learning and risk minimization. By observing conditional modes in truncated data, the author provide consistent mean estimates for original non-truncated data and further establish the theoretical foundation for MR. To ensure the consistency of estimation, [Lee \(1989, 1993\)](#) stipulate that the model must have constant adjustment parameters and symmetric error density distributions around zero. This specification endows the proposed MR estimator with robust properties, enabling it to adapt robustly to the characteristics of the data. [Yao et al. \(2012\)](#) demonstrate the consistency of non-parametric MR estimators under symmetric noise, although MR does not necessarily require the assumption of symmetric noise in practical applications. [Kemp and Silva \(2012\)](#) show the consistency and asymptotic normality of linear MR estimators under asymmetric noise when the mode is globally unique. [Yao and Li \(2014\)](#) propose a kernel-based objective function with bandwidth approaching zero to achieve consistent mode estimation for skewed data, enforcing a linear regression. Since the research by [Yao and Li \(2014\)](#), discussions on MR theory and applications grow, as evidenced by works such as those by [Chen et al. \(2016\)](#), [Yao and Xiang \(2016\)](#), [Krief \(2017\)](#), [Chen \(2018\)](#), [Ota et al. \(2019\)](#), [Zhou and Huang \(2019\)](#), [Yang et al. \(2024\)](#), as well as [Ullah et al. \(2023\)](#) and the references therein. Recently, the research on MR extends into the realm of autoregression, as seen [Ullah et al. \(2022\)](#) and [Jing and Cho \(2025\)](#).

In this paper, we aim to contribute to this rapidly growing literature by proposing a Dynamic Modal ARDL-EC (MARDL-EC) representation, where we can simultaneously handle long-run (cointegration) relationships and associated short-run dynamics in a fully parameterized setting. We expect all optimal estimation properties to be obtained in a manner akin to the combined asymptotic properties of the QARDL-EC model by [Cho et al. \(2015\)](#) and the asymptotic properties of MR by [Yao and Li \(2014\)](#). We provide asymptotic theory for the estimation and testing of MARDL models with non-stationary regressors. The short-run dynamic parameters and long-run cointegration parameters of the MARDL estimator are shown to follow (mixed) normal distributions in the asymptotic sense. We also demonstrate that the null distribution of Wald statistics testing short-run and long-run parameter restrictions weakly converges to a chi-square distribution. Through Monte Carlo simulation studies, we find overall simulation results, focusing on the empirical size and power of statistics, provide strong support for theoretical predictions.

In the empirical part, we propose to analyze the dynamic relationship between confirmed COVID-19

cases and deaths by MARDL model, analyzing both long-run and short-run scenarios. We apply MARDL methods to fit global data and COVID-19 data from two sets of European countries, and use Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coverage Quantile Function (CQF) proposed by Jing and Cho (2025), evaluated here at the 0.5 coverage level, as metrics to measure the performance of ARDL, QARDL and MARDL. Through this empirical application, we demonstrate that the in-sample fit based on MARDL achieves a superior CQF compared to the other two methods, and we highlight its advantage in describing the cyclical behavior of COVID-19 deaths.

This paper makes the following key contributions:

- Firstly, we introduce the innovative concept of MR into the ARDL model. The proposed MARDL model not only develops asymptotic theory but also simultaneously analyzes short-run dynamics and long-run cointegration relationships. To validate our theoretical findings, Monte Carlo simulations are conducted, demonstrating that the Wald test statistics for both short-run and long-run parameter estimates adhere to the chi-square distribution.
- Secondly, we analyze the dynamic relationship between COVID-19 confirmed cases and deaths during the pre-vaccination and post-vaccination periods. We employ rolling windows for multi-round estimation, capturing real-time variations and trends in key variable coefficients in the COVID-19 data by analyzing daily confirmed cases and deaths. We also analyze the performance of these variables across different vaccination periods. This experiment provides effective dynamic insights for pandemic planning and decision-making.
- Lastly, we compare and analyze the dynamic relationships in two sets of European countries (Belgium and Germany, Denmark and Norway) that implement different stringency levels of interventions during the pre-vaccination and post-vaccination periods. This underscores the necessity of government response policies to ensure citizens' confidence in government governance capabilities.

The remaining sections of this paper are divided into four parts. In Section 2, we introduce the MARDL model and derive the asymptotic distribution of both the short-run and the long-run estimator. We extend the inference for MARDL model in Section 2.2. Section 3 evaluates the finite-sample performance via Monte Carlo simulations. Section 4 consists of empirical analysis where we utilize the MARDL Model to examine the dynamic relationship and performance of daily confirmed COVID-19 cases and deaths data in various vaccination periods across the globe and among two sets of European countries with different stringency levels of response strategies, and aims to provide valuable and accurate insights for policymakers and decision-makers. Lastly, Section 5 discusses our capabilities and associated limitations, along with prospects for future work.

Before proceeding to the asymptotic theorem for the estimators, it is convenient to introduce some notations that will be used in the remaining part of this section. Parameters without a subscript are used to indicate generic notations constituting the parameter space. Parameters labeled with subscripts or superscripts “*” characterize elements within the data generating process. Symbols “ $\rightsquigarrow$ ”, “ $\xrightarrow{\mathbb{P}}$ ”, and “ $\lfloor a \rfloor$ ” represent “weak convergence”, “convergence in probability”, and “the smallest integer greater than a ”, respectively. The matrix $[a_{ij}]$ with $i = 1, \dots, m, j = 1, \dots, n$ is an $m \times n$ matrix with the element a_{ij} , and ι_ℓ signifies an $\ell \times 1$ vector of ones. We let $D_n(\cdot) = D(\cdot) + o_{\mathbb{P}}(w_n)$ (resp. $O_{\mathbb{P}}(w_n)$) if $\sup_{x \in \mathcal{X}} |D_n(x) - D(x)| = o_{\mathbb{P}}(w_n)$ (resp. $O_{\mathbb{P}}(w_n)$). We express that $f(n) = o(g(n))$ if for all $c > 0$, there exists some $k > 0$ (independent of n) such that $0 \leq f(n) < cg(n)$ for all $n \geq k$. Furthermore, **for any real matrix A , including a column vector, we use the Frobenius norm $\|A\| = [\text{tr}(AA')]^{1/2}$; for a vector this is exactly its Euclidean norm.** We also abbreviate stochastic integrals by omitting differentials. For example, $\int \tilde{\mathcal{B}}_w(r) \tilde{\mathcal{B}}_w(r)' dr$ and $\int \tilde{\mathcal{B}}_w(r) d\mathcal{B}_\phi(r)$ are abbreviated as $\int \tilde{\mathcal{B}}_w \tilde{\mathcal{B}}_w'$ and $\int \tilde{\mathcal{B}}_w d\mathcal{B}_\phi$, respectively. All other notations conform to standard conventions.

2 Estimation and Inference of MARDL

Cho et al. (2015) extend the ARDL approach to quantile regression and jointly analyze the short- and long-run cointegrating relationships across a range of quantiles, which is known as QARDL. In this section, we extend the QARDL analysis to the context of modal regression to attain the goal of this study.

The QARDL analysis allows for different cointegrating coefficients depending on percentile. Specifically it is defined as

$$y_t = \alpha_*(\tau) + \sum_{j=1}^p \phi_{j*}(\tau) y_{t-j} + \sum_{j=0}^q \theta_{j*}(\tau)' \mathbf{x}_{t-j} + u_t(\tau),$$

which is commonly referred to as QARDL(p, q). Here, $\Delta \mathbf{x}_t$ is a k -dimensional stationary ergodic process with zero population mean ($k \in \mathbb{N}$); the error term $u_t(\tau)$ is defined as $y_t - Q_\tau(y_t | \mathcal{F}_{t-1})$, where $Q_\tau(y_t | \mathcal{F}_{t-1})$ is the conditional quantile function on $\mathcal{F}_{t-1}$; $\mathcal{F}_{t-1}$ is the smallest σ -field generated by $\{\mathbf{x}'_t, y_{t-1}, \mathbf{x}'_{t-1}, \dots\}$ and p and q are the QARDL lag orders.

We define MARDL in parallel to QARDL. If we let $f(\cdot | \mathcal{F}_{t-1})$ be the conditional PDF of y_t given $\mathcal{F}_{t-1}$ we suppose that $\text{mode}(y_t | \mathcal{F}_{t-1}) := \arg \max_y f(y | \mathcal{F}_{t-1})$ is unique and further that $\text{mode}(y_t | \mathcal{F}_{t-1})$ is given as

$$\text{mode}(y_t | \mathcal{F}_{t-1}) = \alpha_* + \sum_{j=1}^p \phi_{j*} y_{t-j} + \sum_{j=0}^q \theta'_{j*} \mathbf{x}_{t-j},$$

so that y_t can be decomposed as follows:

$$y_t = \alpha_* + \sum_{j=1}^p \phi_{j*} y_{t-j} + \sum_{j=0}^q \theta'_{j*} \mathbf{x}_{t-j} + u_t, \quad (1)$$

which we call MARDL(p, q). Here, $u_t := y_t - \text{mode}(y_t | \mathcal{F}_{t-1})$ is defined as the modal error.

Remarks 1. The modal error u_t satisfies $\text{mode}(u_t | \mathcal{F}_{t-1}) = 0$ by construction. Provided that the density function of u_t , denoted as $f_u(\cdot)$, is smooth, this implies that $f_u^{(1)}(0) = 0$. That is, the mode is a stationary point of $f_u(\cdot)$. $\square$

2.1 Environment and Parameter Estimation

In this section we propose an estimation procedure for the long- and short-run parameters defined by MARDL(p, q).

2.1.1 Error-Correction Representation

A modal cointegrating relationship can be established by the error-correction (EC) representation in parallel to [Engle and Granger \(1987\)](#). Specifically, we can convert MARDL(p, q) into the following:

$$\Delta y_t = \alpha_* + \zeta_* (y_{t-1} - \beta'_* \mathbf{x}_{t-1}) + \sum_{j=1}^{p-1} \varphi_{j*} \Delta y_{t-j} + \sum_{j=0}^{q-1} \psi'_{j*} \Delta \mathbf{x}_{t-j} + u_t, \quad (2)$$

where the parameters in (2) are defined as follows:

$$\zeta_* := \sum_{i=1}^p \phi_{i*} - 1, \quad \beta_* := \gamma_* \left(1 - \sum_{i=1}^p \phi_{i*} \right)^{-1}, \quad \gamma_* := \sum_{j=0}^q \theta_{j*}, \quad \varphi_{j*} := - \sum_{h=j+1}^p \phi_{h*},$$

$\theta_j^* := - \sum_{h=j+1}^q \theta_{h*}$ for $j \neq 0$, and $\theta_0^* := \theta_{0*}$. Here, β_* captures the long-run relationship between y_t and $\mathbf{x}_t$, and $\varphi_* := (\varphi_{1*}, \dots, \varphi_{p-1*})'$ and $\psi_* := (\psi'_{0*}, \dots, \psi'_{q-1*})'$ describe the short-run dynamics between the variables.

We envisage the long- and short-run relationships posited by EC in an alternative way. We first rewrite (1) as follows:

$$y_t = \alpha_* + \sum_{j=0}^{q-1} \mathbf{w}'_{t-j} \boldsymbol{\delta}_{j*} + \mathbf{x}'_t \gamma_* + \sum_{j=1}^p \phi_{j*} y_{t-j} + u_t \quad (3)$$

Here, we let $\mathbf{w}_t := \Delta \mathbf{x}_t$ for notational simplicity, and for each $j = 0, \dots, q-1$, we let $\boldsymbol{\delta}_{j*} := - \sum_{i=j+1}^q \theta_{i*}$.

We further reformulate (3) as follows:

$$y_t = \mu_* + \mathbf{x}'_t \boldsymbol{\beta}_* + r_t, \quad (4)$$

where $\mu_* := \alpha_* (1 - \sum_{i=1}^p \phi_{i*})^{-1}$ and $r_t := \sum_{j=0}^{\infty} \mathbf{w}'_{t-j} \boldsymbol{\xi}_{0,j*} + \sum_{j=0}^{\infty} \rho_{j*} u_{t-j}$, such that $\{\pi_{0*}, \pi_{1*}, \dots\}$ is a set of coefficients satisfying

$$(1 - L)^{-1} \left(\frac{\sum_{j=0}^q \boldsymbol{\theta}_{j*} L^j}{1 - \sum_{j=1}^p \phi_{j*} L^j} - \frac{\sum_{j=0}^q \boldsymbol{\theta}_{j*}}{1 - \sum_{j=1}^p \phi_{j*}} \right) \equiv \sum_{j=0}^{\infty} \pi_{j*} L^j,$$

and for $i = 0, 1, \dots, p$ and $j = 0, 1, \dots$, $\boldsymbol{\xi}_{i,j*} := -\boldsymbol{\beta}_*$, for $j < i$; and $\boldsymbol{\xi}_{i,j*} := \boldsymbol{\pi}_{(j-i)*}$ for $j \geq i$, so that $\boldsymbol{\xi}_{0,j*} = \boldsymbol{\pi}_{j*}$. Likewise, $\{\rho_{0*}, \rho_{1*}, \dots\}$ is a set of coefficients satisfying $\sum_{j=0}^{\infty} \rho_{j*} L^j \equiv (1 - \sum_{j=1}^p \phi_{j*} L^j)^{-1}$. Here, the residual term $\mathcal{R}_t$ represents the collection of serially correlated stationary variables irrelevant to the long-run relationship, and the long-run relationship is captured by $\boldsymbol{\beta}_*$. As clear from its definition, $\boldsymbol{\beta}_*$ is defined by the parameters in the MARDL process, $\boldsymbol{\gamma}_*$ and $\boldsymbol{\phi}_* := (\phi_{1*}, \dots, \phi_{p*})'$. That is,

$$\boldsymbol{\beta}_* := \boldsymbol{\gamma}_* \left(1 - \sum_{i=1}^p \phi_{i*} \right)^{-1}. \quad (5)$$

The short-run parameters in (2) can also be represented using the parameters in MARDL(p, q). If we let $\boldsymbol{\delta}_* := (\boldsymbol{\delta}'_{0*}, \dots, \boldsymbol{\delta}'_{(q-1)*})'$ and

$$\mathbf{R}_1 := \begin{bmatrix} \mathbf{U} & \boldsymbol{\iota}_{p-1} \\ \mathbf{0}'_{p-1} & 1 \end{bmatrix},$$

such that $\mathbf{U}$ is an upper-triangular matrix with unities for the nonzero elements, then it follows that

$$\boldsymbol{\varphi}_* = -\mathbf{R}_1 \boldsymbol{\phi}_* \quad \text{and} \quad \boldsymbol{\psi}_* = \boldsymbol{\delta}_* + (\boldsymbol{\gamma}'_*, \mathbf{0}'_{k(q-1)})'. \quad (6)$$

2.1.2 Estimation and Environment

For the goal of the current study, we estimate the long- and short-run parameters. Specifically, we estimate them in two steps: we first estimate the long-run parameters by estimating $\boldsymbol{\gamma}_*$ and $\boldsymbol{\phi}_*$ and next estimate the short-run parameters using the EC representation.

Before estimating the unknown parameters, we rewrite MARDL(p, q) as follows to discuss its notable features:

$$y_t = \mathbf{g}'_t \boldsymbol{\lambda}_* + \widehat{\mathbf{y}}'_t \boldsymbol{\phi}_* + u_t = \mathbf{z}'_t \boldsymbol{\alpha}_* + u_t \quad (7)$$

where $\mathbf{z}_t := (\mathbf{g}'_t, \widehat{\mathbf{y}}'_t)' := (\mathbf{g}'_t, y_{t-1}, \dots, y_{t-p})' := (1, \mathbf{w}'_t, \dots, \mathbf{w}'_{t-q+1}, \mathbf{x}'_t, y_{t-1}, \dots, y_{t-p})'$, and $\boldsymbol{\alpha}_* := (\boldsymbol{\lambda}'_*, \boldsymbol{\phi}'_*)' := (\boldsymbol{\alpha}_*, \boldsymbol{\delta}'_*, \boldsymbol{\gamma}'_*, \boldsymbol{\phi}'_*)'$, where $\boldsymbol{\alpha}_*$ is an interior point of the compact parameter space $\mathcal{A}$.

Remarks 2. (i) The regressor $\mathbf{z}_t$ contains components with different stationarity orders:

$$\mathbf{z}_t = \left(\underbrace{1, \mathbf{w}'_t, \dots, \mathbf{w}'_{t-q+1}}_{\text{stationary, I(0)}}, \underbrace{\mathbf{x}'_t}_{\text{integrated, I(1)}}, \underbrace{y_{t-1}, \dots, y_{t-p}}_{\text{integrated, I(1)}} \right)'.$$

This mixed structure is a key distinction from the pure stationary setting assumed by Yao and Li (2014), where the regularity condition assumes finite fourth moment for the regressors. Meanwhile this condition fails for $\mathbf{x}_t$ and y_{t-j} because $n^{-1} \sum_{t=1}^n \|\mathbf{x}_t\|^4 \asymp n^2$. By this, the asymptotic theory needs to be separately applied to the stationary and integrated components, as we develop below.

(ii) Each element of $\widehat{\mathbf{y}}_t$ is formed as follows:

$$y_{t-i} = \mu_* + \mathbf{x}'_t \boldsymbol{\beta}_* + \sum_{j=0}^{q-1} \mathbf{w}'_{t-j} \boldsymbol{\xi}_{i,j*} + k_{t,i}, \quad i = 1, 2, \dots, p, \quad (8)$$

where $\boldsymbol{\xi}_{i,j*} := -\boldsymbol{\beta}_*$, if $i > j$; and $\boldsymbol{\xi}_{i,j*} := -\sum_{\ell=j-i}^{\infty} \pi_{\ell*}$, otherwise. Furthermore,

$$k_{t,i} := \begin{cases} -\sum_{j=q-i}^{\infty} \mathbf{w}'_{t-i-j} \boldsymbol{\xi}_{0,j*} + \sum_{j=0}^{\infty} \rho_{j*} u_{t-i-j}, & \text{if } i \leq q; \\ -\sum_{j=0}^{i-q-1} \mathbf{w}'_{t-q-j} \boldsymbol{\beta}_* + \sum_{j=0}^{\infty} \mathbf{w}'_{t-i-j} \boldsymbol{\pi}_{j*} + \sum_{j=0}^{\infty} \rho_{j*} u_{t-i-j}, & \text{if } i > q, \end{cases}$$

and for each i , $k_{t,i}$ is stationary and ergodic. We further note that (8) is the lagged version of (4), and we below use it frequently to examine the asymptotic behavior of our estimator. We rewrite it more compactly: $\widehat{\mathbf{y}}_t = \boldsymbol{\Gamma}'_* \mathbf{g}_t + \mathbf{k}_t$ by letting

$$\boldsymbol{\Gamma}_* := \begin{bmatrix} \mu_* \otimes \iota_{1 \times p} \\ \boldsymbol{\Xi}_* \\ \boldsymbol{\beta}_* \otimes \iota_{1 \times p} \end{bmatrix} := \begin{bmatrix} \mu_* & \mu_* & \cdots & \mu_* \\ \boldsymbol{\xi}_{1,0*} & \boldsymbol{\xi}_{2,0*} & \cdots & \boldsymbol{\xi}_{p,0*} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\xi}_{1,q-1*} & \boldsymbol{\xi}_{2,q-1*} & \cdots & \boldsymbol{\xi}_{p,q-1*} \\ \boldsymbol{\beta}_* & \boldsymbol{\beta}_* & \cdots & \boldsymbol{\beta}_* \end{bmatrix} \quad \text{and} \quad \mathbf{k}_t := \begin{bmatrix} k_{t,1} \\ k_{t,2} \\ \vdots \\ k_{t,p-1} \\ k_{t,p} \end{bmatrix}. \quad (9)$$

□

We now estimate the unknown parameters in (7) by MR. Specifically, we let

$$\hat{\alpha}_n := (\hat{\lambda}'_n, \hat{\phi}'_n)' := (\hat{\alpha}_n, \hat{\delta}'_n, \hat{\gamma}'_n, \hat{\phi}'_n)' := \arg \max_{\alpha} Q_{nh}(\alpha)$$

be the MARDL parameter estimator, where n is the sample size;

$$Q_{nh}(\alpha) := \frac{1}{n} \sum_{t=1}^n \phi_h(y_t - \mathbf{z}'_t \alpha); \quad (10)$$

and $\phi_h(\cdot) := \frac{1}{h} \phi(\cdot/h)$ is a kernel density function symmetric around 0. For the remainder of this study we simply let $\phi(\cdot)$ be the standard normal density for analytical convenience. Throughout this study, $\phi(\cdot)$ denotes the kernel function, whereas ϕ_{j*} denotes autoregressive parameters.

Remarks 3. (i) The maximization of $Q_{nh}(\cdot)$ requires a numerical optimization procedure because its maximum has no closed-form expression. For this, we utilize the modal expectation-maximization (MEM) algorithm proposed by Li et al. (2007). Specifically, the MEM algorithm consists of two steps similarly to the EM algorithm. The E-step computes the weight of each observation. Given a parameter value, say $\alpha^{(g)}$, the posterior conditional probability of the t -th observation is computed as

$$\pi(t|\alpha^{(g)}) := \frac{\phi_h(y_t - \mathbf{z}'_t \alpha^{(g)})}{\sum_{i=1}^n \phi_h(y_i - \mathbf{z}'_i \alpha^{(g)})}.$$

Next, the M-step updates the parameter value using the weight calculated in the E-step by maximizing the following objective function:

$$\alpha^{(g+1)} := \arg \max_{\alpha} \sum_{t=1}^n \left\{ \pi(t|\alpha^{(g)}) \log[\phi_h(y_t - \mathbf{z}'_t \alpha)] \right\}.$$

Yao and Li (2014) demonstrate that $Q_{nh}(\alpha^{(g+1)}) > Q_{nh}(\alpha^{(g)})$, so that if we continue these two steps until $\|\alpha^{(g)} - \alpha^{(g+1)}\|$ becomes less than a prespecified value, we can attain a value quite close to the maximum of $Q_{nh}(\cdot)$.

(ii) The bandwidth is set as $h = 1.6 \cdot n^{-1/7} \cdot \widehat{\text{MAD}}_n$, where

$$\widehat{\text{MAD}}_n := \text{med}_t \left\{ |\hat{r}_t - \text{med}_s \hat{r}_s| \right\}, \quad \text{where} \quad \hat{r}_t := y_t - \hat{\mu}_m - \mathbf{x}'_t \hat{\beta}_m,$$

with $(\hat{\mu}_m, \hat{\beta}_m)$ a preliminary estimator for μ_* and β_* in (4), e.g., least-squares estimate. A first-stage scale estimate is required because the MARDL fit itself presupposes a bandwidth. Here, the constant

1.6 is multiplied by a pragmatic choice. As the cointegration residual $\mathcal{R}_t$ is serially correlated and differs from the modal error u_t , we cannot claim that it is selected by an optimality criterion. It is a numerical number calibrated by [Kemp and Silva \(2012\)](#). $\square$

We now examine the asymptotic behavior of the parameter estimator. For this, we provide the following regularized environment for the MARDL estimator along with the aforementioned assumptions:

Assumption 1. (i) $y_t = \mathbf{z}_t' \boldsymbol{\alpha}_* + u_t$ such that

(i.a) the parameter space $\mathcal{A}$ is compact and $\boldsymbol{\alpha}_*$ is an interior point of $\mathcal{A}$; and

(i.b) $\{u_t\}$ is an IID sequence with PDF $f_u(\cdot)$ and CDF $F_u(\cdot)$ such that $\mathbb{E}|u_t|^{2+\lambda} < \infty$ for some $\lambda \in (0, 1]$.

(ii) The kernel $\phi(\cdot) : \mathbb{R} \mapsto \mathbb{R}^+$ is a probability density function, symmetric about zero, such that

(ii.a) $\phi(\cdot)$ is four times continuously differentiable; $\phi^{(j)}(\cdot)$ is bounded, absolutely integrable, and square-integrable for $j = 1, 2, 3$; and $|t|^4 |\phi^{(j)}(t)| \rightarrow 0$ as $|t| \rightarrow \infty$ for $j = 0, 1, 2, 3$; and

(ii.b) $\int |\phi^{(1)}(t)|^{2+\lambda} dt < \infty$ and $\int t^4 \phi(t) dt < \infty$ for the same λ in (i.b). Let $v^2 := \int [\phi^{(1)}(t)]^2 dt$.

(iii) The PDF $f_u(\cdot)$ of u_t satisfies:

(iii.a) $f_u(\cdot) \in \mathcal{C}^{(3)}(\mathbb{R})$, and f_u and its first three derivatives are uniformly bounded;

(iii.b) $f_u(u) < f_u(0)$ for any $u \neq 0$; and

(iii.c) $f_u^{(2)}(0) < 0$, where $f_u^{(\ell)}(\cdot)$ denotes the ℓ -th order derivative of $f_u(\cdot)$.

(iv) $\mathbf{w}_t := \Delta \mathbf{x}_t \in \mathbb{R}^k$ ($k \in \mathbb{N}$) is a linear multivariate stationary and ergodic process such that

(iv.a) $\mathbb{E}[w_{tj}] = 0$ and $\mathbb{E}[|w_{tj}|^{2+\lambda}] < \infty$ for $j = 1, \dots, k$ for the same λ in (i.b);

(iv.b) the sequences $\{u_t\}$ and $\{\mathbf{w}_t\}$ are independent; and

(iv.c) the components of $\mathbf{x}_t$ are not cointegrated; and

(iv.d) $\mathbb{E}\|(\overline{\mathbf{w}}_t', \mathbf{k}_t')'\|^4 < \infty$.

(v) The roots of $(1 - \sum_{j=1}^p \phi_{j*} L^j)$ lie outside the unit circle, $\sum_{j=0}^{\infty} \|\boldsymbol{\pi}_{j*}\| < \infty$, and for each $i = 0, 1, \dots, p$, $\sum_{j=0}^{\infty} \|\boldsymbol{\xi}_{i,j*}\| < \infty$.

(vi) Let $u_{th} := u_t/h$, $m_h := \mathbb{E}[\phi^{(1)}(u_{th})]$, $\overline{\phi}^{(1)}(u_{th}) := \phi^{(1)}(u_{th}) - m_h$, $\overline{\mathbf{w}}_t := [\mathbf{w}_t', \dots, \mathbf{w}_{t-q+1}']'$, and $\overline{\mathbf{k}}_t := \mathbf{k}_t - \mathbb{E}[\mathbf{k}_t]$. Then $\mathbf{B}_n(\cdot) \rightsquigarrow \mathcal{B}(\cdot) := [\mathcal{B}_w(\cdot)', \mathcal{B}_k(\cdot)', \mathcal{B}_\phi(\cdot)', \mathcal{B}_{\phi \cdot \mathbf{w}}(\cdot)', \mathcal{B}_{\phi \cdot \mathbf{k}}(\cdot)']'$, where

$$\mathbf{B}_n(\cdot) := \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \left[\overline{\mathbf{w}}_t', \overline{\mathbf{k}}_t', \frac{\overline{\phi}^{(1)}(u_{th})}{\sqrt{h}}, \frac{\overline{\phi}^{(1)}(u_{th})}{\sqrt{h}} \overline{\mathbf{w}}_t', \frac{\overline{\phi}^{(1)}(u_{th})}{\sqrt{h}} \overline{\mathbf{k}}_t' \right]',$$

and $\mathcal{B}(\cdot)$ is a multivariate Gaussian process with $\mathbb{E}[\mathcal{B}(\cdot)] = \mathbf{0}$ and, for each $r, \tilde{r} \in (0, 1)$, $\mathbb{E}[\mathcal{B}(r)\mathcal{B}(\tilde{r})'] = \boldsymbol{\Omega}(r, \tilde{r}) := \lim_{n \rightarrow \infty} \mathbb{E}[\mathbf{B}_n(r)\mathbf{B}_n(\tilde{r})']$ is positive definite. $\square$

- Remarks 4.** (i) The conditions in Assumption 1 extend the regularity conditions for modal regression to the context of integrated series (see Kemp and Silva, 2012; Yao and Li, 2014; Cho et al., 2015; Ullah et al., 2022, 2023, among others). A key distinction from the pure stationary setting in Yao and Li (2014) is that their finite fourth moment condition violates in the MARDL context as $n^{-1} \sum_{t=1}^n \|\mathbf{x}_t\|^4 \asymp n^2$. We provide the present set of assumptions to resolve this issue by separating the asymptotic treatment of the stationary series from that of integrated series.
- (ii) Assumption 1 (ii) permits kernels with unbounded support, provided that the stated tail and integrability conditions hold. In particular, the standard normal density satisfies Assumption 1 (ii), and we maintain this choice throughout this study. For the standard normal kernel, $\phi^{(1)}(t) = -t\phi(t)$, so that $v^2 = \int [\phi^{(1)}(t)]^2 dt = \int t^2 \phi^2(t) dt = 1/(4\sqrt{\pi})$. Assumption 1 (ii.b) with $\lambda > 0$ is imposed to satisfy the Lindeberg-Lyapunov requirement to apply central limit theorem to modal score.
- (iii) Assumptions 1 (iii.a) and (iii.b) are imposed to ensure a well-separated unique global mode at zero. Together with the differentiability condition, they imply $f_u^{(1)}(0) = 0$, so the latter need not be assumed. By contrast, the strict curvature condition in Assumption 1 (iii.c) is *not* implied by unimodality. For example, $f_u(u) \propto \exp(-u^4)$ has a unique well-separated mode at zero, but $f_u^{(2)}(0) = 0$. We impose Assumption 1 (iii.c) for model identification. As it turns out $f_u^{(2)}(0) < 0$ is a key curvature condition and plays a role analogous to the quantile density $f_\tau := f(F^{-1}(\tau))$ in Cho et al. (2015). Furthermore the mode is well separated by Assumptions 1 (iii.a) and (iii.b): for any $\varepsilon > 0$, $\sup_{|u| \geq \varepsilon} f_u(u) < f_u(0)$. Indeed, $f_u^{(1)}(\cdot)$ is uniformly bounded by Assumption 1 (iii.a), so that $f_u(\cdot)$ is Lipschitz continuous and hence uniformly continuous. Furthermore, $f_u(\cdot)$ is integrable as it is a PDF, so that it vanishes at infinity from the fact that a uniformly continuous integrable function vanishes at infinity. Therefore there is $M_\varepsilon < \infty$ such that $f_u(u) < f_u(0)/2$ whenever $|u| > M_\varepsilon$, whereas on the compact set $\{\varepsilon \leq |u| \leq M_\varepsilon\}$ the continuous function $f_u(\cdot)$ attains its supremum, which is strictly less than $f_u(0)$ by Assumption 1 (iii.b).
- (iv) Assumption 1 (iv) imposes stationarity and ergodicity on $\mathbf{w}_t = \Delta \mathbf{x}_t$. Under the IID condition Assumption 1 (i.b) and the independence condition Assumption 1 (iv.b), the conditional density of u_t given $\mathcal{F}_{t-1}$ coincides with $f_u(\cdot)$. Furthermore, it follows that $\mathbb{E}[\phi^{(1)}(u_{th})|\mathcal{F}_{t-1}] = m_h$, so that $\{\bar{\phi}^{(1)}(u_{th}), \mathcal{F}_t\}$ is a martingale difference array (MDA). We can deliver its asymptotic behavior by Taylor's expansion. By Assumption 1 (iii) and the symmetry of $\phi(\cdot)$, it follows that

$$m_h = -\frac{h^4}{2} \mu_2 f_u^{(3)}(0) + o(h^4) \rightarrow 0 \quad \text{and}$$

$$\text{var}\left(\frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}}\right) = \frac{1}{h} \int [\phi^{(1)}(u/h)]^2 f_u(u) du - \frac{m_h^2}{h} \rightarrow v^2 f_u(0),$$

as $h \rightarrow 0$, where $\mu_2 := \int t^2 \phi(t) dt$, which equals unity for the standard normal kernel; see Section A.1 in the Online Supplement for this derivation. Therefore, it follows that

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}} = \frac{1}{\sqrt{nh}} \sum_{t=1}^n \left(\phi^{(1)}(u_{th}) - \mathbb{E}[\phi^{(1)}(u_{th})] \right) \overset{A}{\rightsquigarrow} N(0, v^2 f_u(0)),$$

and we can extend this to apply functional central limit theorem (FCLT), viz.,

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}} \rightsquigarrow \mathcal{B}_\phi(\cdot).$$

- (v) Assumption 1 (v) is imposed for the existence of a stable long-run relationship between y_t and x_t .
- (vi) By imposing the positive-definite matrix condition on $\mathbf{\Omega}(r, \tilde{r})$, we do not allow identical or constant sample paths among the processes in $\mathcal{B}(\cdot)$. Under Assumptions 1 (i.b) and (iv.b), every cross-covariance between the $\bar{w}$ -components and the score components of $\mathbf{B}_n(\cdot)$ vanishes identically, so that $\mathcal{B}_w(\cdot)$, $\mathcal{B}_\phi(\cdot)$, and $\mathcal{B}_{\phi \cdot w}(\cdot)$ are mutually independent Gaussian processes. In contrast, k_s is involved with lagged modal errors, so that the covariance between $\mathcal{B}_k(\cdot)$ and $\mathcal{B}_{\phi \cdot k}(\cdot)$ may not be zero.
- (vii) Let $\bar{\mathcal{B}}_w(\cdot)$ denote the first k elements of $\mathcal{B}_w(\cdot)$. By Assumptions 1 (iv) and (vi) and the continuous mapping theorem,

$$\frac{1}{n^2} \sum_{t=1}^n x_t x_t' \rightsquigarrow \int_0^1 \bar{\mathcal{B}}_w \bar{\mathcal{B}}_w'.$$

In addition,

$$\frac{1}{n\sqrt{h}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) x_t \rightsquigarrow \int_0^1 \bar{\mathcal{B}}_w d\mathcal{B}_\phi.$$

Unlike the former, the latter is not a consequence of the continuous mapping theorem. It follows from the MDA property of the centered score together with the fact that x_t is measurable with respect to $\mathcal{F}_{t-1}$; see Lemma A.4 in the Online Supplement. The independence condition in Assumption 1 (iv.b) further implies that the one-sided long-run covariance between the score and Δx_t is zero, so that its second-order bias term does not arise. The stochastic integral is mixed normal, viz.,

$$\int \bar{\mathcal{B}}_w d\mathcal{B}_\phi \sim N\left(\mathbf{0}, v^2 f_u(0) \int \bar{\mathcal{B}}_w \bar{\mathcal{B}}_w'\right)$$

conditional on $\bar{\mathcal{B}}_w(\cdot)$. This mixed normal distribution is one of the key ingredient of the limit theory

for $\hat{\alpha}_n$.

(viii) The independence condition in Assumption 1 (iv.b) is not as strong as it appears. The endogenous regressor problem can be removed by projecting the regression errors on $\Delta \mathbf{x}_t$, a device that carries over directly to the present MARDL context.

(ix) The higher level condition in Assumption 1 (vi) is attainable in a couple of ways. If $\{(\bar{\mathbf{w}}'_t, \bar{\mathbf{k}}'_t)'\}$ is an adapted mixingale of size -1 with uniformly bounded variance, we can apply functional central limit theorem to it in a manner similar to [de Jong and Davidson \(2000\)](#) by noting the Lindeberg condition supplied in Assumption 1 (ii.b) together with Assumptions 1 (i.b) and (iv.a). Alternatively, one can strengthen the summability condition in Assumption 1 (v) by imposing finite fourth moment condition. One of these two conditions does not necessarily imply another, so that we opt to impose the higher level assumption. $\square$

For later use, we define the following limit processes before discussing the limit distribution of $\hat{\alpha}_n$:

$$\tilde{\mathcal{B}}_w(r) := \bar{\mathcal{B}}_w(r) - \int \bar{\mathcal{B}}_w \quad \text{and} \quad \bar{\mathcal{B}}_\phi(r) := \mathcal{B}_\phi(r) - r \mathcal{B}_\phi(1),$$

which satisfy the following identity:

$$\int \bar{\mathcal{B}}_w d\bar{\mathcal{B}}_\phi = \int \tilde{\mathcal{B}}_w d\mathcal{B}_\phi \tag{11}$$

because $\int \tilde{\mathcal{B}}_w = 0$. The right side is mixed normal, viz., $\int \tilde{\mathcal{B}}_w d\mathcal{B}_\phi \sim N\left(0, v^2 f_u(0) \int \tilde{\mathcal{B}}_w \tilde{\mathcal{B}}_w'\right)$ conditional on $\bar{\mathcal{B}}_w(\cdot)$.

2.1.3 Limit Behavior of the Estimator

We now derive the limit distribution of $\hat{\alpha}_n$. For this we apply a third-order Taylor expansion to the first-order condition for $\hat{\alpha}_n$ around α_* , we obtain

$$0 = -\frac{1}{nh^2} \sum_{t=1}^n \phi^{(1)}(u_{th}) \mathbf{z}_t + \frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{z}_t \mathbf{z}_t' (\hat{\alpha}_n - \alpha_*) + \mathbf{r}_n, \tag{12}$$

where $\mathbf{r}_n := -(2nh^4)^{-1} \sum_{t=1}^n \phi^{(3)}(u_{th}^*) \{\mathbf{z}_t'(\hat{\alpha}_n - \alpha_*)\}^2 \mathbf{z}_t$ and u_{th}^* is an intermediate point between u_{th} and $\hat{u}_{th} := (y_t - \mathbf{z}_t' \hat{\alpha}_n)/h$. **The order of $\mathbf{r}_n$ is not used to obtain the convergence rates of $\hat{\alpha}_n$, so that no circularity arises. Lemma A.5 is proved directly from the objective function $\tilde{Q}_{nh}(\cdot)$ itself: it shows that, for every $\eta > 0$, there is a $\theta < \infty$ such that $\tilde{Q}_{nh}(\cdot)$ evaluated on the sphere of radius θ around $\vartheta_\dagger$ falls below**

$Q_{nh}(\boldsymbol{\vartheta}_{\dagger})$ with probability at least $1 - \eta$, which locates a local maximizer inside that sphere. That argument expands the objective function, not its first-order condition, and therefore presupposes nothing about $\mathbf{r}_n$. Only after the rates of Lemma A.5 are available do we use them, together with $\mathbb{E}[\phi^{(3)}(u_{th})] = O(h^4)$ and $\text{var}[\phi^{(3)}(u_{th})] = O(h)$ from Section A.1, to conclude that each component of $\mathbf{r}_n$ is asymptotically negligible relative to the first two terms in (12) when $nh^5 \rightarrow \infty$. Accordingly, (12) is used only to identify the limit distribution, after the rates are in hand. [JSCHO: This is not quite a right approach. Lemma A.5 is obtained under the assumption that $\mathbf{r}_n = o_{\mathbb{P}}(1)$, but here, it is claimed that $\mathbf{r}_n = o_{\mathbb{P}}(1)$ by Lemma A.5. Please fix this tautology.] [The circularity itself is removed by the order of argument stated above. We keep this comment in red because the conclusion still rests on the I_3 bound in the proof of Lemma A.5, which is not yet written out in full.]

Two features of $\mathbf{z}_t$ prevent a direct analysis of (12). First, both stationary and integrated components exist in $\mathbf{z}_t$, so that a common normalization does not apply. Second, the integrated components cannot be analyzed block by block, because every element of $\hat{\mathbf{y}}_t$ is cointegrated with $\mathbf{x}_t$ by (8), so that the Hessian blocks linking $\mathbf{x}_t$ and $\hat{\mathbf{y}}_t$ are of the same order as the diagonal blocks.

We resolve both difficulties by reparametrization. Using $\hat{\mathbf{y}}_t = \boldsymbol{\Gamma}'_t \mathbf{g}_t + \mathbf{k}_t$ from (9), we obtain that

$$\mathbf{z}'_t \boldsymbol{\alpha}_* = \mathbf{g}'_t \boldsymbol{\lambda}_* + \hat{\mathbf{y}}'_t \boldsymbol{\phi}_* = \mathbf{g}'_t (\boldsymbol{\lambda}_* + \boldsymbol{\Gamma}_* \boldsymbol{\phi}_*) + \mathbf{k}'_t \boldsymbol{\phi}_* =: \mathbf{g}'_t \boldsymbol{\varsigma}_{\dagger} + \mathbf{k}'_t \boldsymbol{\phi}_{\dagger}, \quad (13)$$

so that $\boldsymbol{\varsigma}_{\dagger} := \boldsymbol{\lambda}_* + \boldsymbol{\Gamma}_* \boldsymbol{\phi}_*$ and $\boldsymbol{\phi}_{\dagger} = \boldsymbol{\phi}_*$. If we decompose $\boldsymbol{\varsigma} := \boldsymbol{\lambda} + \boldsymbol{\Gamma}_* \boldsymbol{\phi}$ into $(\nu, \mathbf{v}', \boldsymbol{\kappa}')'$ to follow the structure of $\mathbf{g}_t$, then

$$\nu_{\dagger} = \alpha_* + \mu_*' \boldsymbol{\phi}_*, \quad \mathbf{v}_{\dagger} = \boldsymbol{\delta}_* + \boldsymbol{\Xi}_* \boldsymbol{\phi}_*, \quad \text{and} \quad \boldsymbol{\kappa}_{\dagger} = \boldsymbol{\gamma}_* + \boldsymbol{\beta}_* \sum_{j=1}^p \boldsymbol{\phi}_{j*} = \boldsymbol{\beta}_* \quad (14)$$

Therefore, the regressors in (13) are split into a purely stationary vector $\mathbf{j}_t := (1, \overline{\mathbf{w}}'_t, \mathbf{k}'_t)'$ with coefficient $\boldsymbol{\rho}_{\dagger} := (\nu_{\dagger}, \mathbf{v}'_{\dagger}, \boldsymbol{\phi}'_{\dagger})'$, and the single integrated regressor $\mathbf{x}_t$ with the long-run parameter $\boldsymbol{\beta}_*$ itself. Using this if we let $\boldsymbol{\vartheta} := (\boldsymbol{\rho}', \boldsymbol{\kappa}')' := (\nu, \mathbf{v}', \boldsymbol{\phi}', \boldsymbol{\kappa}')'$ and $\tilde{\mathbf{z}}_t := (\mathbf{j}'_t, \mathbf{x}'_t)'$, then $\mathbf{z}'_t \boldsymbol{\alpha}_* = \tilde{\mathbf{z}}'_t \boldsymbol{\vartheta}_{\dagger}$ by (13). Furthermore, from the definitions of $\boldsymbol{\varsigma}$ and $\boldsymbol{\rho}$, there exists a linear bijection relationship between $\boldsymbol{\alpha}$ and $\boldsymbol{\vartheta}$. We further let $\tilde{Q}_{nh}(\cdot)$ be a function defined on the space of $\boldsymbol{\vartheta}$ such that for each $\boldsymbol{\alpha}$, $Q_{nh}(\boldsymbol{\alpha}) = \tilde{Q}_{nh}(\boldsymbol{\vartheta})$. From this, we let

$$\hat{\boldsymbol{\vartheta}}_n := (\hat{\boldsymbol{\rho}}'_n, \hat{\boldsymbol{\kappa}}'_n)' := (\hat{\nu}_n, \hat{\mathbf{v}}'_n, \hat{\boldsymbol{\phi}}'_n, \hat{\boldsymbol{\kappa}}'_n)' := \arg \max_{\boldsymbol{\vartheta}} \tilde{Q}_{nh}(\boldsymbol{\vartheta})$$

in parallel to $\hat{\boldsymbol{\alpha}}_n$. Due to the bijection relationship between $\boldsymbol{\alpha}$ and $\boldsymbol{\vartheta}$, the estimators also exhibit the following

regular relationship:

$$\hat{\nu}_n = \hat{\alpha}_n + \mu_* \mathbf{L}'_p \hat{\phi}_n, \quad \hat{v}_n = \hat{\delta}_n + \Xi_* \hat{\phi}_n, \quad \text{and} \quad \hat{\kappa}_n = \hat{\gamma}_n + \beta_* \sum_{j=1}^p \hat{\phi}_{jn}. \quad (15)$$

In addition, $\hat{\phi}_n$ in $\hat{\alpha}_n$ is identical to $\hat{\phi}_n$ in $\hat{\vartheta}_n$.

Asymptotic Behavior of $\hat{\phi}_n$. Although each y_{t-j} is $I(1)$, the reparameterization (13) shows that the autoregressive parameter ϕ_* is the coefficient of the stationary regressor $\mathbf{k}_t$. Furthermore, (14) implies that the $I(1)$ information carried by $\hat{\mathbf{y}}_t$ is entirely vanished by the long-run relationship. By this simple fact, we can focus on the short-run variables to derive the limit distribution of $\hat{\phi}_n$. Specifically, Lemma A.6 (iii) in the Online Supplement shows that

$$\sqrt{nh^3}(\hat{\rho}_n - \rho_* - b_h \mathbf{e}_1) = \left(\frac{1}{nh^3} \sum_{t=1}^n \phi_t^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \right)^{-1} \frac{1}{\sqrt{nh^3}} \sum_{t=1}^n \phi_t^{(1)}(u_{th}) \mathbf{j}_t + o_{\mathbb{P}}(1) \overset{\Delta}{\sim} N(\mathbf{0}, \Psi^+), \quad (16)$$

where $\mathbf{e}_1 := (1, 0, \dots, 0)'$, $b_h := -h^2 f_u^{(3)}(0)/2f_u^{(2)}(0)$, and $\Psi^+ := \omega_* \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1}$ such that $\omega_* := v^2 f_u(0)/[f_u^{(2)}(0)]^2$. Here, $\hat{\rho}_n$ is obtained by minimizing $\tilde{Q}_n(\cdot)$, but $\hat{\phi}_n$ in $\hat{\rho}_n$ is exactly the same as $\hat{\phi}_n$ minimizing $Q_n(\cdot)$, so that we obtain the limit behavior of the latter by the limit distribution of the former. We also note that the asymptotic approximation of $\hat{\rho}_n$ is defined only by the stationary variables, reflecting the fact that the $I(1)$ information carried by $\hat{\mathbf{y}}_t$ does not influence the limit distribution of $\hat{\rho}_n$. If we let $\mathbf{j}_t = (1, \mathbf{s}_t')'$ by letting $\mathbf{s}_t := (\bar{\mathbf{w}}_t', \mathbf{k}_t')'$, then we can partition Ψ^+ as follows:

$$\Psi^+ := \begin{bmatrix} \psi_{00}^+ & \psi_{0s}^+ \\ \psi_{s0}^+ & \Psi \end{bmatrix} \quad \text{such that} \quad \Psi := \begin{bmatrix} \Psi_{\bar{\mathbf{w}}\bar{\mathbf{w}}} & \Psi_{\bar{\mathbf{w}}\mathbf{k}} \\ \Psi_{\mathbf{k}\bar{\mathbf{w}}} & \Pi_{p \times p} \end{bmatrix}. \quad (17)$$

Using the inverse matrix formula, it is trivial to show that $\Psi = \omega_* (\mathbb{E}[\mathbf{s}_t \mathbf{s}_t'] - \mathbb{E}[\mathbf{s}_t] \mathbb{E}[\mathbf{s}_t']^{-1})^{-1}$ by construction and that $\Pi = \omega_* (\mathbb{E}[\mathbf{k}_t \mathbf{k}_t'] - \mathbb{E}[\mathbf{k}_t \hat{\mathbf{w}}_t'] \mathbb{E}[\hat{\mathbf{w}}_t \hat{\mathbf{w}}_t']^{-1} \mathbb{E}[\hat{\mathbf{w}}_t \mathbf{k}_t'])^{-1}$ with $\hat{\mathbf{w}}_t := (1, \bar{\mathbf{w}}_t')'$, so that $\Psi = \omega_* [\text{var}(\mathbf{s}_t)]^{-1}$ and the asymptotic covariance matrix of $\hat{\phi}_n$ is given as Π . Furthermore, the asymptotic approximation in (16) can be concentrated with respect to $\hat{\phi}_n$. If we let $\hat{\mathbf{h}}_t'$ be the t -th row vector of $\hat{\mathbf{H}} := \mathbf{P}_{\hat{\mathbf{W}}} \mathbf{K}$ with $\mathbf{P}_{\hat{\mathbf{W}}} := \mathbf{I}_n - \hat{\mathbf{W}}(\hat{\mathbf{W}}' \hat{\mathbf{W}})^{-1} \hat{\mathbf{W}}'$ such that $\hat{\mathbf{W}} := [\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_n]'$ and $\mathbf{K} := [\mathbf{k}_1, \dots, \mathbf{k}_n]'$, then it follows that

$$\sqrt{nh}(\hat{\phi}_n - \phi_*) = \left(\frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \hat{\mathbf{h}}_t \hat{\mathbf{h}}_t' \right)^{-1} \frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \hat{\mathbf{h}}_t + o_{\mathbb{P}}(1) \overset{\Delta}{\sim} N(\mathbf{0}, \Pi), \quad (18)$$

from which we obtain the limit behavior of $\widehat{\phi}_n$ in parallel to the stationary modal-regression (Yao and Li 2014).

The following theorem contains the limit behavior:

Theorem 1. *Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$ as $n \rightarrow \infty$, then $\sqrt{nh^3}(\widehat{\phi}_n - \phi_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \mathbf{\Pi})$.* $\square$

Remarks 5. (i) If we let $\mathbf{h}_t := \mathbf{k}_t - \mathbb{E}[\mathbf{k}_t \widehat{\mathbf{w}}_t'] \mathbb{E}[\widehat{\mathbf{w}}_t \widehat{\mathbf{w}}_t']^{-1} \widehat{\mathbf{w}}_t$, it also follows that $\mathbf{\Pi} = \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1}$ by noting that $\mathbf{h}_t = \widehat{\mathbf{h}}_t + o_{\mathbb{P}}(1)$.

(ii) The independence condition between u_t and $\mathbf{j}_t$ given in Assumption 1 (i) plays a central role in deriving (18) from (16).

(iii) If $h \propto n^{-1/7}$, $nh^5 = O(n^{2/7}) \rightarrow \infty$ and $nh^7 = O(1)$. To motivate the rate, we note that the asymptotic MSE of the parameter estimator involves a variance with order $(nh^3)^{-1}$ and a squared bias with order h^4 , so that minimizing the asymptotic MSE with respect to h yields the optimal bandwidth $h \propto n^{-1/7}$ (Yao and Li, 2014; Ullah et al., 2022). Although this choice does *not* satisfy the undersmoothing condition $nh^7 \rightarrow 0$, the smoothing bias is associated only with the intercept estimator under Assumption 1 (iv.b). That is, all slope parameter estimators are free from the smoothing bias meaning that we can obtain their limit distribution using this optimal bandwidth.

(iv) As mentioned above, smoothing bias does not arise in Theorem 1. Under Assumption 1 (iv.b), the conditional density of y_t on $\mathcal{F}_{t-1}$ is obtained from $f_u(\cdot)$ by shifting its location. Smoothing with bandwidth h perturbs the objective function identically at every t , and the common perturbation can only shift the location, viz., the intercept. Therefore, the pseudo-true values of all slope coefficients coincide with the true values. The smoothing bias associated with the intercept estimation is given in (16). Given that $b_h := -\frac{h^2}{2} f_u^{(3)}(0)/f_u^{(2)}(0)$, if $f_u^{(3)}(0) \neq 0$, then it follows that

$$\sqrt{nh^3}(\widehat{\alpha}_n - \alpha_* - b_h) \overset{A}{\rightsquigarrow} N(0, \omega_\alpha), \quad \text{where } \omega_\alpha := \mathbf{s}'_\alpha \mathbf{\Psi}^+ \mathbf{s}_\alpha, \quad \text{and } \mathbf{s}_\alpha := (1, \mathbf{0}'_{kq}, -\mu_* \mathbf{l}'_p)'$$

by noting that $\widehat{\alpha}_n - \alpha_* = \mathbf{s}'_\alpha(\widehat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger)$; see Lemma A.7 (v) in the Online Supplement. Therefore, if $nh^7 = O(1)$, b_h asymptotically survives, but it vanishes if $nh^7 \rightarrow 0$. $\square$

Asymptotic Distribution of $\widehat{\gamma}_n$. For the asymptotic distribution of $\widehat{\gamma}_n$, we let $\widehat{\varsigma}_n := \widehat{\boldsymbol{\lambda}}_n + \mathbf{\Gamma}_* \widehat{\phi}_n$ by following the definition of $\varsigma := \boldsymbol{\lambda} + \mathbf{\Gamma}_* \phi$. Furthermore, $\boldsymbol{\kappa}_\dagger = \boldsymbol{\beta}_*$ by (14), so that if we decompose $\widehat{\varsigma}_n$ into $(\widehat{\nu}_n, \widehat{\mathbf{v}}'_n, \widehat{\boldsymbol{\kappa}}'_n)'$ to follow the structure of $\mathbf{g}_t$, then it follows that $\widehat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_* = (\widehat{\gamma}_n - \gamma_*) + \boldsymbol{\beta}_* \mathbf{l}'_p(\widehat{\phi}_n - \phi_*)$. Furthermore, Lemma A.6 (ii) in the Online Supplement obtains the asymptotic approximation of $\widehat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*$ as

follows:

$$\hat{\kappa}_n - \beta_* = \left[\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right]^{-1} \left[\frac{1}{n^2 h^2} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \right] + o_{\mathbb{P}}((n\sqrt{h^3})^{-1}), \quad (19)$$

where $\tilde{\mathbf{x}}_t'$ is the t -th row vector of $\tilde{\mathbf{X}} := \mathbf{P}_{\hat{\mathbf{W}}} \mathbf{X}$. The limit distribution of $(\hat{\gamma}_n - \gamma_*)$ is obtained by combining (19) with (18). Given Theorem 1, we further show that

$$\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \quad \text{and} \quad \frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \rightsquigarrow \int \tilde{\mathbf{B}}_w d\mathbf{B}_\phi; \quad (20)$$

see Lemmas A.6 (i) and A.4 (v) in the Online Supplement. Here, the second limit can also be written as $\int \tilde{\mathbf{B}}_w d\mathbf{B}_\phi$ by (11). Therefore, two convergence rates are in play: $\hat{\kappa}_n - \beta_* = O_{\mathbb{P}}((n\sqrt{h^3})^{-1})$ and $\hat{\phi}_n - \phi_* = O_{\mathbb{P}}((nh^3)^{-1/2})$, implying that the convergence rate of $\hat{\gamma}_n$ is determined by the $\hat{\phi}_n$ because its convergence rate is slower than that of $\hat{\kappa}_n$. That is,

$$\sqrt{nh^3} (\hat{\gamma}_n - \gamma_*) = -\beta_* \boldsymbol{\nu}_p' \sqrt{nh^3} (\hat{\phi}_n - \phi_*) + O_{\mathbb{P}}(n^{-1/2}). \quad (21)$$

By this and (19), the rate $n\sqrt{h^3}$ can be attached to $\hat{\kappa}_n - \beta_*$.

We provide the limit distributions of the estimators in the following theorem:

Theorem 2. *Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$ as $n \rightarrow \infty$,*

- (i) $n\sqrt{h^3} \{(\hat{\gamma}_n - \gamma_*) + \beta_* \boldsymbol{\nu}_p' (\hat{\phi}_n - \phi_*)\} \overset{A}{\rightsquigarrow} N(\mathbf{0}, \omega_* [\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w']^{-1})$ conditional on $\bar{\mathbf{B}}_w(\cdot)$;
- (ii) $\sqrt{nh^3} (\hat{\gamma}_n - \gamma_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \sigma_\phi^2 \beta_* \beta_*')$, where $\sigma_\phi^2 := \boldsymbol{\nu}_p' \boldsymbol{\Pi} \boldsymbol{\nu}_p$. □

Remarks 6. (i) By (19), Theorem 2 (i) also implies that $n\sqrt{h^3}(\hat{\kappa}_n - \beta_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \omega_* [\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w']^{-1})$ conditional on $\bar{\mathbf{B}}_w(\cdot)$.

(ii) The sum of the rows in $\mathbf{P}_{\hat{\mathbf{W}}} \mathbf{X}$ is identical to zero by construction. This implies that the score in (19) is invariant to recentering of $\phi^{(1)}(u_{th})$, and the smoothing bias associated with the intercept estimation does not influence the score.

(iii) Theorem 2 (ii) implies that Wald inference on γ_* is nonstandard because the rank of the asymptotic covariance matrix is one. We revisit this more specifically in Section 2.2. □

Asymptotic Distribution of $\hat{\beta}_n$. In order to derive the asymptotic distribution of $\hat{\beta}_n$, we exploit the following identity:

$$\hat{\beta}_n - \beta_* \equiv \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n}\right]^{-1} \left[(\hat{\gamma}_n - \gamma_*) + \beta_* \sum_{j=1}^p (\hat{\phi}_{j,n} - \phi_{j*})\right], \quad (22)$$

which follows from (5). The bracketed term on the right is exactly the same as $\hat{\kappa}_n - \beta_*$ in (19), so that if we combine (19) with (22),

$$n\sqrt{h^3}(\hat{\beta}_n - \beta_*) = \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n}\right]^{-1} \left[\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t'\right]^{-1} \left[\frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t\right] + o_{\mathbb{P}}(1).$$

Here, $\sum_j \hat{\phi}_{j,n} \xrightarrow{\mathbb{P}} \sum_j \phi_{j*}$ by Theorem 1, and we can apply (20) to obtain the limit distribution of $\hat{\beta}_n$. The following theorem provides the limit distribution:

Theorem 3. Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$ as $n \rightarrow \infty$,

$$n\sqrt{h^3}(\hat{\beta}_n - \beta_*) \overset{A}{\sim} N\left(\mathbf{0}, \frac{\omega_*}{[1 - \sum_{j=1}^p \phi_{j*}]^2} \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'\right]^{-1}\right)$$

conditional on $\bar{\mathbf{B}}_w(\cdot)$. □

This is the MARDL analogue of Cho et al. (2015, theorem 2).

Asymptotic Distribution of $\tilde{\varphi}_n$. The short-run autoregressive parameters in the EC representation are recovered from ϕ_* via (6): $\varphi_* = -\mathbf{R}_1 \phi_*$. We let the short-run estimator in the EC representation be $\tilde{\varphi}_n := -\mathbf{R}_1 \hat{\phi}_n$ by the plug-in principle. The following theorem delivers its limit behavior:

Theorem 4. Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$, then $\sqrt{nh^3}(\tilde{\varphi}_n - \varphi_*) \overset{A}{\sim} N(\mathbf{0}, \mathbf{R}_1 \mathbf{I} \mathbf{R}_1')$. □

The covariance matrix $\mathbf{R}_1 \mathbf{I} \mathbf{R}_1'$ is positive definite because $\mathbf{R}_1$ has full row rank.

Asymptotic Distribution of $\tilde{\psi}_n$. If we let $\mathbf{S} := [\mathbf{I}_k, \mathbf{0}_{k \times k(q-1)}]'$, $\psi_* = \delta_* + \mathbf{S} \gamma_*$ by (6). We therefore estimate ψ_* by $\tilde{\psi}_n := \hat{\delta}_n + \mathbf{S} \hat{\gamma}_n$. Furthermore, (13) gives $\mathbf{v}_\dagger = \delta_* + \Xi_* \phi_*$, so that (21) and $\hat{\delta}_n - \delta_* = (\hat{\mathbf{v}}_n - \mathbf{v}_\dagger) - \Xi_*(\hat{\phi}_n - \phi_*)$ delivers

$$\sqrt{nh^3}(\tilde{\psi}_n - \psi_*) = [\mathbf{I}_{kq} \ : \ -\mathbf{C}] \sqrt{nh^3} \begin{bmatrix} \hat{\mathbf{v}}_n - \mathbf{v}_\dagger \\ \hat{\phi}_n - \phi_* \end{bmatrix} + O_{\mathbb{P}}(n^{-1/2}), \quad \text{where } \mathbf{C} := \Xi_* + \mathbf{S} \beta_* \iota_p', \quad (23)$$

because $\hat{\kappa}_n - \beta_* = O_{\mathbb{P}}((n\sqrt{h^3})^{-1})$. Therefore, all components of $\tilde{\psi}_n$ converge at the rate of $\sqrt{nh^3}$. Given this, (16) implies that

$$\sqrt{nh^3} \begin{bmatrix} \hat{\mathbf{v}}_n - \mathbf{v}_{\dagger} \\ \hat{\phi}_n - \phi_* \end{bmatrix} = \left(\frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{s}_t \mathbf{s}_t' \right)^{-1} \frac{1}{\sqrt{nh^3}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{s}_t + o_{\mathbb{P}}(1) \stackrel{A}{\sim} N(\mathbf{0}, \Psi),$$

so that we can obtain the limit distributions of $\hat{\delta}_n$ and $\tilde{\psi}_n$ using this asymptotic distribution.

The following theorem contains the limit distribution of the estimators:

Theorem 5. *Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$,*

- (i) $\sqrt{nh^3} (\hat{\delta}_n - \delta_*) \stackrel{A}{\sim} N(\mathbf{0}, \Omega_{\delta})$, where $\Omega_{\delta} := [\mathbf{I}_{kq} \dot{\vdash} -\Xi_*] \Psi [\mathbf{I}_{kq} \dot{\vdash} -\Xi_*]'$;
- (ii) $\sqrt{nh^3} (\tilde{\psi}_n - \psi_*) \stackrel{A}{\sim} N(\mathbf{0}, \Omega_{\psi})$, where $\Omega_{\psi} := [\mathbf{I}_{kq} \dot{\vdash} -\mathbf{C}] \Psi [\mathbf{I}_{kq} \dot{\vdash} -\mathbf{C}]'$. □

Remarks 7. (i) Both Ω_{δ} and Ω_{θ} are positive definite by construction.

- (ii) The limit distribution of $\tilde{\psi}_n$ is not a mixed-normal distribution but a normal distribution. □

2.2 Parameter Inference

In this section, we develop inference procedures using the estimators given in Section 2.1.

Testing the Long-Run Parameter β_* . We suppose the following hypotheses on the long-run parameter β_* :

$$\mathbb{H}_0^{(1)} : \mathbf{Q}\beta_* = \mathbf{q} \quad \text{versus} \quad \mathbb{H}_1^{(1)} : \mathbf{Q}\beta_* \neq \mathbf{q},$$

where $\mathbf{Q}$ is an $r \times k$ matrix with $\text{rank}(\mathbf{Q}) = r$ and $\mathbf{q}$ is an $r \times 1$ vector. In order to test the hypotheses, we apply Wald test principle by letting the following be a test statistic:

$$\mathcal{W}_n(\beta) := h^3 (\mathbf{Q}\hat{\beta}_n - \mathbf{q})' \left(\hat{\omega}_n \mathbf{Q} \left[1 - \sum_{j=1}^p \hat{\phi}_{j,n} \right]^{-2} \left(\sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right)^{-1} \mathbf{Q}' \right)^{-1} (\mathbf{Q}\hat{\beta}_n - \mathbf{q}),$$

where

$$\hat{\omega}_n := \frac{v^2 \hat{f}_{un}(0)}{[\hat{f}_{un}^{(2)}(0)]^2} \quad \text{such that} \quad \hat{f}_{un}^{(\ell)}(0) := \frac{1}{n\tilde{h}^{\ell+1}} \sum_{t=1}^n K^{(\ell)}\left(\frac{\hat{u}_t}{\tilde{h}}\right), \quad \ell = 0, 1, 2.$$

Here, $\hat{u}_t := y_t - \mathbf{z}_t' \hat{\alpha}_n$, $K(\cdot)$ is a kernel function, and $\tilde{h}$ is a pilot bandwidth chosen independently of h such that $\tilde{h} \rightarrow 0$, $n\tilde{h} \rightarrow \infty$, and $n\tilde{h}^5 \rightarrow \infty$. From this and the fact that $\max_t |\hat{u}_t - u_t| = o_{\mathbb{P}}(1)$ ¹, it follows that

¹This follows from the fact that $\hat{\alpha}_n$ is consistent for α_* .

$\widehat{f}_u(0) \xrightarrow{\mathbb{P}} f_u(0)$ and $\widehat{f}_u^{(2)}(0) \xrightarrow{\mathbb{P}} f_u^{(2)}(0)$ (see Yao and Li, 2014, section 3.1). In addition, $v^2 := \int [\phi^{(1)}(t)]^2 dt$ is easily computed from the assumed kernel function, letting $\widehat{\omega}_n \xrightarrow{\mathbb{P}} \omega_*$.

The Wald test is defined by exploiting the limit distribution of the long-run parameter estimator. Theorem 3 provides its limit distribution such that its limit covariance matrix is delivered by its sample analog:

$$\frac{\widehat{\omega}_n}{[1 - \sum_{j=1}^p \widehat{\phi}_{j,n}]^2} \frac{1}{n^2} \sum_{t=1}^n \widetilde{\mathbf{x}}_t \widetilde{\mathbf{x}}_t' \rightsquigarrow \frac{\omega_*}{[1 - \sum_{j=1}^p \phi_{j*}]^2} \int \widetilde{\mathbf{B}}_w \widetilde{\mathbf{B}}_w'.$$

From this it follows that $\mathcal{W}_n(\boldsymbol{\beta}) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(1)}$.

We contain the limit behavior of the Wald test in the following corollary:

Corollary 1. *Given Assumption 1, if $\text{rank}(\mathbf{Q}) = r$, $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$,*

- (i) $\mathcal{W}_n(\boldsymbol{\beta}) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(1)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\boldsymbol{\beta}) \geq s_n^{(1)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(1)}$, for any $s_n^{(1)} = o(n^2 h^3)$. □

Corollary 1 follows from Theorem 3 under the null and alternative. Unlike the level coefficient γ_* , the Wald test does not suffer from singular matrix problem as the weight matrix is nonsingular with probability 1.

Testing the Short-Run Parameter ϕ_* . We suppose the following hypotheses:

$$\mathbb{H}_0^{(2)} : \mathbf{R}\phi_* = \mathbf{r} \quad \text{versus} \quad \mathbb{H}_1^{(2)} : \mathbf{R}\phi_* \neq \mathbf{r},$$

where $\mathbf{R}$ is an $r \times p$ matrix with $\text{rank}(\mathbf{R}) = r$, and $\mathbf{r}$ is an $r \times 1$ vector. By Theorem 1, $\widehat{\phi}_n$ is asymptotically normal. Using this fact, we let the following be the Wald test:

$$\mathcal{W}_n(\phi) := nh^3 (\mathbf{R}\widehat{\phi}_n - \mathbf{r})' \left(\mathbf{R}\widehat{\Pi}_n \mathbf{R}' \right)^{-1} (\mathbf{R}\widehat{\phi}_n - \mathbf{r}),$$

where $\widehat{\Pi}_n$ is a consistent estimator for $\Pi := \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1}$. For example, a feasible estimator can be obtained by replacing the population moments in Π with their sample analogs: $\widehat{\Pi}_n = \widehat{\omega}_n (n^{-1} \sum_{t=1}^n \ddot{\mathbf{h}}_t \ddot{\mathbf{h}}_t')^{-1}$, where $\ddot{\mathbf{h}}_t := \widehat{\mathbf{k}}_t - \sum_{t=1}^n \widehat{\mathbf{k}}_t \widehat{\mathbf{w}}_t' (\sum_{t=1}^n \widehat{\mathbf{w}}_t \widehat{\mathbf{w}}_t')^{-1} \widehat{\mathbf{w}}_t$ such that $\widehat{\mathbf{k}}_t := [\widehat{k}_{t,1}, \dots, \widehat{k}_{t,p}]'$, and for each $i = 1, \dots, p$, $\widehat{k}_{t,i}$ is the residual obtained by regressing $y_{t-i} - \mathbf{x}_t' \widehat{\beta}_n$ on $\widehat{\mathbf{w}}_t = (1, \widehat{\mathbf{w}}_t')'$. By (8), $\widehat{\mathbf{k}}_t$ consistently estimates $\mathbf{k}_t$, so that $\widehat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$ because $\widehat{\omega}_n \xrightarrow{\mathbb{P}} \omega_*$ as discussed above.

We state the limit behaviors of $\mathcal{W}_n(\phi)$ as follows:

Corollary 2. *Given Assumption 1, if $\widehat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$, $\text{rank}(\mathbf{R}) = r$, $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$ as $n \rightarrow \infty$,*

- (i) $\mathcal{W}_n(\phi) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(2)}$;
(ii) $\mathbb{P}(\mathcal{W}_n(\phi) \geq s_n^{(2)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(2)}$, for any $s_n^{(2)} = o(nh^3)$. □

Corollary 2 follows from Theorem 1.

Testing the Short-Run Parameter γ_* . Inference on γ_* is fundamentally different from inferring β_* or ϕ_* . The main reason for this is in the structural feature of the modal cointegrating system. By Theorem 2 (ii), the rank of the asymptotic covariance matrix of $\hat{\gamma}_n$ is unity. Consequently, we cannot apply the standard Wald test principle to $\hat{\gamma}_n$ to test the following standard hypotheses:

$$\mathbb{H}_0^{(3)} : \mathbf{Q}\gamma_* = \mathbf{q} \quad \text{versus} \quad \mathbb{H}_1^{(3)} : \mathbf{Q}\gamma_* \neq \mathbf{q}. \quad (24)$$

Alternative testing procedures can be developed in a couple of ways. We note that

$$\mathbf{Q}(\hat{\gamma}_n - \gamma_*) = \mathbf{Q}(\hat{\kappa}_n - \beta_*) - (\mathbf{Q}\beta_*)' \iota_p'(\hat{\phi}_n - \phi_*) \quad (25)$$

by (19), and each component on the right side has different convergence rate from each other as discussed above. Specifically, the convergence rate of $\hat{\phi}_n$ is slower than that of $\hat{\kappa}_n$, and we can develop testing procedures exploiting both components on the right side separately. First, we suppose that $\mathbf{Q}\beta_* = \mathbf{0}$, so that (25) reduces to $\mathbf{Q}(\hat{\gamma}_n - \gamma_*) = \mathbf{Q}(\hat{\kappa}_n - \beta_*)$, and the limit distribution of $(\hat{\kappa}_n - \beta_*)$ is given in Theorem 2 (i). We let the Wald test be defined as follows:

$$\mathcal{W}_n^{(0)}(\gamma) := h^3(\mathbf{Q}\hat{\gamma}_n - \mathbf{q})' \left(\hat{\omega}_n \mathbf{Q} \left[\sum_{t=1}^n \tilde{x}_t \tilde{x}_t' \right]^{-1} \mathbf{Q}' \right)^{-1} (\mathbf{Q}\hat{\gamma}_n - \mathbf{q}).$$

Second, if $\mathbf{Q}\beta_* \neq \mathbf{0}$, the second term in the right side of (25) dominates, so that it follows that $\sqrt{nh^3} \mathbf{Q}(\hat{\gamma}_n - \gamma_*) \overset{A}{\sim} N(\mathbf{0}, \sigma_\phi^2 (\mathbf{Q}\beta_*)(\mathbf{Q}\beta_*)')$ by Theorem 2 (ii). The rank of the asymptotic covariance matrix is unity, so that defining a nondegenerate Wald test is attainable only when $\text{rank}(\mathbf{Q}) = 1$. We cannot test multiple restrictions jointly using the limit distribution of $\hat{\gamma}_n$. If we let the first row of $\mathbf{Q}$ and $\mathbf{q}$ be $\mathbf{q}_1'$ and q_1 , respectively such that $\mathbf{q}_1' \beta_* \neq 0$, then Wald test can be defined as follows:

$$\mathcal{W}_n^{(1)}(\gamma) := \frac{nh^3 (\mathbf{q}_1' \hat{\gamma}_n - q_1)^2}{\hat{\sigma}_{\phi n}^2 (\mathbf{q}_1' \hat{\beta}_n)^2}, \quad \text{where} \quad \hat{\sigma}_{\phi n}^2 := \iota_p' \hat{\Pi}_n \iota_p.$$

Here, $\hat{\Pi}_n$ is already defined while defining $\mathcal{W}_n(\phi)$. We collect the asymptotic behaviors of the Wald tests as follows:

Corollary 3. Given Assumption 1, let $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$.

- (i) If $\mathbf{Q}\beta_* = \mathbf{0}$ and $\text{rank}(\mathbf{Q}) = r$, then
 - (i.a) $\mathcal{W}_n^{(0)}(\gamma) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(3)}$;
 - (i.b) $\mathbb{P}(\mathcal{W}_n^{(0)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(3)}$ for any $s_n = o(n^2h^3)$;
- (ii) If $\mathbf{Q}\beta_* \neq \mathbf{0}$, $\text{rank}(\mathbf{Q}) = 1$, and $\hat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$, then
 - (ii.a) $\mathcal{W}_n^{(1)}(\gamma) \overset{A}{\sim} \chi_1^2$ under $\mathbb{H}_0^{(3)}$;
 - (ii.b) $\mathbb{P}(\mathcal{W}_n^{(1)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(3)}$ for any $s_n = o(nh^3)$. □

Remarks 8. (i) $\mathbf{Q}\beta_* = \mathbf{0}$ or $\neq \mathbf{0}$ is a critical condition to apply Corollary 3. Before testing $\mathbb{H}_0^{(3)}$, it is necessary to test $\mathbf{Q}\beta_* = \mathbf{0}$ using $\mathcal{W}_n(\beta)$.

(ii) The form of $\mathcal{W}_n^{(0)}(\gamma)$ is parallel to that of $\mathcal{W}_n(\beta)$. This follows from the identity $\gamma_* \equiv (1 - \sum_{j=1}^p \phi_{j*})\beta_*$. Given that $\sum_{j=1}^p \phi_{j*} < 1$, imposing $\mathbf{Q}\gamma_* = \mathbf{0}$ is equivalent to letting $\mathbf{Q}\beta_* = \mathbf{0}$, so that Corollary 3 implicitly assumes that $\mathbb{H}_0^{(3)} : \mathbf{Q}\beta_* = \mathbf{0}$, and $\mathcal{W}_n^{(0)}(\gamma)$ is defined by taking the weight matrix of $\mathcal{W}_n(\beta)$ as its weight matrix.

(iii) In case $\mathbf{Q}\beta_* \neq \mathbf{0}$ but $\text{rank}(\mathbf{Q}) = r > 1$, then we may apply $\mathcal{W}_n^{(1)}(\gamma)$ by replacing $\mathbf{q}_1$ and q_1 with each row of $\mathbf{Q}$ and $\mathbf{q}$, respectively, so that r -number of Wald tests can be separately applied in total. □

Testing the Short-Run Parameter φ_* . In order to test the short-run autoregressive parameters $\varphi_* = -\mathbf{R}_1\phi_*$, we suppose the following hypotheses:

$$\mathbb{H}_0^{(4)} : \mathbf{N}\varphi_* = \mathbf{p} \quad \text{versus} \quad \mathbb{H}_1^{(4)} : \mathbf{N}\varphi_* \neq \mathbf{p},$$

where $\mathbf{N}$ is $r \times (p-1)$ with $\text{rank}(\mathbf{N}) = r \leq p-1$ and $\mathbf{p}$ is $r \times 1$. Since $\tilde{\varphi}_n = -\mathbf{R}_1\hat{\phi}_n$ is asymptotically normal by Theorem 4, we define the following Wald test:

$$\mathcal{W}_n(\varphi) := nh^3(\mathbf{N}\tilde{\varphi}_n - \mathbf{p})'(\mathbf{N}\mathbf{R}_1\hat{\Pi}_n\mathbf{R}_1'\mathbf{N}')^{-1}(\mathbf{N}\tilde{\varphi}_n - \mathbf{p}).$$

We contain the asymptotic behaviors of the Wald test in the following corollary:

Corollary 4. Given Assumption 1, if $\hat{\Pi}_n \xrightarrow{\mathbb{P}} \Pi$, $\text{rank}(\mathbf{N}) = r$, $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$, then

- (i) $\mathcal{W}_n(\varphi) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(4)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\varphi) \geq s_n^{(4)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(4)}$, for any $s_n^{(4)} = o(nh^3)$. □

As Corollary 4 follows from Theorem 4, we omit the proof.

Testing the Short-Run Parameter ψ_* . We test the following hypotheses on ψ_* :

$$\mathbb{H}_0^{(5)} : \mathbf{F}\psi_* = \mathbf{f} \quad \text{versus} \quad \mathbb{H}_1^{(5)} : \mathbf{F}\psi_* \neq \mathbf{f},$$

where $\mathbf{F}$ is $r \times (kq)$ with $\text{rank}(\mathbf{F}) = r$ and $\mathbf{f}$ is $r \times 1$. By Theorem 5 (iii), $\tilde{\psi}_n$ converges at $\sqrt{nh^3}$ to a normal limit. So, we let the following the Wald test to test these hypotheses:

$$\mathcal{W}_n(\psi) := nh^3(\mathbf{F}\tilde{\psi}_n - \mathbf{f})'(\mathbf{F}\hat{\Omega}_{\psi n}\mathbf{F}')^{-1}(\mathbf{F}\tilde{\psi}_n - \mathbf{f}),$$

where $\hat{\Omega}_{\psi n}$ is a consistent estimator for $\Omega_\psi := [\mathbf{I}_{kq} \vdots -\mathbf{C}] \Psi [\mathbf{I}_{kq} \vdots -\mathbf{C}]'$. For example, a feasible estimator can be defined as $\hat{\Omega}_{\psi n} := [\mathbf{I}_{kq} \vdots -\hat{\mathbf{C}}_n] \hat{\Psi}_n [\mathbf{I}_{kq} \vdots -\hat{\mathbf{C}}_n]'$, where $\hat{\mathbf{C}}_n := \hat{\Xi}_n + \mathbf{S}\hat{\beta}_n\boldsymbol{\nu}'_p$, and $\hat{\Psi}_n$ and $\hat{\Xi}_n$ are constructed to estimate $\Psi := \omega_*(\mathbb{E}[\mathbf{s}_t\mathbf{s}'_t] - \mathbb{E}[\mathbf{s}_t]\mathbb{E}[\mathbf{s}_t']^{-1})$ and Ξ_* consistently such that $\mathbf{s}_t := (\bar{\mathbf{w}}'_t, \mathbf{k}'_t)'$ as follows:

$$\hat{\Psi}_n := \hat{\omega}_n \left(\frac{1}{n} \sum_{t=1}^n \hat{\mathbf{s}}_t \hat{\mathbf{s}}'_t \right)^{-1} \quad \text{and} \quad \hat{\Xi}_n := [\hat{\xi}_{1,n}, \dots, \hat{\xi}_{p,n}],$$

where $\hat{\mathbf{s}}_t := (\bar{\mathbf{w}}'_t, \mathbf{k}'_t)'$, and $\hat{\xi}_{i,n}$ is the OLS coefficient of $\bar{\mathbf{w}}_t$ that is obtained by regressing $y_{t-i} - \mathbf{x}'_{t-i}\hat{\beta}_n$ on $\bar{\mathbf{w}}_t := (1, \bar{\mathbf{w}}'_t)'$ for $i = 1, \dots, p$. Both $\hat{\Psi}_n$ and $\hat{\Xi}_n$ are the sample analogs of Ψ and Ξ_* , respectively, so that $\hat{\mathbf{C}}_n := \hat{\Xi}_n + \mathbf{S}\hat{\beta}_n\boldsymbol{\nu}'_p \xrightarrow{\mathbb{P}} \mathbf{C}$, and $\hat{\Omega}_{\psi n} \xrightarrow{\mathbb{P}} \Omega_\psi$.

We contain the limit behaviors of the Wald test in the following corollary:

Corollary 5. *Given Assumption 1, if $\hat{\Omega}_{\psi n} \xrightarrow{\mathbb{P}} \Omega_\psi$, $\text{rank}(\mathbf{F}) = r$, $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$ as $n \rightarrow \infty$, then*

- (i) $\mathcal{W}_n(\psi) \overset{A}{\sim} \chi_r^2$ under $\mathbb{H}_0^{(5)}$;
- (ii) $\mathbb{P}(\mathcal{W}_n(\psi) \geq s_n^{(5)}) \xrightarrow{\mathbb{P}} 1$ under $\mathbb{H}_1^{(5)}$, for any $s_n^{(5)} = o(nh^3)$. □

As Corollary 5 follows from Theorem 5, we omit the proof.

3 Monte Carlo Simulations

In this section, we examine the properties of the Wald tests defined in Section 2.2 to corroborate the asymptotic theory on the estimators and test statistics.

3.1 Simulation Using MARDL Model

We focus on estimating an MARDL model. To this end, we examine MARDL (1, 1):

$$y_t = \alpha_{1*} + \phi_{1*}y_{t-1} + \theta_{0*}x_t + \theta_{1*}x_{t-1} + u_t,$$

where $x_t := x_{t-1} + w_t$ such that $w_t := \rho_* r_t + \sqrt{1 - \rho_*^2} r_{t-1}$ and $r_t \sim_{\text{iid}} U(-0.5, 0.5)$. We set the parameter values at $\alpha_{1*} = 0, \theta_{0*} = 0.5, \theta_{1*} = 0.5, \phi_{1*} = 0.5$, and $\rho_* = 0.5$ to examine the empirical rejection rates of the Wald tests under the null and alternative hypotheses. By construction, $\gamma_{1*} = 1$ and $\beta_{1*} = 2$.

Given this, we estimate the unknown parameters in MARDL(1,1) model. For this, we further consider three distinct data generating processes (DGPs) by assuming three different scenarios for u_t that is assumed to be independent of x_t :

- DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$, so that $\mathbb{E}(u_t) = 0$, $\text{median}(u_t) = 0$, and $\text{mode}(u_t) = 0$.
- DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$. By this, $\mathbb{E}(u_t) = 0.0833$, $\text{median}(u_t) = 0.0618$, and $\text{mode}(u_t) = 0$.
- DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$, so that $\mathbb{E}(u_t) = 1$, $\text{median}(u_t) = 0.6741$, and $\text{mode}(u_t) = 0$.

We examine the finite sample performance of the Wald tests. Null simulation results are reported in Table 1. Hypothesis systems are specified as follows: first, the long-run parameter β_* is hypothesized as follows: $\mathbb{H}_0(\beta) : \beta_* = 2$ versus $\mathbb{H}_1(\beta) : \beta_* \neq 2$, which we test by $\mathcal{W}_n(\beta)$. Second, the short-run parameters ϕ_* and γ_* are hypothesized as follows: $\mathbb{H}_0(\phi) : \phi_* = 0.5$ versus $\mathbb{H}_1(\phi) : \phi_* \neq 0.5$, which we test by $\mathcal{W}_n(\phi)$. In addition, $\mathbb{H}_0(\gamma) : \gamma_* = 1$ versus $\mathbb{H}_1(\gamma) : \gamma_* \neq 1$. As $k = 1$ and $\beta_* \neq 0$ under $\mathbb{H}_0(\gamma)$, this null has to be tested by $\mathcal{W}_n^{(1)}(\gamma)$. Simulation results are obtained by repeating 3000 independent experiments and we report the empirical rejection rates of $\mathcal{W}_n(\beta)$, $\mathcal{W}_n(\phi)$, and $\mathcal{W}_n^{(1)}(\gamma)$ in Table 1 for seven different sample sizes, $n = 200, 400, 600, 800, 1,000, 2000$, and $3,000$, when the significance level is 5%.

Next, we report power simulation results in Table 2. For simulation, we use the closer fixed false null values $\beta_* = 1.95$, $\phi_* = 0.35$, and $\gamma_* = 0.70$, respectively. Thus, we test $\mathbb{H}_0(\beta) : \beta_* = 1.95$ versus $\mathbb{H}_1(\beta) : \beta_* \neq 1.95$, $\mathbb{H}_0(\phi) : \phi_* = 0.35$, and $\mathbb{H}_0(\gamma) : \gamma_* = 0.70$ against their corresponding two-sided alternatives. As before we repeat independent experiments 3000 times for $n = 200, 400, 600, 800, 1,000, 2,000, 3000$, and report the empirical rejection rates of $\mathcal{W}_n(\beta)$, $\mathcal{W}_n(\phi)$, and $\mathcal{W}_n^{(1)}(\gamma)$ when the significance level is 5%.

Table 2 now displays the power paths under alternatives closer to the true parameter values. For DGPs 1 and 3, the rejection frequencies for $\mathcal{W}_n(\phi)$ and $\mathcal{W}_n^{(1)}(\gamma)$ rise from about 35–48% at $n = 200$ to about 97–99% at $n = 3,000$, while the power of $\mathcal{W}_n(\beta)$ also rises substantially. Under DGP 2, the powers increase more rapidly because the finite-sample dispersion is smaller. These results show the increase in power with the sample size without using the table to compare divergence rates across tests, since the fixed deviations

from the true values differ across the three restrictions.

In addition to the simulations, we report additional simulation results in the Online Supplement. We examine the finite sample bias and mean square error (MSE) of the parameter estimators and compare their finite sample distributions with the asymptotic distribution; see Section A.4.1 in the Online Supplement.

3.2 Simulation Using MARDL-EC Representation

We extend our simulation by considering MARDL-EC model estimation. For this we suppose MARDL(2, 1) given as

$$y_t = \alpha_{2*} + \phi_{2*}y_{t-1} + \phi_{3*}y_{t-2} + \theta_{2*}x_t + \theta_{3*}x_{t-1} + u_t,$$

where $x_t := x_{t-1} + w_t$, $w_t := \rho_* r_t + \sqrt{1 - \rho_*^2} r_{t-1}$, $r_t \sim_{\text{iid}} U(-0.5, 0.5)$. We set the parameter values at $\alpha_{2*} = 0$, $\theta_{2*} = 0.5$, $\theta_{3*} = 0.5$, $\phi_{2*} = 0.5$, $\phi_{3*} = 0$ and $\rho_* = 0.5$. By this, MARDL-EC(2,1) is obtained as

$$\Delta y_t = \alpha_{3*} + \zeta_* (y_{t-1} - \beta_{2*} x_{t-1}) + \varphi_* \Delta y_{t-1} + \psi_* \Delta x_t + u_t,$$

letting $\gamma_{2*} = 1$ and $\beta_{2*} = 2$ by construction, and the short-run parameters are obtained as $\zeta_* = -0.5$, $\varphi_* = 0$ and $\psi_* = 0.5$, as described in Section 2. Given this, we assume the three different DGPs for u_t in Section 3.1 and conduct simulations.

We examine the performance of the Wald tests defined by the MARDL-EC parameter estimators. For this we iterate 3,000 independent experiments by letting $n = 200, 400, 600, 800, 1000, 2000$, and 3000, as before. We also hypothesize the parameters for null simulation as follows: $\mathbb{H}_0(\varphi) : \varphi_* = 0$ versus $\mathbb{H}_1(\varphi) : \varphi_* \neq 0$ which is tested by $\mathcal{W}_n(\varphi)$. Next, we consider $\mathbb{H}_0(\psi) : \psi_* = 0.5$ versus $\mathbb{H}_1(\psi) : \psi_* \neq 0.5$ to be tested by $\mathcal{W}_n(\psi)$.

We report the null simulation results in Table 3 that reports the rejection rates at the significance level of 5%. All the empirical levels of the Wald tests trend to the nominal level as n grows, implying that the Wald tests are asymptotically well approximated by the chi-square distribution with 1 degree of freedom.

We report the power simulation results in Table 4. For the power examination, we use the closer false null $\mathbb{H}_0(\varphi) : \varphi_* = -0.15$ versus $\mathbb{H}_1(\varphi) : \varphi_* \neq -0.15$, which is tested by $\mathcal{W}_n(\varphi)$. We retain the economically interpretable restriction $\mathbb{H}_0(\psi) : \psi_* = 0$ versus $\mathbb{H}_1(\psi) : \psi_* \neq 0$, corresponding to no short-run effect. The total number of experiments is 3000. After moving the false null for φ_* closer to its true value of zero, the rejection frequencies of $\mathcal{W}_n(\varphi)$ rise gradually from 41.13%, 57.80%, and 35.33% at $n = 200$ to 91.50%, 99.63%, and 95.23% at $n = 3,000$ under DGPs 1–3, respectively. The power of $\mathcal{W}_n(\psi)$ also increases with n under the unchanged no-effect restriction, although the increase is slower under DGP 3 because of its

greater finite-sample dispersion.

In addition to the simulations, we report additional simulation results in the Online Supplement. We examine the finite sample bias and MSE of the parameter estimators and compare their finite sample distributions with the asymptotic distribution; see Section [A.4.2](#) in the Online Supplement.

4 Empirical Application to COVID-19 Data

In this section we apply the MARDL methodology to global COVID-19 data and extend its application to analyze the government responses to COVID-19.

4.1 Global Pandemic Data Analysis

In this section we focus on the global data analysis of COVID-19 confirmed cases and deaths.

4.1.1 Motivation and Model Set-Up

As [Fernández-Villaverde and Jones \(2020\)](#) point out, analyzing the COVID-19 pandemic and devising effective pandemic response measures is exceptionally challenging. When dealing with pandemic outbreaks the primary task for scientists and policy makers is in gaining a deep understanding of the pandemic trends and the evolution of the crisis through both short- and long-run analyses.

Short- and long-run analyses are motivated by different objectives. Short-run predictive analysis plays a crucial role in resource allocation, immediate healthcare planning, and the formulation of interventions like social distancing and lockdowns as noted by [Petropoulos et al. \(2022\)](#). Meanwhile, [Ramazi et al. \(2021\)](#) point out that long-run trajectory analysis is frequently applied across various critical domains, including planning intensive care unit capacity, selecting clinical trial locations, ensuring ventilator supplies, and addressing vaccine production during ongoing pandemics, and even formulating national economic policy strategies.

While examining the short- and long-run relationships between COVID-19 cases and deaths, we note the following two features. First, COVID-19 vaccination significantly and positively altered the course of the pandemic on a global scale. Besides the individual-level effects on disease risk and progression, vaccines also have secondary benefits in slowing down transmission rates and reducing related incidence and mortality rates. For example, prioritizing older individuals in vaccination proved the most effective in reducing mortality (see [Dinleyici et al., 2021](#)). The vaccination indeed offered protective and therapeutic effects against long-term COVID-19, and it remains the cornerstone of the response to the pandemic, given the continued outbreak of COVID-19.

Despite the success of the vaccination, achieving widespread public support for vaccination was still challenging. Vaccine hesitancy and refusal continued partly due to the misinformation leading to suboptimal vaccination rates (see [Machingaidze and Wiysonge, 2021](#)). Some countries continued to rely on non-pharmaceutical interventions due to significant heterogeneity in vaccination coverage. The reasons for vaccine hesitancy varied by context. [Lazarus et al. \(2023\)](#) identified more than 70 influencing factors, and many of them are time- and context-specific such as low education level, distrust in science and government and misinformation.

Second, global data analysis is key to the analysis of COVID-19, as COVID-19 is a global pandemic. Nevertheless, global data analysis on COVID-19 is scarce. During the COVID-19 period, the clinical and socio-economic impacts of safe and effective COVID-19 vaccines were significant to secure public health. However, empirical analysis on the clinical and economic effects of rapidly deploying COVID-19 vaccines on a global scale are limited, although impacts of COVID-19 vaccines on public health and economic variables have been examined using local data (see [Di Fusco et al., 2022](#), for example). Only a few studies employ empirical data to assess global public health impacts, particularly with limited data on the direct and indirect economic effects of COVID-19 vaccine administration (see [Yang et al., 2023](#), for example).

We contribute to the literature by focusing on both long- and short-run dynamic relationships between global COVID-19 confirmed cases and deaths. In particular, we apply MARDL methodology to investigate the short- and long-run dynamic relationships while taking the effects of vaccination into account. When compared to the mean and median regressions for stationary series, MR is better for predicting periodic peaks and valleys more effectively because the variation range of MR is broader and closer to the actual cyclic patterns. This feature is appealing even when examining integrated series. For example, [Ullah et al. \(2022\)](#) and [Jing and Cho \(2025\)](#) demonstrate that MR better fits COVID-19 data and produces forecasts more accurate than those obtained by mean and median regressions.

For the purpose, we establish a model accommodating both long- and short-run relationships between the variables. We first suppose that $\Delta d_t = \alpha_* - \zeta_*(d_t^* - d_{t-1}) + e_t$, where d_t and d_t^* represent the currently observed and predicted levels of confirmed cases at time t , respectively; $|\zeta_*|$ is a smoothing coefficient between 0 and 1 that describes the degree of death smoothing. It is widely accepted that there exists a long-run relationship between the number of COVID-19 deaths and confirmed cases, represented as $d_t^* = \beta_* c_t$, where β_* is the case fatality rate. We find that d_t and c_t are non-stationary, consistent with the empirical findings in [Ullah et al. \(2022\)](#) and [Jing and Cho \(2026\)](#). Therefore, $d_t^* = \beta_* c_t$ indeed represents the cointegration relationship between d_t and c_t .² Importantly, there may be a serial correlation in e_t . Therefore, we extend

²These results are confirmed through unit root and cointegration tests, which are not reported here for brevity.

the model as follows: $\Delta d_t = \alpha_* + \zeta_* d_{t-1} + \gamma_* c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*} \Delta d_{t-j} + \sum_{j=0}^{q-1} \omega_{j*} \Delta c_{t-j} + \varepsilon_t$. Here, ε_t is assumed to have no serial correlation by letting the lag orders p and q be sufficiently large. However, it should be noted that Δc_t may be simultaneously correlated with the error term ε_t . In such a case, we can control this correlation by projecting ε_t onto Δc_t as follows: $\varepsilon_t = \nu_* \Delta c_t + u_t$, where u_t is constructed to be uncorrelated with Δc_t . Substituting this projection, we obtain the following ARDL(p, q):

$$\Delta d_t = \alpha_* + \zeta_*(d_{t-1} - \beta_* c_{t-1}) + \sum_{j=1}^{p-1} \varphi_{j*} \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*} \Delta c_{t-j} + u_t, \quad (26)$$

where $\beta_* := -\frac{\gamma_*}{\zeta_*}$, $\psi_{0*} := \omega_{0*} + \nu_*$, and $\psi_{j*} := \omega_{j*}$ for $j = 1, \dots, q-1$.

Remarks are warranted for the parameters in (26). First, if $\nu_* \neq 0$, (26) cannot determine the simultaneous causality relationship between Δd_t and Δc_t . Otherwise, the changes in the number of COVID-19 cases can immediately cause the changes in the number of COVID-19 deaths under some mild regularity conditions, but not vice versa. Second, we can interpret the new coefficient ψ_{j*} on Δc_{t-j} as the impulse response parameter. Third, all other parameters including β_* and ζ_* in (26) can be estimated by excluding endogeneity factors in the regression because Δc_t is included in the regression.

The following four parameters are of our special interest. First, the EC model parameter ζ_* measures the smoothing mortality. Second, the long-run (cointegration) coefficient β_* measures the fatality rate. Third, $\varphi_* := \sum_{j=1}^{p-1} \varphi_{j*}$ measures the momentum effect of death, which quantifies the cumulative impact of the past death growth on the current death growth, viz., $\sum_{j=1}^{p-1} \partial \Delta d_t / \partial \Delta d_{t-j}$. Finally, $\psi_* := \sum_{j=0}^{q-1} \psi_{j*}$ measures the impulse response of death growth to new confirmed cases, which quantifies the cumulative impact of the current and past growth of confirmed cases on the current death growth, viz., $\sum_{j=0}^{q-1} \partial \Delta d_t / \partial \Delta c_{t-j}$. Based on the existing experience and theoretical analysis, our prior assumptions for these four parameters are as follows: $\zeta_* < 0$, $0 < \beta_* < 1$, $\varphi_* \leq 0$, and $\psi_* \geq 0$.

Using the global pandemic data, we estimate the parameters of interests and investigate their implications in three dimensions. First, we estimate ARDL(p, q) using three different estimation methods: ARDL, QARDL, and MARDL and explore the implications targeted by each estimation. Second, we divide the timeline into two sub-intervals so that we can clearly compare the short- and long-run dynamic relationships between confirmed cases and deaths during the pre-vaccination and post-vaccination periods. By this investigation, we desire to explore the global effect of vaccination on public health. Finally, we investigate the temporal patterns in the evolving relationship between COVID-19 confirmed cases and deaths as vaccination becomes more widespread. For this we use the rolling estimation method.

4.1.2 Empirical Analysis

We analyze the daily global COVID-19 data and report the consequences of the analysis in terms of the three dimensions. The global pandemic data and their source are described in the Online Supplement; see Section A.5.1. The sample ranges from December 8, 2020 to December 7, 2022, and Figure 1 illustrates the global COVID-19 deaths and cases. We split the whole sample into two sub-periods: pre-vaccination period ranging from December 8, 2020 to December 7, 2021, and post-vaccination period ranging from December 8, 2021 to December 7, 2022, respectively. This sample division is made by noting that the first dose of the COVID-19 vaccine is administrated in non-trial setting on December 8, 2020, and an estimated 55.9% of the global population has received at least one dose of the COVID-19 vaccine; 45.5% has received two doses; and 4.3% has received booster doses by December 8, 2021 (see Apergis, 2023). For model estimations, we select the orders p and q by Bayesian Information Criterion (BIC). As a result, $p = 6$ and $q = 2$ for the pre-vaccination period, and $p = 7$ and $q = 1$ for the post-vaccination period.

Comparison of the Three Model Estimations For comparison purpose, we specify QARDL and MARDL models as counterparts of ARDL (p, q):

$$\Delta d_t = \alpha_*(\tau) + \zeta_*(\tau)d_{t-1} + \gamma_*(\tau)c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*}(\tau)\Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}(\tau)\Delta c_{t-j} + u_t(\tau),$$

$$\Delta d_t = \alpha_*^M + \zeta_*^M d_{t-1} + \gamma_*^M c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*}^M \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}^M \Delta c_{t-j} + u_t^M.$$

Next, we estimate ARDL, QARDL, and MARD models by mean regression, quantile regression with the quantile level 0.5, and MR, respectively. The Root MSE (RMSE), Mean Absolute Error (MAE), and Coverage Quantile Function (CQF) results are reported in Table 5 for the three ARDL methods applied to the global data.

We observe that the MR method yields the smallest CQF values for both periods. Particularly, during the post-vaccination periods, MARDL achieves the minimum values for RMSE, MAE, and CQF defined as the median of $\{|\hat{y}_t - y_t|\}$. This indicates that the in-sample fit obtained by MR captures the cyclical behavior of daily COVID-19 deaths more closely than those obtained by the mean and median regressions. When a time series exhibits multiple outliers and skewed distributions, the robustness and efficiency achieved by mean and median regression are compromised. Meanwhile, estimating the mode of the conditional distribution is not influenced by the outliers or skewed distribution, letting MR track the cyclical behaviors of the time series

more successful than mean and median regressions. Figure 2 illustrates the actual and fitted values of the global daily new deaths per million across the two vaccination periods.

Analysis of Vaccination Effects We examine the MARDL estimation results applied to the global data. The estimation results for the four parameters of interest are contained in the first panel of Table 6 for the two different data sets. We summarize the quantitative results as follows:

- (a) The EC coefficients ζ_*^M for the pre- and post-vaccination periods are -0.0535 and -0.1743, respectively indicating adjustment rates of approximately 5.35% and 17.43%. This suggests that the system is returning to the long-run equilibrium state more rapidly, with the daily confirmed cases and daily deaths converging at a faster pace towards their long-run relationship.
- (b) The long-run fatality rate β_*^M is 0.0571 during the pre-vaccination period, but it decreases to 0.0036 during the post-vaccination period. This result is consistent with the findings of [Watson et al. \(2022\)](#).
- (c) The momentum effect parameters φ_*^M for death growth are estimated to be -2.0325 and -4.2241 for the pre- and post-vaccination periods, respectively. This suggests that the change in newly confirmed death counts are negatively correlated with the change in current death counts, and the impact is negatively strengthened by vaccination, meaning that past deaths further reduce current deaths by vaccination.
- (d) The impact reaction parameter ψ_*^M of death growth to the growth of confirmed cases has reduced from 0.0096 to 0.0006 between the pre- and post-vaccination periods, implying that the association between the changes in deaths and cases has been weakened by vaccination. $\square$

Overall, the magnitudes and signs of all four key coefficients generally correspond with our initial assumptions corroborating the vaccination effect.

Analysis by Rolling Window Approach We adopt a robust rolling window estimation method and examine the pandemic evolution. After letting the window length be fixed at 266 days, we implement the rolling window method by iteratively sliding the estimation window forward by one day until reaching the end of the sample, estimating MARDL(6,2) and MARDL(7,1) for the pre- and post-vaccination periods. Figure 3 shows the evolution of the four parameter estimators obtained by the rolling window method. We can summarize the estimation results as follows:

- (a) The rolling estimation vividly illustrates significant temporal variation of EC model coefficient ζ_* . We observe that during the early stages before vaccination, the adjustment speed fluctuates between 1% and 6%, but this range gradually increases to 3% to 5%. After vaccination, there is an upward trend in the overall convergence speed, rising from 12.5% to 40%. These findings align with the overall

analysis results discussed above.

- (b) The rolling window estimation for β_*^M reveals meaningful patterns. During the early stages of the pre-vaccination period, the long-run fatality rate shows an ascending trend, followed by a substantial decrease and stabilizing at around 0.05. This finding broadly supports the findings of the literature. In the initial months of 2021, due to the necessity of establishing infrastructures for vaccine distribution some regions deferred the second dose of vaccination, leading to a relatively minor impact of vaccination on the fatality rate. Furthermore, due to the limited vaccine supply, many countries prioritized vaccination for high-risk individuals, including healthcare workers and the elderly. Such strategies yielded direct protective effects but had a relatively small impact on the spread of the pandemic. However, from the mid-2021 onwards, countries with ample vaccine supplies began large-scale vaccination campaigns for adults and children. Simultaneously, to maintain high-level protection, some regions also introduced booster shots. These measures contributed to lowering the case fatality rate (see [Imai et al., 2021](#); [Wouters et al., 2021](#), for example). The rolling window mode estimates in the post-vaccination period exhibit a clear declining trend, stabilizing around 0.002 in the mid-to-late period.
- (c) Turning to the rolling mode estimates for φ_*^M , we observe fluctuating temporal patterns. During the early stages of the pre-vaccination period, φ_*^M displays a minor upward trend with negative values ranging from approximately -2.1 to -1.6, implying that changes in the growth of past deaths are negatively associated with the change in current deaths. In the post-vaccination period, φ_*^M briefly rises from -2.7 to -2.3 before stabilizing to around -3.7. This suggests that changes in the growth of past deaths remain negatively correlated with the change in current deaths with amplified magnitude. This is due to the fact that higher death counts at previous periods is linked to the current death counts in a manner to decrease the current deaths. As the vaccination became more effective, current death count decreases, letting the growth of past deaths be further negatively correlated with the change in current deaths.
- (d) Finally, during the pre-vaccination period, ψ_*^M gradually decreases from an early level of 0.012 to stabilize at around 0.011, after reaching a level as low as 0.009. In the post-vaccination period, δ_*^M fluctuates in the early stages between 0.0006 and 0.0009, then stabilizes at around 0.0007. This observation implies that the vaccination made the change in current deaths become less correlated with the change in past and current confirmed cases. □

4.2 Local Pandemic Data Analysis

In this section we focus on the local data analysis of COVID-19 confirmed cases and deaths.

4.2.1 Motivation

Governmental response policies play a pivotal role in controlling disease transmission and reducing mortality rates. These measures include isolation, social distancing, mask usage, travel restrictions, and so on. Numerous studies have shown that lockdowns and containment policies had a positive effect on reducing COVID-19 infections and deaths, particularly when implemented early.

Despite the widely recognized importance of interventions, the formulation and implementation of specific strategies against COVID-19 pose multiple challenges and uncertainties. For instance, a comprehensive consideration of factors is required when deciding when to initiate or lift restrictions and determining the intensity of interventions. These factors include public health, socio-economic considerations, public response, and the need to strike a balance between various interests and risks.

In this perspective, it is vital to accumulate the experiences from the previous pandemic. From this, we aim to quantitatively analyze the significant impact of government response policies on COVID-19. We attain this by comparing COVID-19 control policies among countries by utilizing OxCGR index to track the government's overall responses during the pre- and post-vaccination periods. This index reflects the strictness of policies related to measures such as lockdowns and economic policies. We particularly examine whether government response policies during the pre- and post-vaccination periods have an impact on confirmed cases and deaths in countries with lower and higher population densities.

For our empirical analysis, we select four European countries: Belgium, Denmark, Germany, and Norway. We can classify Belgium and Germany as countries with higher population densities, but Denmark and Norway can be classified as countries with relatively lower population densities.

Descriptive statistics related to COVID-19 of the four countries are provided in the Online Supplement. In addition to the population density, the countries in each group share many similarities in terms of per capita GDP, proportion of elderly population, smoking rate, and hospital bed capacity; see Section A.5.2 in the Online Supplement.

The governmental responses of the four countries to COVID-19 are not the same among themselves and different between the periods before and after the vaccination. All European Union countries simultaneously commenced COVID-19 vaccination campaigns on December 27th, 28th, and 29th, 2020, and Figure 4 presents the overall government response index for the two sets of the countries. During the pre- and post-

vaccination periods, Norway and Germany exhibited stricter measures compared to Denmark and Belgium respectively.

We investigate the short- and long-run relationship between COVID-19 deaths and confirmed cases using MARDL. Figure 5 shows the time plots of COVID-19 deaths and cases for the four countries. As the four countries with a similar counterpart implemented different response policies, we expect different estimation results among the four countries and between the periods before and after the vaccination, from which we desire to shed light on the experiences of government response policies on COVID-19. All data used for our investigation are detailed in the Online Supplement; see Section A.5.1 in the Online Supplement.

4.2.2 Empirical Analysis

Our local data analysis is conducted in parallel to the global data analysis. That is, we first estimate ARDL, QARDL, and MARDL and ensure the advantage of using MARDL over ARDL and QARDL. Next, we analyze the estimation results by focusing on the four parameters of our interest.

Table 5 reports the RMSE, MAE, and CQF results obtained from the three models applied to the pre- and post-vaccination periods of the four countries. The analysis indicates that the MR method most often achieves the lowest CQF values across both periods as for the global data analysis. This aspect is reinforced by comparing the fitted series with the actual series of the deaths. Figures 6 and 7 present the in-sample fit for the two sets of the countries, and the cyclical patterns of the deaths are precisely tracked by MARDL. From this we exploit MARDL to analyze COVID-19 data statistically along with its implications on the government responses.

We contain the MARDL estimation results in Table 6 for the two sets of the four countries across two distinct periods. We summarize the quantitative results as follows:

- (a) The adjustment speeds $|\zeta_*^M|$ of the four countries exhibit an increasing trend during both pre- and post-vaccination periods, consistent with the global pattern. Belgium increases from 5.14% to 11.89%, Germany from 9.88% to 17.79%, and Norway from 9.69% to 12.07%. The EC model coefficient result of Denmark is not significant for the pre-vaccination period, but becomes significant for the post-vaccination period. All these results are consistent with those obtained from the global data.
- (b) The success of overall government response policies should be assessed over the long run. We therefore compare the long-run fatality rate β_*^M for the two sets of the countries and analyze the impact of the government responses during the pre- and post-vaccination periods.

First, we compare Belgium and Germany. During the pre-vaccination period, the fatality rates are 0.0080 and 0.0086, respectively, aligning with their overall government response indices. For instance

during the pre-vaccination period, Germany implemented lockdown measures starting from November 2020, which lasted until March 2021. Meanwhile, Belgium tightened its preventive measures, including border policies, starting from December 2020 and temporarily banned all non-essential travel from January 27 to April 1, 2021. During the post-vaccination period, the fatality rates decrease to 0.0013 for both Belgium and Germany. This reduction is attributed to the higher overall government response index of Germany during the post-vaccination period. In 2022, Germany maintained relatively strict pandemic control policies and gradually eased the measures from February 2022 until all pandemic-related regulations were lifted in April 2023. Belgium began easing entry policies from March 11, 2022, transitioning from an orange to a yellow alert level. From May 23, 2022, the Belgian federal government officially suspended the COVID-19 alert level, further relaxing pandemic control measures and entry policies. During the post-vaccination period, Germany's fatality rate decreases to the same level as Belgium's.

Second, we compare Denmark and Norway. During the pre- and post-vaccination periods, Denmark consistently exhibited higher fatality rates than Norway. In the pre-vaccination period, Denmark has a rate of 0.0091, while Norway has a rate of 0.0025. During the post-vaccination period, Denmark's rate decreases to 0.0010, and Norway's rate decreases to 0.0007. This trend aligns with the overall government response indices, indicating that Norway's pandemic control policies were stricter than Denmark's for both pre- and post-vaccination periods. Denmark reinstated partial COVID-19 preventive measures at the end of November 2020 to control the COVID-19 spread. From March 2021, it gradually relaxed the measures in different regions. In contrast, Norway enhanced nationwide pandemic control measures starting from March 25, 2021, and introduced regulations for the domestic use of COVID-19 health certificates in June 2021. Starting from December 2021, Norway implemented a series of nationwide new pandemic control policies. Meanwhile, Denmark re-introduced partial COVID-19 preventive measures at the end of November 2021 to address a new wave of infections during the winter. Denmark's government no longer classified COVID-19 as a critical societal threat after January 31, 2022 and lifted all domestic pandemic restrictions and revoked the mandatory mask-wearing and vaccine passport policies on February 1, 2022, becoming the first EU country to remove all pandemic restrictions. Similarly, on February 12, 2022, Norway also lifted most pandemic control policies.

- (c) The momentum parameters φ_*^M of death growth for Germany, Denmark, and Norway show changes during both pre- and post-vaccination periods that align with global trends. For Germany, these parameters are -2.1072 and -2.4030, for Denmark -1.8727 and -2.3689, and for Norway -4.2421 and -5.0767. This signifies a negative correlation and an enhanced impact of changes in newly confirmed

death cases on the current death count. In the case of Belgium, the momentum effect during the post-vaccination period differs from the pre-vaccination period, and they are -2.2320 and -1.1020 respectively. This aligns with the significant decrease in Belgium's overall government response index during the post-vaccination period.

- (d) The impact reaction of the change in deaths to the changes in confirmed cases ψ_*^M demonstrates a consistent trend with global data for Belgium and Germany during the post-vaccination period, showing a decrease compared to the pre-vaccination phase. Specifically, for Belgium, the estimates are 0.0003 and 0.0001, and Germany's estimates are 0.01773 and 0.0025, respectively. Interestingly, for Denmark and Norway, the estimates are negative when they are significant. During both pre- and post-vaccination periods, the estimates are -0.0003 and -0.0001 for Denmark and -0.0009 and -0.0003 for Norway, but the estimates are not significant for the pre-vaccination periods. This might be attributed to the fact that in Denmark and Norway, the changes in deaths are not attributed to the changes in confirmed cases any longer. Other causes could be more relevant to the changes in deaths. $\square$

5 Conclusion

This paper proposes the MARDL method to study the dynamic relationship between COVID-19 daily confirmed cases and deaths during pre- and post-vaccination periods. The study consists of four main steps. Firstly, we introduce an asymptotic theory for the estimators of short-run dynamics and long-run (cointegrating) parameters for MARDL and MARDL-EC representation, which asymptotically follow a (mixture) normal distribution. Through Monte Carlo simulation, we observe that the null distribution of Wald statistics for testing parameter restrictions weakly converges to a chi-squared distribution, strongly supporting theoretical predictions. Secondly, we apply the MARDL method to the analysis of COVID-19 cases in the global context and in two sets of European countries. Through the analysis and comparison of key variable coefficients, we assess the dynamic relationships in confirmed cases and deaths during pre- and post-vaccination periods, as well as the impact of government response policies on the case fatality rate.

Our empirical analysis yields several key findings. Firstly, whether using global data or data from the two sets of European countries, the MARDL method consistently outperforms in terms of CQF when it comes to COVID-19 data. This suggests that MARDL performs better in capturing the typical magnitude of the fitting error. Secondly, when comparing the key variable coefficients (EC coefficients, the long-run case fatality rate, the momentum effect parameters of death growth, and the impact reaction of confirmed deaths to confirmed cases growths) in the pre- and post-vaccination period using global data, we find that

compared to the pre-vaccination period, the post-vaccination period exhibits an increase in the adjustment speed, a decrease in the confirmed death rate, a negative correlation with the momentum effect in death growths, and a decrease in the impact response to new confirmed cases. Thirdly, through a comparison of the strictness of government response policies in the two sets of European countries, we observe that countries with stricter government response policies have lower confirmed death rates both in the pre-vaccination and post-vaccination periods. This finding holds significant economic implications, indicating that vaccination and government response policies can effectively reduce the number of confirmed deaths from COVID-19.

The research methods employed in this study can extend to analyzing data from other countries, exploring different infectious diseases, and examining various periods. However, it is important to note that the focus of this study is primarily on the impact of vaccination and government response policies on the number of confirmed cases and deaths from COVID-19. Numerous other factors can influence the spread of COVID-19. For instance, cultural practices, weather conditions, and more can all play significant roles in the transmission of the pandemic. Exploring these interrelationships may be a potential direction for future research, contributing to a more comprehensive understanding of and response to the mechanisms behind infectious disease transmission.

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DGP	Stat. \ n	200	400	600	800	1,000	2,000	3,000
1	$\mathcal{W}_n(\phi)$	12.03	8.33	7.23	6.83	6.17	6.03	5.77
	$\mathcal{W}_n^{(1)}(\gamma)$	12.90	8.03	7.10	6.80	6.53	5.90	5.67
	$\mathcal{W}_n(\beta)$	10.73	8.03	6.57	6.67	7.57	5.77	5.67
2	$\mathcal{W}_n(\phi)$	10.80	7.57	5.93	7.63	7.23	5.47	6.73
	$\mathcal{W}_n^{(1)}(\gamma)$	10.30	7.57	5.90	7.20	6.93	5.60	6.57
	$\mathcal{W}_n(\beta)$	10.53	8.87	7.77	7.33	7.97	7.00	5.83
3	$\mathcal{W}_n(\phi)$	6.67	5.00	4.90	5.13	4.90	4.83	4.67
	$\mathcal{W}_n^{(1)}(\gamma)$	11.43	6.77	6.13	5.93	5.67	5.03	4.60
	$\mathcal{W}_n(\beta)$	7.83	5.80	5.07	4.73	4.90	4.40	4.37

Table 1: EMPIRICAL LEVELS OF $\mathcal{W}_n(\phi)$, $\mathcal{W}_n^{(1)}(\gamma)$, AND $\mathcal{W}_n(\beta)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) Number of replications: 3,000. (ii) DGP: $y_t = \alpha_{1*} + \phi_{1*}y_{t-1} + \theta_{0*}x_t + \theta_{1*}x_{t-1} + u_t$, with $(\alpha_{1*}, \phi_{1*}, \theta_{0*}, \theta_{1*}) = (0, 0.5, 0.5, 0.5)$, $x_t = x_{t-1} + w_t$, $w_t = 0.5r_t + \sqrt{0.75}r_{t-1}$, and $r_t \sim_{\text{iid}} U(-0.5, 0.5)$. (iii) The null values are $\phi_* = 0.5$, $\gamma_* = 1$, and $\beta_* = 2$. (iv) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

DGP	Stat. \ n	200	400	600	800	1,000	2,000	3,000
1	$\mathcal{W}_n(\phi)$	44.10	56.53	65.03	71.93	78.60	92.83	97.20
	$\mathcal{W}_n^{(1)}(\gamma)$	48.43	62.13	70.70	76.93	81.27	93.37	97.20
	$\mathcal{W}_n(\beta)$	20.23	32.03	46.70	58.03	67.87	92.07	98.80
2	$\mathcal{W}_n(\phi)$	98.70	99.97	100.0	100.0	100.0	100.0	100.0
	$\mathcal{W}_n^{(1)}(\gamma)$	99.27	99.93	100.0	100.0	100.0	100.0	100.0
	$\mathcal{W}_n(\beta)$	70.97	93.63	98.40	99.93	100.0	100.0	100.0
3	$\mathcal{W}_n(\phi)$	34.60	52.23	63.70	72.67	79.17	94.43	98.50
	$\mathcal{W}_n^{(1)}(\gamma)$	40.53	55.20	66.10	73.47	80.00	93.93	97.50
	$\mathcal{W}_n(\beta)$	8.43	9.20	10.90	14.73	19.77	45.17	65.70

Table 2: EMPIRICAL POWERS OF $\mathcal{W}_n(\phi)$, $\mathcal{W}_n^{(1)}(\gamma)$, AND $\mathcal{W}_n(\beta)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) The number of replications, DGP, model, and error distributions are as in Table 1. (ii) The fixed false null values are $\phi_* = 0.35$, $\gamma_* = 0.70$, and $\beta_* = 1.95$, respectively. (iii) Under DGP 3, the finite-sample dispersion of $\hat{\beta}_n$ is markedly larger than under DGPs 1 and 2, so the rejection frequency of $\mathcal{W}_n(\beta)$ increases more slowly; it rises monotonically from 8.43% to 65.70% over the reported sample sizes.

DGP	Stat. \ n	200	400	600	800	1,000	2,000	3,000
1	$\mathcal{W}_n(\varphi)$	12.67	8.47	8.30	6.93	7.17	6.33	6.43
	$\mathcal{W}_n(\psi)$	13.73	9.33	8.83	7.70	7.43	6.27	5.37
2	$\mathcal{W}_n(\varphi)$	11.77	9.30	7.87	7.70	7.70	7.20	7.37
	$\mathcal{W}_n(\psi)$	11.73	8.40	8.93	7.73	8.07	7.53	7.13
3	$\mathcal{W}_n(\varphi)$	8.50	5.90	5.33	5.20	4.83	5.40	4.43
	$\mathcal{W}_n(\psi)$	8.90	6.93	5.30	5.20	4.17	4.60	4.77

Table 3: EMPIRICAL LEVELS OF $\mathcal{W}_n(\varphi)$ AND $\mathcal{W}_n(\psi)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) Number of replications: 3,000. (ii) DGP: $y_t = \alpha_{2*} + \phi_{2*}y_{t-1} + \phi_{3*}y_{t-2} + \theta_{2*}x_t + \theta_{3*}x_{t-1} + u_t$, with $(\alpha_{2*}, \theta_{2*}, \theta_{3*}, \phi_{2*}, \phi_{3*}) = (0, 0.5, 0.5, 0.5, 0)$ and the process for x_t as in Table 1. (iii) The EC representation is $\Delta y_t = \alpha_{3*} + \zeta_*(y_{t-1} - \beta_{2*}x_{t-1}) + \varphi_*\Delta y_{t-1} + \psi_*\Delta x_t + u_t$. (iv) The true null values are $\varphi_* = 0$ and $\psi_* = 0.5$. (v) The three error distributions are as in Table 1.

DGP	Stat. \ n	200	400	600	800	1,000	2,000	3,000
1	$\mathcal{W}_n(\varphi)$	41.13	50.20	57.17	62.27	66.00	83.87	91.50
	$\mathcal{W}_n(\psi)$	57.73	71.97	80.73	87.50	90.60	97.90	99.60
2	$\mathcal{W}_n(\varphi)$	57.80	75.50	83.90	89.10	92.90	98.80	99.63
	$\mathcal{W}_n(\psi)$	100.0	100.0	100.0	100.0	100.0	100.0	100.0
3	$\mathcal{W}_n(\varphi)$	35.33	47.63	57.67	66.30	72.40	90.23	95.23
	$\mathcal{W}_n(\psi)$	15.93	17.47	19.47	22.80	25.37	36.10	47.07

Table 4: EMPIRICAL POWERS OF $\mathcal{W}_n(\varphi)$ AND $\mathcal{W}_n(\psi)$ (LEVEL OF SIGNIFICANCE: 5%). Notes: (i) The number of replications, DGP, ECM, and error distributions are as in Table 3. (ii) The fixed false null values are $\varphi_* = -0.15$ and $\psi_* = 0$, respectively. (iii) Under DGP 3, the finite-sample dispersion of $\tilde{\psi}_n$ is markedly larger than under DGPs 1 and 2, so the rejection frequency of $\mathcal{W}_n(\psi)$ increases more slowly; it rises monotonically from 15.93% to 47.07% over the reported sample sizes.

Country	Period	Model	RMSE	MAE	CQF	(p, q)
World	Pre-vaccination	ARDL	0.1182	0.0650	0.0439	(6,2)
		QARDL	0.1196	0.0634	0.0429	
		MARDL	0.1208	0.0641	0.0414	
	Post-vaccination	ARDL	0.1025	0.0527	0.0307	(7,1)
		QARDL	0.1073	0.0487	0.0229	
		MARDL	0.1139	0.0499	0.0213	
Belgium	Pre-vaccination	ARDL	0.5141	0.3606	0.2432	(6,1)
		QARDL	0.5189	0.3567	0.2300	
		MARDL	0.5463	0.3690	0.2132	
	Post-vaccination	ARDL	0.4139	0.2962	0.2039	(4,1)
		QARDL	0.4164	0.2942	0.2064	
		MARDL	0.4199	0.2956	0.2031	
Germany	Pre-vaccination	ARDL	0.5191	0.3171	0.1878	(8,6)
		QARDL	0.5801	0.2797	0.1100	
		MARDL	0.6425	0.2936	0.0958	
	Post-vaccination	ARDL	0.2725	0.1913	0.1322	(9,8)
		QARDL	0.2905	0.1823	0.1176	
		MARDL	0.3071	0.1877	0.1049	
Denmark	Pre-vaccination	ARDL	0.5471	0.3151	0.1904	(5,1)
		QARDL	0.5511	0.3115	0.1813	
		MARDL	0.5795	0.3226	0.1699	
	Post-vaccination	ARDL	1.1454	0.6449	0.3726	(6,1)
		QARDL	1.1704	0.6222	0.3696	
		MARDL	1.1629	0.6281	0.3224	
Norway	Pre-vaccination	ARDL	0.6193	0.3689	0.2051	(8,4)
		QARDL	0.6986	0.3139	0.0187	
		MARDL	0.6898	0.3223	0.0207	
	Post-vaccination	ARDL	1.1754	0.5351	0.1511	(7,9)
		QARDL	1.2840	0.4171	0.0061	
		MARDL	1.3831	0.4239	0.0060	

Table 5: THE QUANTITATIVE RESULTS OF DIFFERENT ESTIMATION METHODS. This table evaluates the efficacy of the three different ARDL methods using the COVID-19 data. The lags are determined by Bayesian Information Criterion (BIC). All three measures are computed using the in-sample observations. RMSE and MAE are the root mean and the mean of the absolute fitting errors, and CQF is their median.

Country	Period	ζ_*^M	β_*^M	φ_*^M	ψ_*^M
World	Pre-vaccination	-0.0535** (0.0203)	0.0571** (0.0199)	-2.0325*** (0.3854)	0.0096*** (0.0022)
	Post-vaccination	-0.1743*** (0.0510)	0.0036*** (0.0007)	-4.2241*** (0.5056)	0.0006*** (0.0002)
Belgium	Pre-vaccination	-0.0514** (0.0172)	0.0080*** (0.0019)	-2.2320*** (0.2112)	0.0008*** (0.0002)
	Post-vaccination	-0.1189*** (0.0279)	0.0013*** (0.0002)	-1.2210*** (0.1177)	0.0003*** (0.0001)
Germany	Pre-vaccination	-0.0988* (0.0457)	0.0086** (0.0030)	-2.1072* (0.5231)	0.0148* (0.0064)
	Post-vaccination	-0.1779** (0.0607)	0.0013*** (0.0001)	-2.4030*** (0.6095)	0.0025** (0.0009)
Denmark	Pre-vaccination	-0.1333 (0.0777)	0.0091* (0.0044)	-1.8727*** (0.2482)	-0.0002 (0.0004)
	Post-vaccination	-0.1213** (0.0417)	0.0010*** (0.0002)	-2.3689*** (0.3703)	-0.0003* (0.0001)
Norway	Pre-vaccination	-0.1534*** (0.0134)	0.0025*** (0.0003)	-4.2421*** (0.3080)	0.0005 (0.0013)
	Post-vaccination	-0.3351*** (0.0701)	0.0007*** (0.0002)	-5.0767*** (0.0722)	-0.0004*** (0.0002)

Table 6: MARDL ESTIMATION RESULTS. Notes: (i) * indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$. (ii) Figures in parentheses are standard errors, and those of the long-run coefficient are calculated using the delta method. (iii) MARDL estimation results are based on the following model $\Delta d_t = \alpha_*^M + \zeta_*^M d_{t-1} + \gamma_*^M c_{t-1} + \sum_{j=1}^{p-1} \varphi_{j*}^M \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}^M \Delta c_{t-j} + u_t^M = \alpha_*^M + \zeta_*^M (d_{t-1} - \beta_*^M c_{t-1}) + \sum_{j=1}^{p-1} \varphi_{j*}^M \Delta d_{t-j} + \sum_{j=0}^{q-1} \psi_{j*}^M \Delta c_{t-j} + u_t^M$.

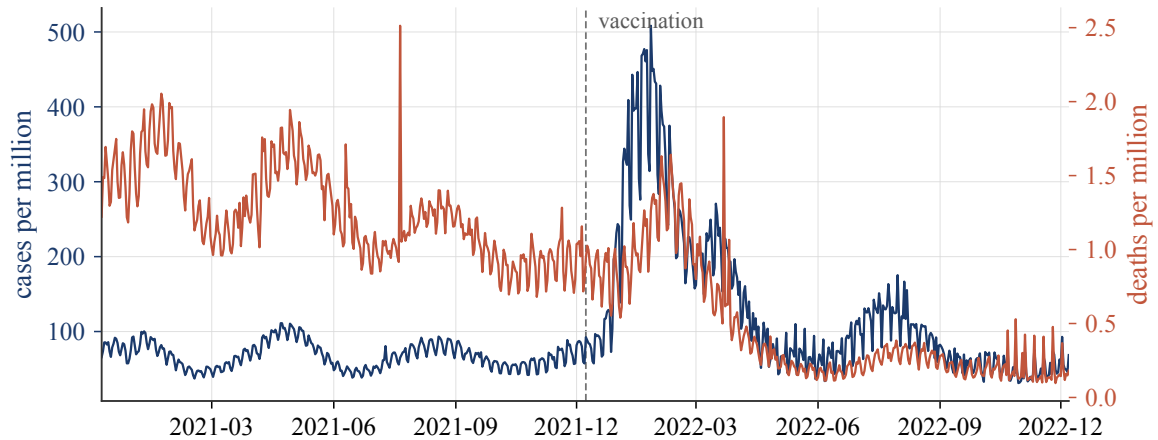


Figure 1: DAILY GLOBAL COVID-19 CASES AND DEATHS PER MILLION. Notes: (i) The sample ranges from December 8, 2020 to December 7, 2022. (ii) The left axis measures new cases per million and the right axis measures new deaths per million. (iii) The dashed vertical line at December 8, 2021 separates the pre- and post-vaccination periods. (iv) Source: Our World in Data; see Section A.5.1 in the Online Supplement.

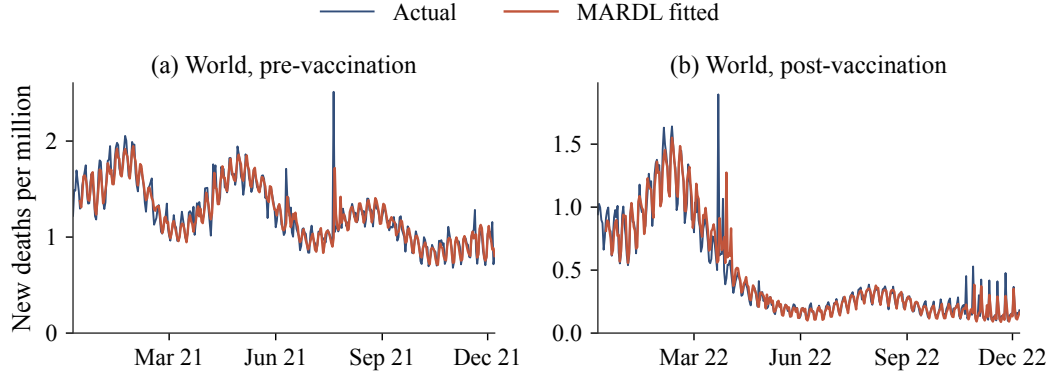


Figure 2: ACTUAL AND MARDL FITTED DAILY NEW DEATHS PER MILLION USING THE GLOBAL DATA. Notes: (i) The in-sample fitted values are drawn together with the actual values. (ii) Panel (a) covers the pre-vaccination period, and panel (b) covers the post-vaccination period. (iii) Lag orders are those in Table 5. (iv) The vertical axis measures new deaths per million people; fitted values are truncated below at zero. (v) Source: Our World in Data; see Section A.5.1 in the Online Supplement.

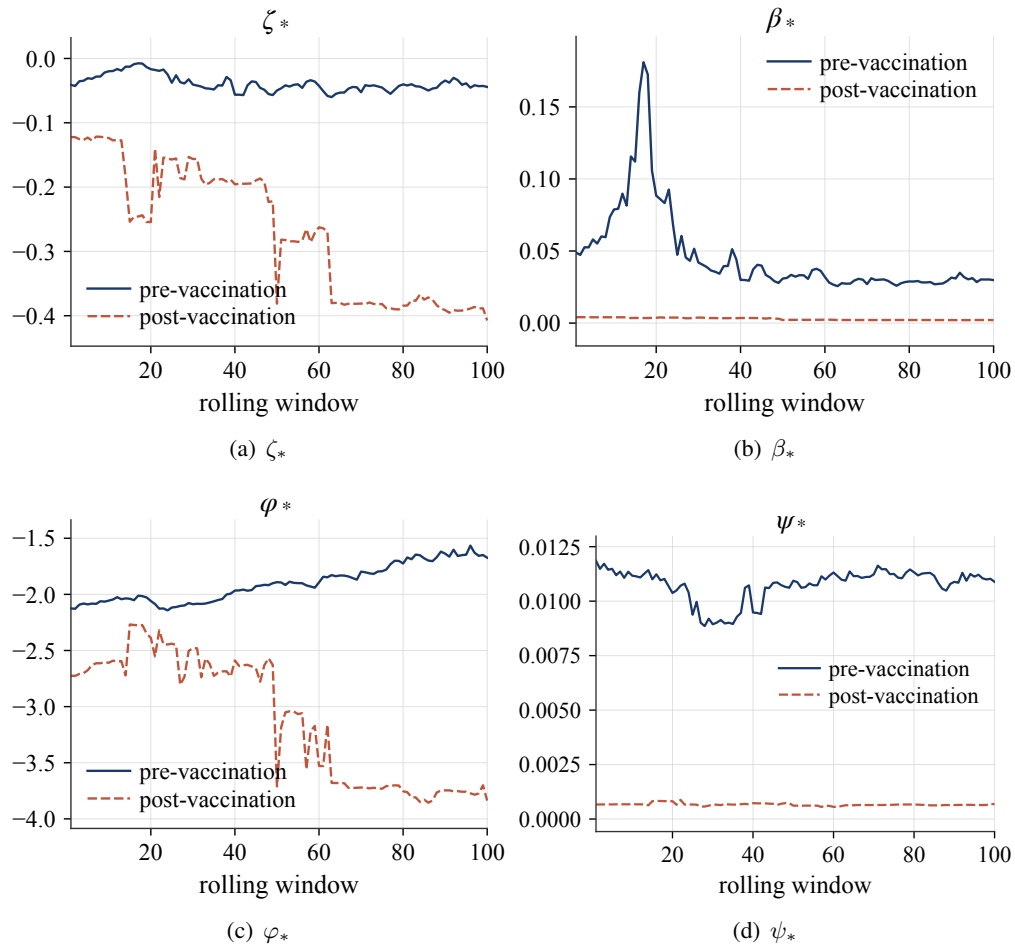


Figure 3: PARAMETER ESTIMATES OF ζ_* , β_* , φ_* AND ψ_* USING THE ROLLING WINDOW METHOD. The window length is 266 days. Each panel overlays the pre- and post-vaccination periods.

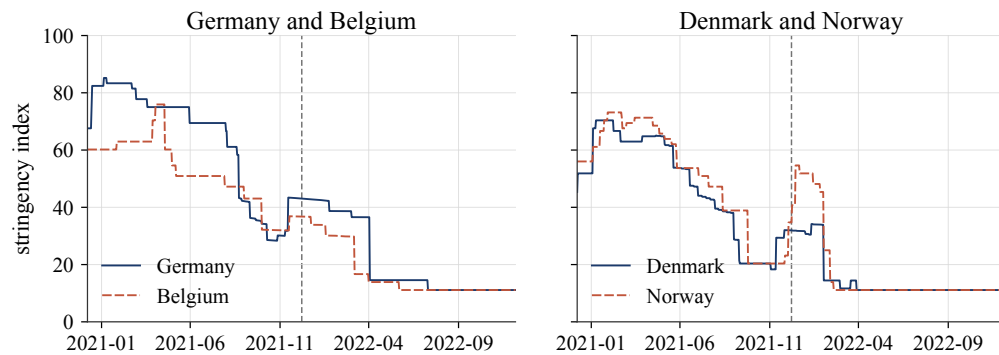


Figure 4: STRINGENCY INDEX FOR THE TWO SETS OF EUROPEAN COUNTRIES. Higher values indicate stricter measures. The data is sourced from <https://www.bsg.ox.ac.uk/research/covid-19-government-response-tracker>.

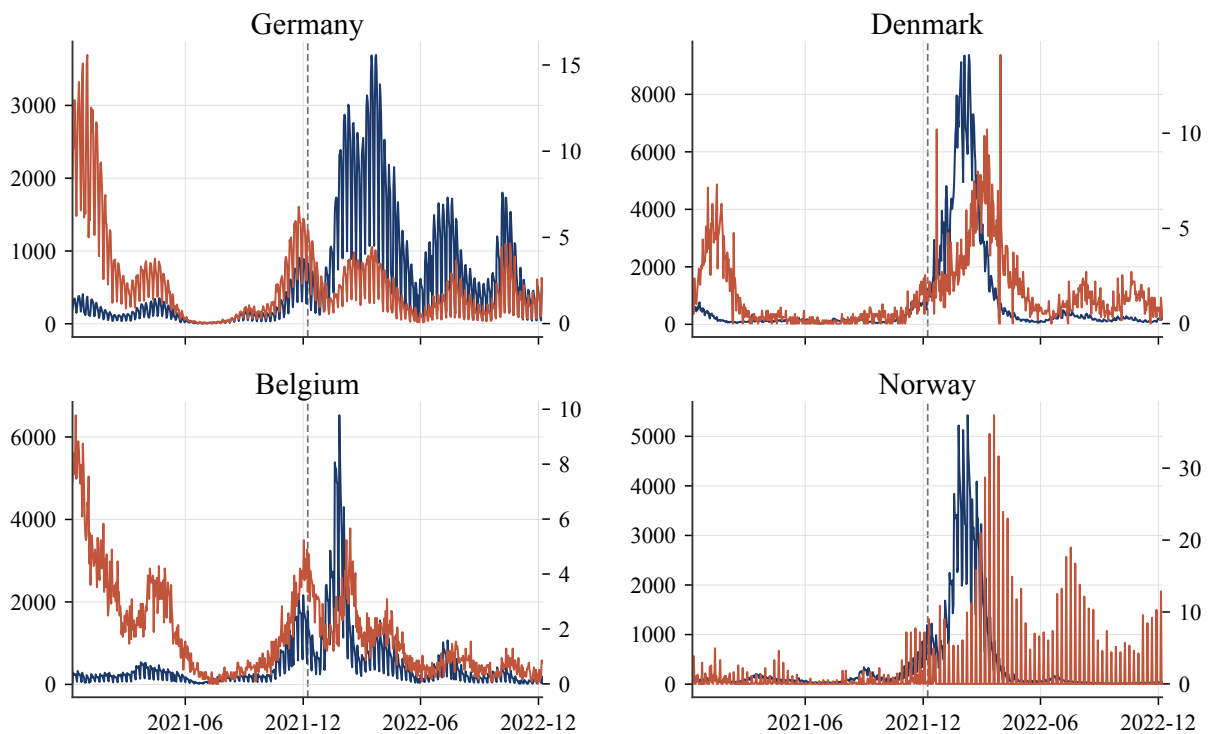


Figure 5: DAILY COVID-19 CASES AND DEATHS PER MILLION FOR THE FOUR COUNTRIES. Notes: (i) Germany and Belgium are on the left; Denmark and Norway are on the right. (ii) The sample ranges from December 8, 2020 to December 7, 2022. (iii) The left axis measures new cases per million and the right axis measures new deaths per million. (iv) The dashed vertical line at December 8, 2021 separates the pre- and post-vaccination periods. (v) Source: Our World in Data; see Section A.5.1 in the Online Supplement.

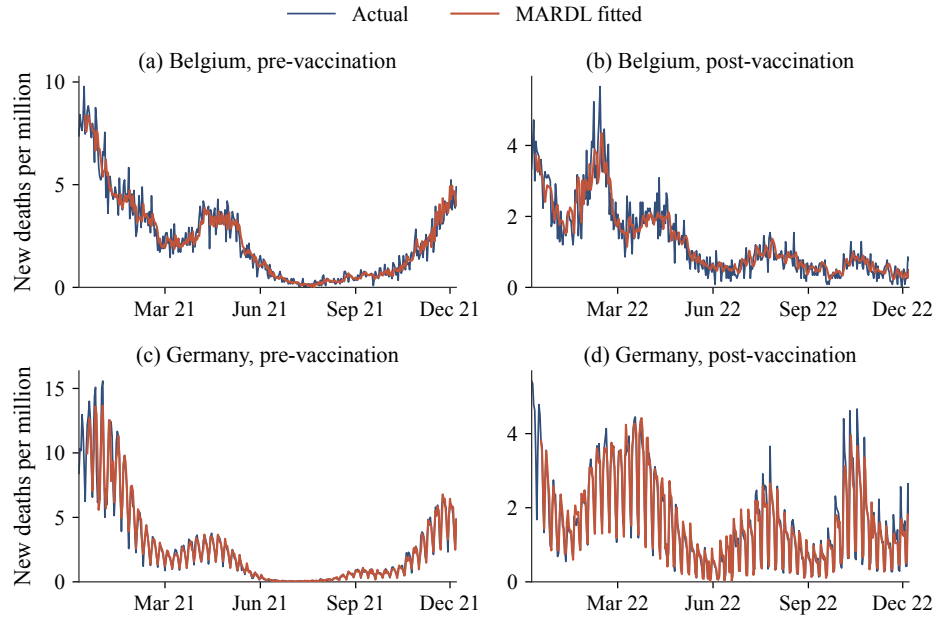


Figure 6: ACTUAL AND MARDL FITTED DAILY NEW DEATHS PER MILLION: BELGIUM AND GERMANY. Notes: (i) Panels (a) and (c) cover the pre-vaccination period and panels (b) and (d) the post-vaccination period. (ii) The vertical scale differs across panels. (iii) Conventions, periods, lag orders and source are the same as in Figure 2.

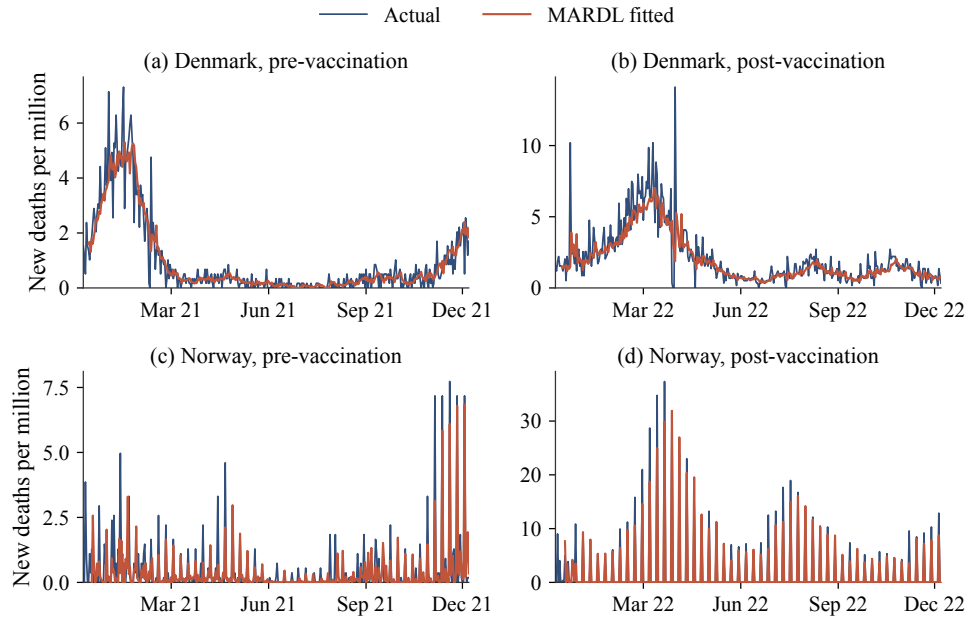


Figure 7: ACTUAL AND MARDL FITTED DAILY NEW DEATHS PER MILLION: DENMARK AND NORWAY. Notes: (i) Panels (a) and (c) cover the pre-vaccination period and panels (b) and (d) the post-vaccination period. (ii) The vertical scale differs across panels, and the truncation at zero binds often for Norway, where deaths are reported in weekly batches. (iii) Conventions, periods, lag orders and source are the same as in Figure 2.

Online Supplement for

‘Modal Estimation of Autoregressive Distributed Lag Model Applied to COVID-19 Pandemic Data’

by

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This Online Supplement is an Appendix that supplements the claims, simulations and empirical analysis provided in the main text.

A Appendix

This supplementary Appendix comprises five sections. Section A.1 derives the moments of the modal score and Section A.2 provides auxiliary lemmas to facilitate the proofs of the main claims. Section A.3 proves the main claims, and Sections A.4 and A.5 provide supplementary simulations and empirical analyses.

A.1 Moments of the Modal Score

In this section, we derive the moments of the modal score. Throughout, we let $\mu_2 := \int t^2 \phi(t) dt$, which equals unity for the standard normal kernel.

We first derive the approximation of m_h . By the change of variables, if we let $u = hs$,

$$m_h = \mathbb{E}[\phi^{(1)}(u_{th})] = \int \phi^{(1)}(u/h) f_u(u) du = h \int \phi^{(1)}(s) f_u(hs) ds.$$

Assumption 1 (iii.a) permits the following expansion:

$$f_u(hs) = f_u(0) + hs f_u^{(1)}(0) + \frac{1}{2} h^2 s^2 f_u^{(2)}(0) + \frac{1}{6} h^3 s^3 f_u^{(3)}(0) + o(h^3 s^3), \quad (\text{A.1})$$

and for $j = 0, 1, 2, 3$, $\int |s|^j |\phi^{(1)}(s)| ds$ is finite for the standard normal kernel. For a general kernel $\int |s|^j |\phi^{(1)}(s)| ds$ is finite by Assumption 1 (ii.a) if $j \leq 2$; and $\int |s|^3 |\phi^{(1)}(s)| ds$ is finite if for some $\delta > 0$ $|t|^{4+\delta} |\phi^{(1)}(t)| \rightarrow 0$. Integrating by parts, we obtain that

$$\int \phi^{(1)}(s) ds = 0, \quad \int s \phi^{(1)}(s) ds = -1, \quad \int s^2 \phi^{(1)}(s) ds = 0, \quad \text{and} \quad \int s^3 \phi^{(1)}(s) ds = -3\mu_2,$$

where the first and third integrals vanish because $\phi^{(1)}(\cdot)$ and $(\cdot)^2\phi^{(1)}(\cdot)$ are odd functions. As $f_u^{(1)}(0) = 0$ by the first-order condition, the contributions driven by $O(h)$ and $O(h^2)$ drop out, so that it follows that

$$m_h = h \left\{ -\frac{1}{2}h^3\mu_2 f_u^{(3)}(0) + o(h^3) \right\} = -\frac{h^4}{2}\mu_2 f_u^{(3)}(0) + o(h^4). \quad (\text{A.2})$$

Second, we derive the approximation of $\text{var}(\bar{\phi}^{(1)}(u_{th})/\sqrt{h})$. By the same change of variables, we note that

$$\mathbb{E}[\{\phi^{(1)}(u_{th})\}^2] = \int [\phi^{(1)}(u/h)]^2 f_u(u) du = h \int [\phi^{(1)}(s)]^2 f_u(hs) ds.$$

Therefore, if we combine this with (A.1), then

$$\begin{aligned} & \int [\phi^{(1)}(s)]^2 f_u(hs) ds \\ &= \int [\phi^{(1)}(s)]^2 \left[f_u(0) + hs f_u^{(1)}(0) + \frac{1}{2}h^2 s^2 f_u^{(2)}(0) + \frac{1}{6}h^3 s^3 f_u^{(3)}(0) + o(h^3 s^3) \right] ds \end{aligned}$$

Here, $\int [\phi^{(1)}(s)]^2 s ds = 0$ and $\int [\phi^{(1)}(s)]^2 s^3 ds = 0$, because $[\phi^{(1)}(\cdot)]^2$ is even from the fact that $\phi^{(1)}(\cdot)$ is odd. Furthermore, $v^2 := \int [\phi^{(1)}(s)]^2 ds < \infty$ by Assumption 1 (ii.b) and $\int s^2 [\phi^{(1)}(s)]^2 ds < \infty$ by Assumption 1 (ii.a), so that it follows that

$$\mathbb{E}[\{\phi^{(1)}(u_{th})\}^2] = h \int [\phi^{(1)}(s)]^2 f_u(hs) ds = h v^2 f_u(0) \{1 + O(h^2)\}. \quad (\text{A.3})$$

Given this, $m_h^2 = O(h^8)$ from (A.2), so that

$$\text{var}\left(\frac{\bar{\phi}^{(1)}(u_{th})}{\sqrt{h}}\right) = \frac{1}{h} \left(\mathbb{E}[\{\phi^{(1)}(u_{th})\}^2] - m_h^2 \right) \rightarrow v^2 f_u(0),$$

as $h \rightarrow 0$.

Third, we derive the corresponding moments of the second- and third-order kernel derivatives, which we use for Lemmas A.5 and A.7 given below. For $j = 2, 3$, the same change of variables gives $\mathbb{E}[\phi^{(j)}(u_{th})] = h \int \phi^{(j)}(s) f_u(hs) ds$, and repeated integrations by parts yield the following:

$$\begin{aligned} \int \phi^{(2)}(s) ds &= 0, \quad \int s \phi^{(2)}(s) ds = 0, \quad \int s^2 \phi^{(2)}(s) ds = 2, \quad \int s^3 \phi^{(2)}(s) ds = 0, \\ \int \phi^{(3)}(s) ds &= 0, \quad \int s \phi^{(3)}(s) ds = 0, \quad \int s^2 \phi^{(3)}(s) ds = 0, \quad \int s^3 \phi^{(3)}(s) ds = -6, \end{aligned} \quad (\text{A.4})$$

where the odd-order integrals vanish because $\phi^{(2)}(\cdot)$ is even and $\phi^{(3)}(\cdot)$ is odd. [JSCHO: Needs more

explanations on this sentence.] Substituting (A.1) into ???, we obtain

$$\mathbb{E}[\phi^{(2)}(u_{th})] = h^3 f_u^{(2)}(0) + o(h^4) = h^3 f_u^{(2)}(0)\{1 + o(h)\} \quad \text{and}$$

$$\mathbb{E}[\phi^{(3)}(u_{th})] = -h^4 f_u^{(3)}(0) + o(h^4) = O(h^4) \quad (\text{A.5})$$

using $f_u^{(1)}(0) = 0$. Here the relative error is not $O(h^2)$ but $o(h)$. Under Assumption 1 (iii.a), $f_u(\cdot)$ is three times continuously differentiable, so that the Taylor remainder in (A.1) is of $o(h^3 s^3)$, and the third-order term itself drops out because $\int s^3 \phi^{(2)}(s) ds = 0$.

Finally, the second moment of $\phi^{(j)}(u_{th})$ is obtained similarly to the second moment of $\phi^{(1)}(u_{th})$ for $j = 1, 2, 3$. That is,

$$\mathbb{E}[\{\phi^{(j)}(u_{th})\}^2] = h \int [\phi^{(j)}(s)]^2 f_u(hs) ds = h v_j^2 f_u(0)\{1 + O(h^2)\}, \quad v_j^2 := \int [\phi^{(j)}(s)]^2 ds. \quad (\text{A.6})$$

Here, $O(h^2)$ and $O(h^4)$ terms drop out as $[\phi^{(j)}(\cdot)]^2$ is even. Because $\mathbb{E}[\phi^{(2)}(u_{th})]^2 = O(h^6)$ and $\mathbb{E}[\phi^{(3)}(u_{th})]^2 = O(h^8)$, they are of smaller order than h , so that $\text{var}[\phi^{(2)}(u_{th})] = O(h)$ and $\text{var}[\phi^{(3)}(u_{th})] = O(h)$. We use these features while proving of Lemma A.5. Note that $v_1^2 = v^2$ by construction.

A.2 Auxiliary Lemmas

Before proving the main claims in the text, we first provide some preliminary lemmas and prove the main claims efficiently.

Lemma A.1. *Given Assumption 1,*

- (i) $\sum_{t=1}^n \mathbf{w}_{t-i} = O_{\mathbb{P}}(\sqrt{n})$ for $i = 0, 1, 2, \dots$;
- (ii) $\sum_{t=1}^n \bar{\mathbf{k}}_t = O_{\mathbb{P}}(\sqrt{n})$ and $\sum_{t=1}^n \mathbf{k}_t = O_{\mathbb{P}}(n)$. □

Lemma A.2. *Given Assumption 1,*

- (i) $n^{-1} \sum_{t=1}^n \mathbf{w}_{t-i} \mathbf{w}_{t-j}' \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{w}_{t-i} \mathbf{w}_{t-j}']$ for $i, j = 0, 1, \dots, q-1$;
- (ii) $\sum_{t=1}^n \mathbf{w}_{t-i} \mathbf{x}_t' = O_{\mathbb{P}}(n)$ and $\sum_{t=1}^n u_{t-i} \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2})$ for $i = 0, 1, \dots$;
- (iii) $n^{-1} \sum_{t=1}^n \mathbf{k}_t \mathbf{k}_t' \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{k}_t \mathbf{k}_t']$;
- (iv) $n^{-1} \sum_{t=1}^n \mathbf{k}_t \mathbf{w}_{t-i}' \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{k}_t \mathbf{w}_{t-i}']$ for $i = 0, 1, \dots, q-1$;
- (v) $n^{-3/2} \sum_{t=1}^n \mathbf{k}_t \mathbf{x}_t' \rightsquigarrow \mathbb{E}[\mathbf{k}_t] \int_0^1 \bar{\mathbf{B}}_w'$. □

Lemma A.3. *Given Assumption 1,*

- (i) $n^{-3/2} \sum_{t=1}^n \mathbf{x}_t \rightsquigarrow \int_0^1 \bar{\mathbf{B}}_w$;

$$(ii) \ n^{-2} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t' \rightsquigarrow \int_0^1 \overline{\mathbf{B}}_w \overline{\mathbf{B}}_w'.$$

Remarks A.1. (i) Lemmas A.1 to A.3 are the MARDL analogues of lemmas A1 to A3 in [Cho et al. \(2015\)](#) and they are established by applying FCLT and the continuous mapping theorem to Assumption 1 (iv) and (vi).

(ii) Lemma A.1 describes the growth rates of the sums of the stochastic processes $\mathbf{w}_{t-i}$ and $\mathbf{k}_t$, which are $O_{\mathbb{P}}(\sqrt{n})$ and $O_{\mathbb{P}}(n)$, respectively. Lemma A.2 further explores the asymptotic behaviors between the co-products of the stochastic processes in Lemma A.1, revealing their limits. Lemma A.3 characterizes the asymptotic weak limits of the sums of $\mathbf{x}_t$ and its square. The normalization rates are $n^{3/2}$ and n^2 for $\sum \mathbf{x}_t$ and $\sum \mathbf{x}_t \mathbf{x}_t'$, respectively, reflecting the natures of integrated series $\mathbf{x}_t$, being the source of the super-consistency of the long-run parameter estimators in Theorem 2 (i) and Theorem 3. $\square$

Let $\tilde{\mathbf{x}}_t'$ be the t -th row vector of $\tilde{\mathbf{X}} := \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X} = [\mathbf{I} - \widehat{\mathbf{W}}(\widehat{\mathbf{W}}'\widehat{\mathbf{W}})^{-1}\widehat{\mathbf{W}}']\mathbf{X}$.

Lemma A.4. Given Assumption 1,

(i)

$$\mathbf{D}_{\mathbf{G}}^{-1} \mathbf{G}' \mathbf{G} \mathbf{D}_{\mathbf{G}}^{-1} \rightsquigarrow \begin{bmatrix} 1 & \mathbf{0}' & \int_0^1 \overline{\mathbf{B}}_w' \\ \mathbf{0} & \mathbb{E}[\overline{\mathbf{w}}_t \overline{\mathbf{w}}_t'] & \mathbf{0} \\ \int_0^1 \overline{\mathbf{B}}_w & \mathbf{0} & \int_0^1 \overline{\mathbf{B}}_w \overline{\mathbf{B}}_w' \end{bmatrix},$$

where $\mathbf{D}_{\mathbf{G}} := \text{diag}([\sqrt{n} \boldsymbol{\iota}'_{1+qk}, n \boldsymbol{\iota}'_k]')$, $\mathbf{G} := [\mathbf{g}_1, \dots, \mathbf{g}_n]'$, and $\mathbf{g}_t := [1, \overline{\mathbf{w}}_t', \mathbf{x}_t']'$;

(ii) $h^{-1/2} \mathbf{D}_{\mathbf{G}}^{-1} \mathbf{G}' \overline{\boldsymbol{\Phi}}^{(1)}(\mathbf{u}_h) \rightsquigarrow [\mathcal{B}_{\phi}(1), \mathcal{B}_{\phi \cdot \mathbf{w}}(1)', \int \overline{\mathbf{B}}_w' d\mathcal{B}_{\phi}(r)]'$, where $\overline{\boldsymbol{\Phi}}^{(1)}(\mathbf{u}_h) := [\overline{\phi}^{(1)}(u_{1h}), \dots, \overline{\phi}^{(1)}(u_{nh})]'$;

(iii) $\mathbf{D}_{\mathbf{H}}^{-1} \mathbf{G}' \mathbf{k} = O_{\mathbb{P}}(1)$, where $\mathbf{D}_{\mathbf{H}} := \text{diag}([n \boldsymbol{\iota}'_{1+qk}, n^{3/2} \boldsymbol{\iota}'_k]')$ and $\mathbf{k} := [\mathbf{k}_1, \dots, \mathbf{k}_n]'$;

(iv) $n^{-2} \tilde{\mathbf{X}}' \tilde{\mathbf{X}} = n^{-2} \mathbf{X}' \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X} = n^{-2} \mathbf{X}' [\mathbf{I} - \widehat{\mathbf{W}}(\widehat{\mathbf{W}}'\widehat{\mathbf{W}})^{-1}\widehat{\mathbf{W}}']\mathbf{X} \rightsquigarrow \int_0^1 \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$;

(v) $(n^2 h)^{-\frac{1}{2}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \rightsquigarrow \int \tilde{\mathbf{B}}_w d\mathcal{B}_{\phi}$;

(vi) $(nh)^{-\frac{1}{2}} \sum_{t=1}^n \overline{\phi}^{(1)}(u_{th}) \mathbf{j}_t \overset{A}{\rightsquigarrow} N(\mathbf{0}, v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'])$. $\square$

Remarks A.2. (i) Lemma A.4 is the MARDL counterpart of corollary A1 in [Cho et al. \(2015\)](#) that derives the weak limits of the normalized matrices. Lemma A.4 (ii) is the key result establishing the FCLT for the modal regression score $\phi^{(1)}(u_{th})/\sqrt{h}$ with other integrated regressors. In particular the normalized and centered modal score $h^{-1/2} \overline{\phi}^{(1)}(u_{th})$ plays the role parallel to the quantile score $\psi_{\tau}(u_t) := \tau - \mathbb{1}\{u_t < 0\}$ in the QARDL framework. Both scores are MDAs with respect to $\mathcal{F}_t$ and the limiting variance $v^2 f_u(0)$ of the former corresponds to $\tau(1 - \tau)$ of the latter. The essential difference is in the fact that the modal score is a triangular array indexed by the bandwidth, so that all weak convergence results in Lemma A.4 are array limits driven by Assumption 1 (vi).

(ii) Lemmas A.4 (ii) and (v) provide the weak limits of the modal score along with other integrated series and the weak limit in Lemma A.4 (v) is the source of the mixed normal limits in Theorem 2 (i) and Theorem 3. Lemmas A.4 (iv)–(vi) constitute the weak limits in Lemma A.6 given below, from which the asymptotic distributions of all parameter estimators in Theorems 1 to 5 are derived. $\square$

Lemma A.5. *Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$,*

- (i) $\|\hat{\rho}_n - \rho_\dagger\| = O_{\mathbb{P}}((nh^3)^{-1/2})$, where $\hat{\rho}_n := (\hat{\nu}_n, \hat{\nu}'_n, \hat{\phi}'_n)'$;
- (ii) $\|\hat{\kappa}_n - \beta_*\| = O_{\mathbb{P}}((n^2h^3)^{-1/2})$;
- (iii) $\|\hat{\alpha}_n - \alpha_*\| = O_{\mathbb{P}}((nh^3)^{-1/2})$. $\square$

Remarks A.3. (i) By Lemma A.5, the convergence rates of the estimators differ between the parameters due to the mixed stationarity structure of z_t . For λ_* and ϕ_* , their estimators are influenced by two distinct sources. First, the standard deviation of the modal estimator along the stationary directions introduces an estimation error with convergence rate $(nh^3)^{-1/2}$. Second, the smoothing bias generated by the bandwidth in the modal kernel objective function introduces another estimation error with convergence rate h^2 . Under the given bandwidth condition $nh^7 = O(1)$, the latter is dominated by the former, since $h^2 = O((nh^3)^{-1/2})$, leading to the same convergence rate for the estimators of λ_* and ϕ_* . Here, $\hat{\rho}_n$ estimates $\rho_\dagger = (\nu_\dagger, \nu'_\dagger, \phi'_\dagger)'$ such that $\nu_\dagger = \alpha_* + \mu_* \iota'_p \phi_*$, $\nu_\dagger = \delta_* + \Xi_* \phi_*$, and $\phi_\dagger = \phi_*$ by (14). Therefore, $\hat{\rho}_n$ is a linear transformation of $\hat{\lambda}_n$ and $\hat{\phi}_n$ as given in (15). Consequently the convergence rate of $\hat{\rho}_n$ is determined by those of $\hat{\lambda}_n$ and $\hat{\phi}_n$.

(ii) For the long-run parameter γ_* associated with the integrated regressor x_t , $\hat{\kappa}_n$ exhibits a rate $(n^2h^3)^{-1/2}$ faster than that of $\hat{\rho}_n$, reflecting that $\sum_{t=1}^n x_t x'_t = O_{\mathbb{P}}(n^2)$. This is the MARDL analogue of the $O_{\mathbb{P}}(n^{-1})$ convergence rate of the long-run parameter estimator in Cho et al. (2015, theorem 2). The additional $h^{3/2}$ factor is introduced from the modal bandwidth. In addition, each of $\hat{\alpha}_n$, $\hat{\delta}_n$, $\hat{\gamma}_n$, and $\hat{\phi}_n$ converges at the rate $(nh^3)^{-1/2}$ by Lemma A.5 (iii).

(ii) Estimating the density function by the smoothing kernel introduces a smoothing bias, which is computed as $b_h := -h^2 f_u^{(3)}(0)/(2f_u^{(2)}(0))$; see the proof of Lemma A.6 (iii). Therefore, $b_h = O(h^2)$ which is also $O((nh^3)^{-1/2})$ if $nh^7 = O(1)$. By this, the smoothing bias can influence the asymptotic distributions of $\hat{\rho}_n$ and $\hat{\alpha}_n$. Lemma A.6 and Theorem 1 given below examine the influence of the smoothing bias to the asymptotic distributions. $\square$

Let e_1 denote the first standard basis vector, viz., $e' = (1, \mathbf{0}'_{(kq+p) \times 1})'$.

Lemma A.6. *Given Assumption 1, if $h \rightarrow 0$, $nh^5 \rightarrow \infty$, and $nh^7 = O(1)$, as $n \rightarrow \infty$,*

(i)

$$\frac{1}{nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{j}_t \mathbf{j}_t' \xrightarrow{\mathbb{P}} f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'],$$

$$\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w', \quad \text{and} \quad \frac{1}{n^{3/2} h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \mathbf{j}_t \tilde{\mathbf{x}}_t' \xrightarrow{\mathbb{P}} \mathbf{0};$$

(ii)

$$\begin{aligned} n\sqrt{h^3} (\hat{\kappa}_n - \beta_*) &= \left[\frac{1}{n^2 h^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right]^{-1} \left[\frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t \right] + o_{\mathbb{P}}(1) \\ &\stackrel{A}{\sim} N\left(\mathbf{0}, \omega_* \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1}\right) \end{aligned}$$

conditional on $\tilde{\mathbf{B}}_w(\cdot)$;

(iii) $\sqrt{nh^3} (\hat{\rho}_n - \rho_{\dagger} - b_h \mathbf{e}_1) \stackrel{A}{\sim} N(\mathbf{0}, \Psi^+)$;

(iv) the weak limits in (ii) and (iii) are independent; and

(v) $\sqrt{nh^3} ((\hat{\mathbf{v}}_n - \mathbf{v}_{\dagger})', (\hat{\phi}_n - \phi_*)')' \stackrel{A}{\sim} N(\mathbf{0}, \Psi)$. □

Let $\ddot{\mathbf{x}}_t$ be the t -th row vector of $\ddot{\mathbf{X}} := \mathbf{P}_{\mathbf{J}} \mathbf{X}$, where $\mathbf{P}_{\mathbf{J}} := (\mathbf{I} - \mathbf{J}(\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}')$ such that $\mathbf{J} := [\mathbf{j}_1, \dots, \mathbf{j}_n]'$.

[JSCHO: Using this definition, prove A.7 (i.a) and (i.b).]

Lemma A.7. (i) (i.a) $(n^2 h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} (\ddot{\mathbf{x}}_t \ddot{\mathbf{x}}_t' - \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t') = o_{\mathbb{P}}(1)$; [JSCHO: prove this using the above definition.]

(i.b) $(n\sqrt{h})^{-1} \sum_{t=1}^n \phi_t^{(1)} (\ddot{\mathbf{x}}_t - \tilde{\mathbf{x}}_t) = o_{\mathbb{P}}(1)$; [JSCHO: prove this using the above definition.]

(ii) $(n^{3/2} h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t' \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w'$;

(iii) $\sum_{t=1}^n \phi_t^{(2)} \bar{\mathbf{w}}_t \mathbf{x}_t' = O_{\mathbb{P}}(n\sqrt{h})$;

(iv) $\sum_{t=1}^n \phi_t^{(2)} \mathbf{k}_t \mathbf{x}_t' = O_{\mathbb{P}}(n\sqrt{h})$;

(v) $\sqrt{nh^3} (\hat{\alpha}_n - \alpha_* - b_h) \stackrel{A}{\sim} N(0, \omega_{\alpha})$;

(vi) $(n^{3/2} h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}_t' = o_{\mathbb{P}}(1)$. □

Remarks A.4. (i) The smoothing bias b_h is associated with only the intercept estimator when $f_u^{(3)}(0) \neq 0$.

This bias vanishes if the undersmoothing condition holds: $nh^7 \rightarrow 0$. However, the bandwidth for optimizing the asymptotic mean squared error (AMSE) does not satisfy the undersmoothing condition as pointed out by Yao and Li (2014). If $h \propto n^{-1/7}$, then it holds that $nh^5 \rightarrow \infty$ for the consistency of the estimator, but it does not satisfy $nh^7 \rightarrow 0$ for bias elimination.

(ii) Lemma A.6 (ii) states that the long-run direction is estimated at the rate $n\sqrt{h^3}$ with a mixed normal limit, and both features have a simple source. The rate is faster than the stationary rate $\sqrt{nh^3}$ by

a factor of $\sqrt{n}$ because the regressor is integrated, viz., $\sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' = O_{\mathbb{P}}(n^2)$ rather than $O_{\mathbb{P}}(n)$ so that the same amount of score information is divided by a much larger Hessian. The limit is mixed normal, rather than normal, because that Hessian does not converge to a constant. That is $n^{-2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$. Conditionally on this limit, the score is Gaussian, so that the ratio is asymptotically mixed normal. This is the modal counterpart of the mixed normal limit for the long-run coefficient in [Cho et al. \(2015\)](#) with ω_* replacing $\tau(1 - \tau)/f_\tau^2$. Because the mixing variate cancels in a studentized statistic, the Wald test of Section 2.2 remains asymptotically chi-squared under the null. $\square$

Proof of Lemma A.1: Assumption 1 (vi) yields that $n^{-1/2} \sum_{t=1}^n \bar{\mathbf{w}}_t \rightsquigarrow \mathbf{B}_w(1)$ and $n^{-1/2} \sum_{t=1}^n \bar{\mathbf{k}}_t \rightsquigarrow \mathbf{B}_k(1)$. Additionally, it follows that $n^{-1} \sum_{t=1}^n \mathbf{k}_t \xrightarrow{\mathbb{P}} \mathbb{E}[\mathbf{k}_t]$ by the ergodic theorem. This completes the proof. $\square$

Proof of Lemma A.2: (i) Utilizing the ergodic theorem given Assumption 1 (iv), Lemma A.2 (i) follows.

(ii) We follow the proof of Proposition 18.1(d) in [Hamilton \(2020, pp. 562–563\)](#). Specifically, $\mathbf{u}_{t-s}$ and $\boldsymbol{\xi}_{t-1}$ in [Hamilton \(2020\)](#) correspond to $\mathbf{w}_{t-i}$ and $\mathbf{x}_t$ of here. Moreover, by Assumptions 1 (iv) and (vi), $\sum_{t=1}^n \mathbf{w}_{t-i} \mathbf{x}_t' = O_{\mathbb{P}}(n)$ and $n^{-1/2} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \mathbf{w}_t \rightsquigarrow \bar{\mathbf{B}}_w(\cdot)$. By the continuous mapping theorem and Assumptions 1 (i) and (iv), it follows that $\sum_{t=1}^n u_{t-i} \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2})$. Specifically, $\sum_{t=1}^n u_{t-i} \mathbf{x}_t = \mathbb{E}[u_t] \sum_{t=1}^n \mathbf{x}_t + \sum_{t=1}^n (u_{t-i} - \mathbb{E}[u_t]) \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2}) + O_{\mathbb{P}}(n)$ by Lemma A.3 (i).

(iii)–(v) The proofs follow from the stationarity and ergodicity of $\{k_{t,i}\}$, which is established by Assumption 1 (v) and [Brockwell and Davis \(1991, theorem 4.4.1\)](#) combined with Lemma A.2 (ii) and Assumptions 1 (iv) and (vi). The detailed arguments are identical to the proof of Lemma A2 in [Cho et al. \(2015\)](#). $\square$

Proof of Lemma A.3: (i and ii) By Assumptions 1 (iv) and (vi), $n^{-1/2} \sum_{t=1}^{\lfloor n(\cdot) \rfloor} \mathbf{w}_t \rightsquigarrow \bar{\mathbf{B}}_w(\cdot)$, and this implies that $n^{-1/2} \mathbf{x}_{\lfloor nr \rfloor} = n^{-1/2} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{w}_t \rightsquigarrow \bar{\mathbf{B}}_w(r)$ with $n^{-1/2} \mathbf{x}_0 = o_{\mathbb{P}}(1)$. We next apply the continuous mapping theorem and Lemma 3.1(e) of [Phillips and Durlauf \(1986, lemma 3.1.e\)](#) to obtain the desired results. $\square$

Proof of Lemma A.4: (i) By Lemmas A.1 (i), A.2 (i) and (ii), it follows that $\{n^{-1} \sum_{t=1}^n \bar{\mathbf{w}}_t, n^{-1} \sum_{t=1}^n \bar{\mathbf{w}}_t \bar{\mathbf{w}}_t'\}$ $n^{-3/2} \sum_{t=1}^n \bar{\mathbf{w}}_t \mathbf{x}_t' \} \xrightarrow{\mathbb{P}} \{\mathbf{0}, \mathbb{E}[\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t'], \mathbf{0}\}$. Furthermore, Lemma A.3 provides the limits of $n^{-3/2} \sum_{t=1}^n \mathbf{x}_t$ and $n^{-2} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t'$. All the convergences are jointly achieved because each of them is a continuous functional of $\mathbf{B}_n(\cdot)$, whose weak limit is given in Assumption 1 (vi).

(ii) The first two components are equal to the third and fourth components of $\mathbf{B}_n(1)$ in Assumption 1 (vi). For the final component, we let $\varepsilon_{nt} := \bar{\phi}^{(1)}(u_{th})/\sqrt{h}$, noting that $\{\varepsilon_{nt}, \mathcal{F}_t\}$ is an MDA

with $\sup_{n,t} \mathbb{E}|\varepsilon_{nt}|^{2+\lambda} < \infty$ by Assumption 1 (ii.b) and (iii.a). Furthermore, Assumption 1 (vi) implies that $n^{-1/2}(\mathbf{x}_{[n(\cdot)]}, \sum_{t \leq [n(\cdot)]} \varepsilon_{nt}) \rightsquigarrow (\bar{\mathbf{B}}_w(\cdot), \mathcal{B}_\phi(\cdot))$, so that $n^{-1} \sum_{t=1}^n (n^{-1/2} \mathbf{x}_t) \varepsilon_{nt} \rightsquigarrow \int \bar{\mathbf{B}}_w d\mathcal{B}_\phi$ by Hansen (1992, theorem 3.1). Here, the contribution of $\mathbf{x}_0$ is asymptotically negligible, because $n^{-1} h^{-1/2} \mathbf{x}_0 \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) = \mathbf{x}_0 O_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1)$.

(iii) We first note that $\mathbf{D}_H^{-1} \mathbf{G}' \mathbf{k} = [n^{-1} \sum_{t=1}^n \mathbf{k}_t, n^{-1} \sum_{t=1}^n (\bar{\mathbf{w}}_t \mathbf{k}_t')', n^{-3/2} \sum_{t=1}^n (\mathbf{x}_t \mathbf{k}_t')']'$, and each element is $O_{\mathbb{P}}(1)$ by Lemmas A.1 (ii), A.2 (iv), and A.2 (v), respectively. Therefore, $\mathbf{D}_H^{-1} \mathbf{G}' \mathbf{k} = O_{\mathbb{P}}(1)$.

(iv) We first note that

$$n^{-2} \mathbf{X}' \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X} = n^{-2} \mathbf{X}' \mathbf{X} - (n^{-3/2} \mathbf{X}' \widehat{\mathbf{W}}) (n^{-1} \widehat{\mathbf{W}}' \widehat{\mathbf{W}})^{-1} (n^{-3/2} \widehat{\mathbf{W}}' \mathbf{X}).$$

Given this, Lemma A.3 (ii) implies that $n^{-2} \mathbf{X}' \mathbf{X} \rightsquigarrow \int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w'$. Furthermore, we note that $n^{-1} \widehat{\mathbf{W}}' \widehat{\mathbf{W}} \xrightarrow{\mathbb{P}} \text{diag}(1, \mathbb{E}[\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t'])$ and $n^{-3/2} \widehat{\mathbf{W}}' \mathbf{X} \rightsquigarrow [\int \bar{\mathbf{B}}_w, \mathbf{0}]'$ by (i), so that

$$n^{-2} \mathbf{X}' \mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X} \rightsquigarrow \int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w' - \int \bar{\mathbf{B}}_w \int \bar{\mathbf{B}}_w' = \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'.$$

(v) We note that $\sum_{t=1}^n \phi^{(1)}(u_{th}) \tilde{\mathbf{x}}_t = \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \tilde{\mathbf{x}}_t$, because the rows of $\mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X}$ sum to zero. Therefore,

$$\begin{aligned} \frac{1}{n\sqrt{h}} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \tilde{\mathbf{x}}_t &= \frac{1}{n\sqrt{h}} \mathbf{X}' \bar{\boldsymbol{\Phi}}^{(1)}(\mathbf{u}_h) - (n^{-3/2} \mathbf{X}' \widehat{\mathbf{W}}) (n^{-1} \widehat{\mathbf{W}}' \widehat{\mathbf{W}})^{-1} \left(\frac{1}{\sqrt{nh}} \widehat{\mathbf{W}}' \bar{\boldsymbol{\Phi}}^{(1)}(\mathbf{u}_h) \right) \\ &\rightsquigarrow \int \bar{\mathbf{B}}_w d\mathcal{B}_\phi - \left[\int \bar{\mathbf{B}}_w, \mathbf{0} \right] \text{diag}[1, \mathbb{E}[\bar{\mathbf{w}}_t \bar{\mathbf{w}}_t']] \begin{bmatrix} \mathcal{B}_\phi(1) \\ \mathcal{B}_{\phi \cdot \mathbf{w}}(1) \end{bmatrix} \\ &= \int \bar{\mathbf{B}}_w d\mathcal{B}_\phi - \int \bar{\mathbf{B}}_w \mathcal{B}_\phi(1) \end{aligned}$$

by (i) and (ii). We further note that $\int \bar{\mathbf{B}}_w d\mathcal{B}_\phi - \int \bar{\mathbf{B}}_w \cdot \mathcal{B}_\phi(1) = \int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi$ using (11).

(vi) Given that $\mathbf{j}_t := (1, \bar{\mathbf{w}}_t', \mathbf{k}_t')'$, $(nh)^{-1/2} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{j}_t$ is a fixed linear map of the $\mathcal{B}_{\phi \cdot}$, $\mathcal{B}_{\phi \cdot \mathbf{w}}$ - and $\mathcal{B}_{\phi \cdot \mathbf{k}}$ -components of $\mathbf{B}_n(1)$, so that it is asymptotically normal by Assumption 1 (vi). Furthermore, its covariance matrix converges to $v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$ because $\mathbb{E}[\{\bar{\phi}^{(1)}(u_{th})\}^2 | \mathcal{F}_{t-1}]$ is constant under Assumptions (i.b) and (iv.b), so that $h^{-1} \mathbb{E}[\{\bar{\phi}^{(1)}(u_{th})\}^2 \mathbf{j}_t \mathbf{j}_t'] = h^{-1} \mathbb{E}[\{\bar{\phi}^{(1)}(u_{th})\}^2] \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'] \rightarrow v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$ by (A.3). This completes the proof. $\blacksquare$

Proof of Lemma A.5: If we let $\boldsymbol{\vartheta} := (\boldsymbol{\rho}', \boldsymbol{\kappa}')'$ and $\tilde{\mathbf{z}}_t := (\mathbf{j}_t', \mathbf{x}_t')'$, then $\mathbf{z}_t' \boldsymbol{\alpha}_* = \tilde{\mathbf{z}}_t' \boldsymbol{\vartheta}_\dagger$ by (13). Furthermore from the definitions of $\boldsymbol{\varsigma} := \boldsymbol{\lambda} + \boldsymbol{\Gamma}_* \boldsymbol{\phi}$ and $\boldsymbol{\rho} := (\nu, \mathbf{v}', \boldsymbol{\phi}')'$, there exists a linear bijection relationship between $\boldsymbol{\alpha}$ and $\boldsymbol{\vartheta}$. Using this fact, we let $\tilde{Q}_{nh}(\cdot)$ be a function defined on the space of $\boldsymbol{\vartheta}$ such that for each $\boldsymbol{\alpha}$

$$Q_{nh}(\alpha) = \tilde{Q}_{nh}(\vartheta).$$

To show the given claims, we let $\tilde{\mathbf{D}}_n := \text{diag}(\sqrt{nh^3} \mathbf{I}_{1+kq+p}, n\sqrt{h^3} \mathbf{I}_k)$ and show that for any $\eta > 0$ there exists $\theta < \infty$ such that

$$\mathbb{P} \left\{ \sup_{\|c\|=\theta} \tilde{Q}_{nh}(\vartheta_{\dagger} + \tilde{\mathbf{D}}_n^{-1}c) < \tilde{Q}_{nh}(\vartheta_{\dagger}) \right\} \geq 1 - \eta, \quad (\text{A.7})$$

following [Fan and Li \(2001\)](#). For this we approximate $\tilde{Q}_{nh}(\vartheta_{\dagger} + \tilde{\mathbf{D}}_n^{-1}c)$ by a third-order expansion:

$$\tilde{Q}_{nh}(\vartheta_{\dagger} + \tilde{\mathbf{D}}_n^{-1}c) - \tilde{Q}_{nh}(\vartheta_{\dagger}) = I_1 + I_2 + I_3,$$

where

$$I_1 := -\frac{1}{nh^2} \sum_{t=1}^n \phi^{(1)}(u_{th}) c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t, \quad I_2 := \frac{1}{2nh^3} \sum_{t=1}^n \phi^{(2)}(u_{th}) (c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t)^2, \quad \text{and}$$

$$I_3 := -\frac{1}{6nh^4} \sum_{t=1}^n \phi^{(3)}(u_{th}^*) (c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t)^3,$$

where, for each t , u_{th}^* denotes an intermediate point between u_{th} and $u_{th} - c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t$, viz., the value of the standardized error at the perturbed parameter $\vartheta_{\dagger} + \tilde{\mathbf{D}}_n^{-1}c$. It arises from the exact third-order Taylor expansion with Lagrange remainder, so that $|u_{th}^* - u_{th}| \leq |c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t|$ and u_{th}^* is $\mathcal{F}_{t-1}$ -measurable given u_{th} .

We examine each term on the right side. First, we examine I_1 . For this we decompose $\phi^{(1)}(u_{th}) = \bar{\phi}^{(1)}(u_{th}) + m_h$ with $m_h = O(h^4)$. Therefore,

$$I_1 = -\frac{1}{nh^2} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t + \frac{1}{nh^2} \sum_{t=1}^n m_h c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t.$$

Here, we note that $\tilde{z}_t = (\mathbf{j}'_t, \mathbf{x}'_t)'$ and that for given c ,

$$\frac{1}{nh^2} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t = \frac{1}{nh^2} c' \tilde{\mathbf{D}}_n^{-1} \sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) (\mathbf{j}'_t, \mathbf{x}'_t)' = O_{\mathbb{P}}((nh^3)^{-1}),$$

because $\sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{j}_t = O_{\mathbb{P}}(\sqrt{nh})$ and $\sum_{t=1}^n \bar{\phi}^{(1)}(u_{th}) \mathbf{x}_t = O_{\mathbb{P}}(n\sqrt{h})$ by Lemma [A.4](#) (ii). In addition we note that $\tilde{z}_t = (\mathbf{j}'_t, \mathbf{x}'_t)' = (1, \bar{\mathbf{w}}'_t, \mathbf{k}'_t, \mathbf{x}'_t)'$, so that

$$\frac{1}{nh^2} \sum_{t=1}^n m_h c' \tilde{\mathbf{D}}_n^{-1} \tilde{z}_t = \frac{m_h}{nh^2} c' \tilde{\mathbf{D}}_n^{-1} \sum_{t=1}^n (1, \bar{\mathbf{w}}'_t, \mathbf{k}'_t, \mathbf{x}'_t)' = O_{\mathbb{P}}((n^{-1}h)^{1/2}),$$

because $m_h = O(h^4)$, $\sum_{t=1}^n 1 = O(n)$, $\sum_{t=1}^n \bar{w}_t = O_{\mathbb{P}}(\sqrt{n})$, $\sum_{t=1}^n \mathbf{k}_t = O_{\mathbb{P}}(\sqrt{n})$, and $\sum_{t=1}^n \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2})$ by Lemma A.1. [JSCHO: $\sum_{t=1}^n \mathbf{k}_t = O_{\mathbb{P}}(n)$? Check how this modification influences the other results.] We further note that $O_{\mathbb{P}}((nh^3)^{-1}) = O_{\mathbb{P}}((n^{-1}h)^{1/2}) + o_{\mathbb{P}}(1)$, given that $nh^7 = O_{\mathbb{P}}(1)$. Therefore $I_1 = O_{\mathbb{P}}((nh^3)^{-1})$.

Second, we examine I_2 . We first note that

$$I_2 = \frac{1}{2nh^3} \sum_{t=1}^n \left\{ \left[\phi^{(2)}(u_{th}) - \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] \right] + \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] \right\} (\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t)^2.$$

Because $u_t \perp \mathcal{F}_{t-1}$, it follows that $\mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] = h^3 f_u^{(2)}(0) \{1 + o(h)\}$ uniformly by (A.5). Therefore

$$\begin{aligned} \frac{1}{2nh^3} \sum_{t=1}^n \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}] (\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t)^2 &= \frac{f_u^{(2)}(0)}{2} \frac{1}{n} \sum_{t=1}^n (\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t)^2 \{1 + o_{\mathbb{P}}(1)\} \\ &= \frac{f_u^{(2)}(0)}{2nh^3} \mathbf{c}' \ddot{\mathbf{D}}_n^{-1} \left(\sum_{t=1}^n \tilde{\mathbf{z}}_t \tilde{\mathbf{z}}_t' \right) \ddot{\mathbf{D}}_n^{-1} \mathbf{c} \{1 + o_{\mathbb{P}}(1)\}, \end{aligned}$$

where $\ddot{\mathbf{D}}_n := h^{-3/2} \tilde{\mathbf{D}}_n$. We further note that

$$\mathbf{M}_n := \ddot{\mathbf{D}}_n^{-1} \left(\sum_{t=1}^n \tilde{\mathbf{z}}_t \tilde{\mathbf{z}}_t' \right) \ddot{\mathbf{D}}_n^{-1} \rightsquigarrow \mathbf{M} := \begin{bmatrix} \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'] & \mathbb{E}[\mathbf{j}_t] \int \bar{\mathbf{B}}_w' \\ \int \bar{\mathbf{B}}_w \mathbb{E}[\mathbf{j}_t]' & \int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w' \end{bmatrix}$$

by Lemmas A.1, A.2 (i, iii, iv), A.3, and A.4 (i). Because $\mathbb{E}[\mathbf{j}_t]$ is the first column of $\mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$, the Schur complement of $\mathbf{M}$ is $\int \bar{\mathbf{B}}_w \bar{\mathbf{B}}_w' - \int \bar{\mathbf{B}}_w \mathbb{E}[\mathbf{j}_t]' \mathbb{E}[\mathbf{j}_t] \int \bar{\mathbf{B}}_w' = \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$, which is positive definite a.s. by Assumption 1 (iv.c) and (vi), so that $\mathbf{M}$ is positive definite with probability 1.

Next, we examine the MDA part of I_2 . If we write $\bar{\phi}_t^{(2)} := \phi^{(2)}(u_{th}) - \mathbb{E}[\phi^{(2)}(u_{th})|\mathcal{F}_{t-1}]$ and $a_t := \mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t$, then $\{\bar{\phi}_t^{(2)} a_t^2, \mathcal{F}_t\}$ is an MDA, because a_t is $\mathcal{F}_{t-1}$ -measurable. By (A.6), $\mathbb{E}[(\bar{\phi}_t^{(2)})^2] = O(h)$, so that

$$\text{var} \left(\frac{1}{2nh^3} \sum_{t=1}^n \bar{\phi}_t^{(2)} a_t^2 \right) = \frac{1}{4n^2 h^6} \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(2)})^2 a_t^4] = \frac{O(h)}{4n^2 h^6} \sum_{t=1}^n \mathbb{E}[a_t^4].$$

Along the stationary directions, $a_t = \theta (nh^3)^{-1/2} \bar{u}_t$ for a unit vector contraction $\bar{u}_t$ of $\mathbf{j}_t$, and $\mathbb{E}[\bar{u}_t^4] < \infty$ by Assumption 1 (iv.d). [JSCHO: ???] Along the integrated direction, $\mathbf{x}_t = O_{\mathbb{P}}(\sqrt{n})$ is offset by the factor n^{-1} in $\tilde{\mathbf{D}}_n^{-1}$, giving the same order. Hence $\sum_{t=1}^n \mathbb{E}[a_t^4] = O(\theta^4 (nh^6)^{-1})$ and the standard deviation of the display is $O(\theta^2 h^{1/2} n^{-3/2} h^{-6})$. Relative to $(nh^3)^{-1} \theta^2$, this is of order $(nh^5)^{-1/2} = o(1)$ under $nh^5 \rightarrow \infty$. The MDA part of I_2 is therefore negligible relative to its systematic part. [JSCHO: The verbal explanation is not clear. State this part more clearly.]

Combining the two parts, and recalling $f_u^{(2)}(0) < 0$ from Assumption 1 (iii.c), we obtain

$$I_2 = \frac{f_u^{(2)}(0)}{2nh^3} \mathbf{c}' \mathbf{M}_n \mathbf{c} \{1 + o_{\mathbb{P}}(1)\} \leq -\frac{\kappa_n \theta^2}{(nh^3)}, \quad \text{where} \quad \kappa_n := \frac{1}{2} |f_u^{(2)}(0)| \lambda_{\min}(\mathbf{M}_n) \{1 + o_{\mathbb{P}}(1)\},$$

on a set whose probability tends to one, where $\lambda_{\min}(\cdot)$ denotes the smallest eigenvalue and $\|\mathbf{c}\| = \theta$. Since $\mathbf{M}_n \rightsquigarrow \mathbf{M}$ with $\mathbf{M}$ positive definite with probability one, $\lambda_{\min}(\mathbf{M}_n)$ is bounded away from zero in probability and so is κ_n .

Third, we examine I_3 . We write $\phi^{(3)}(u_{th}^*) = \phi^{(3)}(u_{th}) + \{\phi^{(3)}(u_{th}^*) - \phi^{(3)}(u_{th})\}$. Then, it follows that

$$I_3 = -\frac{1}{6nh^4} \sum_{t=1}^n \phi^{(3)}(u_{th}) (\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t)^3 - \frac{1}{6nh^4} \sum_{t=1}^n \{\phi^{(3)}(u_{th}^*) - \phi^{(3)}(u_{th})\} (\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t)^3.$$

We examine each component on the right side. If we rewrite the first part on the right side as follow:

$$-\frac{1}{6nh^4} \sum_{t=1}^n \left\{ \left[\phi^{(3)}(u_{th}) - \mathbb{E}[\phi^{(3)}(u_{th}) | \mathcal{F}_{t-1}] \right] + \mathbb{E}[\phi^{(3)}(u_{th}) | \mathcal{F}_{t-1}] \right\} (\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t)^3,$$

then the systematic and martingale parts are of order $\theta(nh^3)^{-1/2}$ and $\theta(nh^5)^{-1/2}$ respectively by following the procedure used for I_2 . Here, we use the fact that $\mathbb{E}[\phi^{(3)}(u_{th})] = O(h^4)$ and $\text{var}[\phi^{(3)}(u_{th})] = O(h)$ by (A.5) and (A.6), respectively. Therefore, both terms are $o_{\mathbb{P}}(1)$. For the second part, note that the shift $\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t$ is $\mathcal{F}_{t-1}$ -measurable, and for any $\mathcal{F}_{t-1}$ -measurable a_t , $\mathbb{E}[\phi^{(j)}((u_t - a_t)/h) | \mathcal{F}_{t-1}] \leq h \|f_u\|_{\infty} \int |\phi^{(j)}(s)| ds = O(h)$ uniformly, while $|\phi^{(3)}(u_{th}^*) - \phi^{(3)}(u_{th})| \leq \|\phi^{(4)}\|_{\infty} |\mathbf{c}' \tilde{\mathbf{D}}_n^{-1} \tilde{\mathbf{z}}_t|/h$. Interpolating these two bounds via $\min\{x, 1\} \leq x^{2/3}$ and applying Hölder's inequality with the fourth moment condition of Assumption 1 (iv.d) the second part of I_3 is $o_{\mathbb{P}}(|I_2|)$ provided $nh^5 \rightarrow \infty$. The same computations apply to the integrated block with $\mathbf{x}_t = O_{\mathbb{P}}(\sqrt{n})$ offset by the factor n^{-1} in $\tilde{\mathbf{D}}_n^{-1}$.

Combining the three terms, (A.7) holds for sufficiently large θ , which delivers a local maximizer with $\|\tilde{\mathbf{D}}_n(\hat{\boldsymbol{\vartheta}}_n - \boldsymbol{\vartheta}_*)\| = O_{\mathbb{P}}(1)$, i.e., (i) and (ii). Part (iii) follows from the inverse rotation $\hat{\boldsymbol{\lambda}}_n = \hat{\boldsymbol{\varsigma}}_n - \boldsymbol{\Gamma}_* \hat{\boldsymbol{\phi}}_n$. In particular, $\hat{\boldsymbol{\gamma}}_n - \boldsymbol{\gamma}_* = (\hat{\boldsymbol{\kappa}}_n - \boldsymbol{\beta}_*) - \boldsymbol{\beta}_* \boldsymbol{\iota}_p' (\hat{\boldsymbol{\phi}}_n - \boldsymbol{\phi}_*) = O_{\mathbb{P}}((nh^3)^{-1/2})$. $\blacksquare$

Proof of Lemma A.6: Throughout this proof, we write $\phi_t^{(j)} := \phi^{(j)}(u_{th})$ for notational simplicity and work with the reparameterized model in (13) so that the objective function $Q_{nh}(\cdot)$ is translated into $\tilde{Q}_{nh}(\cdot)$ such that the parameters and regressors are given as $\boldsymbol{\vartheta} = (\boldsymbol{\rho}', \boldsymbol{\kappa}')'$ and $\tilde{\mathbf{z}}_t = (\mathbf{j}_t', \mathbf{x}_t')'$ with $\mathbf{j}_t = (1, \bar{\mathbf{w}}_t', \mathbf{k}_t')'$ respectively. By (14), $\kappa_{\dagger} = \beta_*$, so that $\hat{\boldsymbol{\kappa}}_n$ estimates $\boldsymbol{\beta}_*$. Given this, the first-order condition for $\hat{\boldsymbol{\vartheta}}_n$ is given

as $\sum_{t=1}^n \phi^{(1)}((y_t - \tilde{\mathbf{z}}_t' \hat{\boldsymbol{\theta}}_n)/h) \tilde{\mathbf{z}}_t \equiv \mathbf{0}$, and we expand it around $\boldsymbol{\theta}_\dagger$, obtaining the following block system:

$$\mathbf{s}_j = h^{-1} [\mathbf{T}_{jj} \mathbf{a} + \mathbf{T}_{j\mathbf{X}} \mathbf{b}] + o_{\mathbb{P}}(\sqrt{nh}) \quad \text{and} \quad \mathbf{s}_{\mathbf{X}} = h^{-1} [\mathbf{T}_{\mathbf{X}j} \mathbf{a} + \mathbf{T}_{\mathbf{X}\mathbf{X}} \mathbf{b}] + o_{\mathbb{P}}(n\sqrt{h}), \quad (\text{A.8})$$

where $\mathbf{a} := \hat{\boldsymbol{\rho}}_n - \boldsymbol{\rho}_\dagger$, $\mathbf{b} := \hat{\boldsymbol{\kappa}}_n - \boldsymbol{\kappa}_\dagger$, $\mathbf{s}_j := \sum_{t=1}^n \phi_t^{(1)} \mathbf{j}_t$, $\mathbf{s}_{\mathbf{X}} := \sum_{t=1}^n \phi_t^{(1)} \mathbf{x}_t$, and $\mathbf{T}_{jj} := \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{j}_t'$, $\mathbf{T}_{j\mathbf{X}} := \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{x}_t' = \mathbf{T}_{\mathbf{X}j}'$, and $\mathbf{T}_{\mathbf{X}\mathbf{X}} := \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t \mathbf{x}_t'$.

(i) We first examine the asymptotic behavior of $\mathbf{T}_{jj}$. For this we first note that $\mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] = h^3 f_u^{(2)}(0) \{1 + o(h)\}$ by (A.5). Therefore,

$$\frac{1}{nh^3} \mathbf{T}_{jj} = \frac{1}{nh^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \mathbf{j}_t \mathbf{j}_t' + \frac{1}{nh^3} \sum_{t=1}^n \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] \mathbf{j}_t \mathbf{j}_t'.$$

Here, the second term converges to $f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']$ by the ergodic theorem, and the first term is $o_{\mathbb{P}}(1)$. Specifically, if we let $\bar{\phi}_t^{(2)} := \phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]$, then the following holds. First, $\{\bar{\phi}_t^{(2)}, \mathcal{F}_t\}$ is an MDA by construction, and $\mathbb{E}[(\bar{\phi}_t^{(2)})^2] = O(h)$ by (A.6). Second, $\mathbf{j}_t$ is $\mathcal{F}_{t-1}$ -measurable, so that $\{\bar{\phi}_t^{(2)} \mathbf{j}_t \mathbf{j}_t', \mathcal{F}_t\}$ is also an MDA. Third,

$$\text{var} \left(\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{j}_t \mathbf{j}_t' \right) = \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(2)})^2 \mathbf{j}_t \mathbf{j}_t' \otimes \mathbf{j}_t \mathbf{j}_t'] = O(h) n \mathbb{E}[\|\mathbf{j}_t\|^4] = O(nh),$$

where $\mathbb{E}\|\mathbf{j}_t\|^4 < \infty$ by Assumption 1 (iv.d). Fourth, $\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{j}_t \mathbf{j}_t' = O_{\mathbb{P}}(\sqrt{nh})$ by applying Chebyshev's inequality, so that after dividing by nh^3 the first term is $O_{\mathbb{P}}(\sqrt{nh}/(nh^3)) = O_{\mathbb{P}}((nh^5)^{-1/2}) = o_{\mathbb{P}}(1)$ under $nh^5 \rightarrow \infty$.

We next examine the asymptotic behavior of $\sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t'$. In parallel to $\mathbf{T}_{jj}$,

$$\begin{aligned} \frac{1}{n^2 h^3} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' &= \frac{1}{n^2 h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' + \frac{1}{n^2 h^3} \sum_{t=1}^n \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \\ &= \frac{1}{n^2 h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' + \frac{f_u^{(2)}(0)}{n^2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \{1 + o_{\mathbb{P}}(1)\}. \end{aligned}$$

Here, the second term on the right side weakly converges to $f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$ by Lemma A.4 (iv). For the first term, $\{\bar{\phi}_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t', \mathcal{F}_t\}$ is an MDA because $\tilde{\mathbf{x}}_t$ is $\mathcal{F}_{t-1}$ -measurable, and $\mathbb{E}\|\tilde{\mathbf{x}}_t\|^4 = O(t^2)$ since $\tilde{\mathbf{x}}_t$ is a linear combination of $I(1)$ variables. Hence its variance is $O(h) \sum_{t=1}^n \mathbb{E}\|\tilde{\mathbf{x}}_t\|^4 = O(hn^3)$, so that the sum is $O_{\mathbb{P}}(n^{3/2} \sqrt{h})$ and, after dividing by $n^2 h^3$, it is $O_{\mathbb{P}}(n^{-1/2} h^{-5/2}) = O_{\mathbb{P}}((nh^5)^{-1/2}) = o_{\mathbb{P}}(1)$.

Finally, we examine the asymptotic behavior of $\sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}'_t$. As before, we first obtain that

$$\begin{aligned} \frac{1}{n^{3/2}h^3} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}'_t &= \frac{1}{n^{3/2}h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \mathbf{j}_t \tilde{\mathbf{x}}'_t + \frac{1}{n^{3/2}h^3} \sum_{t=1}^n \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}] \mathbf{j}_t \tilde{\mathbf{x}}'_t \\ &= \frac{1}{n^{3/2}h^3} \sum_{t=1}^n (\phi_t^{(2)} - \mathbb{E}[\phi_t^{(2)} | \mathcal{F}_{t-1}]) \mathbf{j}_t \tilde{\mathbf{x}}'_t + \frac{f_u^{(2)}(0)}{n^{3/2}} \sum_{t=1}^n \mathbf{j}_t \tilde{\mathbf{x}}'_t \{1 + o_{\mathbb{P}}(1)\}. \end{aligned}$$

Here, for the second term, $\tilde{\mathbf{x}}_t$ is orthogonal to $\hat{\mathbf{w}}_t := (1, \bar{\mathbf{w}}'_t)'$ by construction, so that it follows that

$$\frac{1}{n^{3/2}} \sum_{t=1}^n \mathbf{j}_t \tilde{\mathbf{x}}'_t = \frac{1}{n^{3/2}} \sum_{t=1}^n \begin{bmatrix} \hat{\mathbf{w}}_t \tilde{\mathbf{x}}'_t \\ \mathbf{k}_t \tilde{\mathbf{x}}'_t \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ n^{-3/2} \sum_{t=1}^n \mathbf{k}_t \tilde{\mathbf{x}}'_t \end{bmatrix} \rightsquigarrow \begin{bmatrix} \mathbf{0} \\ \mathbb{E}[\mathbf{k}_t] \int \tilde{\mathbf{B}}'_w \end{bmatrix}$$

by applying the proof of Lemma A.2 (v). Furthermore, we note that $\int \tilde{\mathbf{B}}_w \equiv \mathbf{0}$ by construction, implying that the second term on the right converges to zero in probability. For the first term, the same argument applies with $\mathbf{j}_t \tilde{\mathbf{x}}'_t$ in place of $\tilde{\mathbf{x}}_t \tilde{\mathbf{x}}'_t$: its variance is $O(h) \sum_{t=1}^n \mathbb{E}[\|\mathbf{j}_t\|^2 \|\tilde{\mathbf{x}}_t\|^2] = O(hn^2)$ by the Cauchy–Schwarz inequality together with Assumption 1 (iv.d), so that the sum is $O_{\mathbb{P}}(n\sqrt{h})$ and, after dividing by $n^{3/2}h^3$, it is $O_{\mathbb{P}}((nh^5)^{-1/2}) = o_{\mathbb{P}}(1)$.

(ii) If we solve for $\mathbf{b}$ from (A.8) by the Schur complement, it follows that

$$\begin{aligned} \mathbf{b} &= h(\mathbf{T}_{xx} - \mathbf{T}_{xj} \mathbf{T}_{jj}^{-1} \mathbf{T}_{jx})^{-1} (\mathbf{s}_x - \mathbf{P}_n \mathbf{s}_j) + o_{\mathbb{P}}((n\sqrt{h^3})^{-1}) \\ &= h\left(\sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}'_t\right)^{-1} \sum_{t=1}^n \phi_t^{(1)} \ddot{\mathbf{x}}_t + o_{\mathbb{P}}((n\sqrt{h^3})^{-1}), \end{aligned} \quad (\text{A.9})$$

where $\mathbf{P}_n := (\sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t \mathbf{j}'_t)(\sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{j}'_t)^{-1}$ and $\ddot{\mathbf{x}}_t := \mathbf{x}_t - \mathbf{P}_n \mathbf{j}_t$. Here, the second equality is obtained by noting that $\mathbf{s}_x - \mathbf{P}_n \mathbf{s}_G = \sum_{t=1}^n \phi_t^{(1)} (\mathbf{x}_t - \mathbf{P}_n \mathbf{j}_t) = \sum_{t=1}^n \phi_t^{(1)} \ddot{\mathbf{x}}_t$ and $\mathbf{T}_{xx} - \mathbf{T}_{xj} \mathbf{T}_{jj}^{-1} \mathbf{T}_{jx} = \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}'_t = \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}'_t$ because

$$\begin{aligned} \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \ddot{\mathbf{x}}'_t &= \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t (\mathbf{x}_t - \mathbf{P}_n \mathbf{j}_t)' = \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \mathbf{x}'_t - \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \mathbf{j}'_t \mathbf{P}'_n \quad \text{and} \\ \sum_{t=1}^n \phi_t^{(2)} \ddot{\mathbf{x}}_t \mathbf{j}'_t \mathbf{P}'_n &= \sum_{t=1}^n \phi_t^{(2)} (\mathbf{x}_t - \mathbf{P}_n \mathbf{j}_t) \mathbf{j}'_t \mathbf{P}'_n = \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t \mathbf{j}'_t \mathbf{P}'_n - \mathbf{P}_n \left(\sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{j}'_t \right) \mathbf{P}'_n = \mathbf{0}. \end{aligned}$$

Given this, if we replace $\ddot{\mathbf{x}}_t$ with $\tilde{\mathbf{x}}_t$ at the cost of $o_{\mathbb{P}}(1)$ by Lemma A.7 (i), it follows that

$$n\sqrt{h^3} \mathbf{b} = \left[\frac{1}{n^2 h^3} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}'_t \right]^{-1} \left[\frac{1}{n\sqrt{h}} \sum_{t=1}^n \phi_t^{(1)} \tilde{\mathbf{x}}_t \right] + o_{\mathbb{P}}(1).$$

We now note that $(n^2 h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \rightsquigarrow f_u^{(2)}(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w'$ by (i), and $(n^2 h)^{-1/2} \sum_{t=1}^n \phi_t^{(1)} \tilde{\mathbf{x}}_t \rightsquigarrow \int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi$ by Lemma A.4 (v) such that $\int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi \sim N(\mathbf{0}, v^2 f_u(0) \int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w')$ conditional on $\tilde{\mathbf{B}}_w(\cdot)$. Therefore,

$$n\sqrt{h^3} \mathbf{b} \rightsquigarrow \frac{1}{f_u^{(2)}(0)} \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1} \int \tilde{\mathbf{B}}_w d\mathcal{B}_\phi \sim N\left(\mathbf{0}, \omega_* \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1}\right)$$

conditional on $\tilde{\mathbf{B}}_w(\cdot)$, by noting that $\omega_* := v^2 f_u(0) / (f_u^{(2)}(0))^2$.

(iii) From (A.8), we note that $\mathbf{a} = h \mathbf{T}_{jj}^{-1} \mathbf{s}_j - \mathbf{T}_{jj}^{-1} \mathbf{T}_{jx} \mathbf{b} + o_{\mathbb{P}}((nh^3)^{-1/2})$. Given this, we already verified that $\mathbf{b} = O_{\mathbb{P}}((n^2 h^3)^{-1/2})$ in (ii), and $\mathbf{T}_{jj} = O_{\mathbb{P}}(nh^3)$ in (i). Furthermore,

$$\mathbf{T}_{jx} = \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \mathbf{x}_t' = \begin{bmatrix} \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t' \\ \sum_{t=1}^n \phi_t^{(2)} \bar{\mathbf{w}}_t \mathbf{x}_t' \\ \sum_{t=1}^n \phi_t^{(2)} \mathbf{k}_t \mathbf{x}_t' \end{bmatrix} = \begin{bmatrix} O_{\mathbb{P}}(n^{3/2} h^3) \\ O_{\mathbb{P}}(n\sqrt{h}) \\ O_{\mathbb{P}}(n\sqrt{h}) \end{bmatrix},$$

as Lemmas A.7 (ii)–(iv) demonstrate below. Collecting the three blocks, $\mathbf{T}_{jx} = O_{\mathbb{P}}(n^{3/2} h^3) + O_{\mathbb{P}}(n\sqrt{h})$, and since $\mathbf{T}_{jj}^{-1} = O_{\mathbb{P}}((nh^3)^{-1})$ by the first part of (i), we obtain $\mathbf{T}_{jj}^{-1} \mathbf{T}_{jx} = O_{\mathbb{P}}(n^{1/2}) + O_{\mathbb{P}}(n^{-1/2} h^{-5/2})$. Multiplying by $\mathbf{b} = O_{\mathbb{P}}((n^2 h^3)^{-1/2})$ gives $\mathbf{T}_{jj}^{-1} \mathbf{T}_{jx} \mathbf{b} = O_{\mathbb{P}}((nh^3)^{-1/2} \cdot n^{-1/2} \{1 + (nh^5)^{-1/2}\}) = o_{\mathbb{P}}((nh^3)^{-1/2})$ under $nh^5 \rightarrow \infty$, which is the stated equality. Therefore,

$$\mathbf{a} = \mathbf{T}_{jj}^{-1} h \mathbf{s}_j + o_{\mathbb{P}}((nh^3)^{-1/2}). \quad (\text{A.10})$$

We next examine the asymptotic behavior of $\mathbf{s}_j$. We first decompose $\phi_t^{(1)}$ into $\bar{\phi}_t^{(1)} + m_h$ such that $m_h = -\frac{h^4}{2} f_u^{(3)}(0) + o(h^4)$; see Section A.1. From this, we obtain that

$$\mathbf{T}_{jj}^{-1} h \mathbf{s}_j = \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \phi_t^{(1)} \mathbf{j}_t = \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t + \mathbf{T}_{jj}^{-1} h m_h \sum_{t=1}^n \mathbf{j}_t.$$

We examine each term on the right side. First, we focus on the second term on the right side:

$$\mathbf{T}_{jj}^{-1} h m_h \sum_{t=1}^n \mathbf{j}_t = \frac{m_h}{h^2 f_u^{(2)}(0)} \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1} \mathbb{E}[\mathbf{j}_t] \{1 + o_{\mathbb{P}}(1)\}$$

by noting that $(nh^3)^{-1} \mathbf{T}_{jj} = f_u^{(2)}(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t'] + o_{\mathbb{P}}(1)$ and $n^{-1} \sum_{t=1}^n \mathbf{j}_t = \mathbb{E}[\mathbf{j}_t] + o_{\mathbb{P}}(1)$ by (i) and the ergodic theorem, respectively. Given this, we further note that $\mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1} \mathbb{E}[\mathbf{j}_t] = \mathbf{e}_1$ by noting that the first-row element of $\mathbf{j}_1$ is unity. Hence, if we let $b_h := m_h / (h^2 f_u^{(2)}(0)) = -\frac{h^2}{2} f_u^{(3)}(0) / f_u^{(2)}(0)$, then it

follows that

$$\mathbf{T}_{jj}^{-1} h m_h \sum_{t=1}^n \mathbf{j}_t = b_h \mathbf{e}_1 \{1 + o_{\mathbb{P}}(1)\}.$$

Second, we focus on the first term on the right side. We note that

$$\sqrt{nh^3} \mathbf{T}_{jj}^{-1} h \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t = \left[\frac{1}{nh^3} \mathbf{T}_{jj} \right]^{-1} \left[\frac{1}{\sqrt{nh}} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t \right] \stackrel{\Delta}{\sim} \frac{1}{f_u^{(2)}(0)} \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1} N(\mathbf{0}, v^2 f_u(0) \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']).$$

Therefore, the limit distribution is normal with mean zero and asymptotic covariance $\Psi^+ := \omega_* \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1}$.

If we combine these two limits, then it follows that $\sqrt{nh^3}(\mathbf{a} - b_h \mathbf{e}_1) \stackrel{\Delta}{\sim} N(\mathbf{0}, \Psi^+)$.

(iv) The limits in (ii) and (iii) are driven by $(nh)^{-1/2} \sum_{t=1}^n \bar{\phi}_t^{(1)} \mathbf{j}_t$ and $(n\sqrt{h})^{-1} \sum_{t=1}^n \phi_t^{(1)} \tilde{\mathbf{x}}_t$, respectively. Their asymptotic covariance is governed by the limit of $n^{-3/2} \sum_{t=1}^n \mathbf{j}_t \tilde{\mathbf{x}}_t'$, and both statistics are the weighted sums of the same MDA $\{\bar{\phi}_t^{(1)}\}$ such that the first and second statistics have the weights $\mathbf{j}_t$ and $\tilde{\mathbf{x}}_t$, respectively. For such sums, their asymptotic covariance is the limit of the corresponding weighted cross moment, viz., of $(nh)^{-1/2} (n\sqrt{h})^{-1} \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(1)})^2 | \mathcal{F}_{t-1}] \mathbf{j}_t \tilde{\mathbf{x}}_t'$. Furthermore $\mathbb{E}[(\bar{\phi}_t^{(1)})^2 | \mathcal{F}_{t-1}] = h v^2 f_u(0) \{1 + o_{\mathbb{P}}(1)\}$ by (A.6), so that $(nh)^{-1/2} (n\sqrt{h})^{-1} \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(1)})^2 | \mathcal{F}_{t-1}] \mathbf{j}_t \tilde{\mathbf{x}}_t' = n^{-3/2} v^2 f_u(0) \sum_{t=1}^n \mathbf{j}_t \tilde{\mathbf{x}}_t' \{1 + o_{\mathbb{P}}(1)\}$. Hence a zero limit for that cross moment gives zero asymptotic covariance, so that they are asymptotically independent.

(v) By (iii), $\sqrt{nh^3}(\hat{\rho}_n - \rho_{\dagger} - b_h \mathbf{e}_1) \stackrel{\Delta}{\sim} N(\mathbf{0}, \Psi^+)$ with $\rho = (\nu, \mathbf{v}', \phi')'$. The vector in (v) is the sub-vector of $\hat{\rho}_n - \rho_{\dagger}$ obtained by deleting the first coordinate, viz., $\mathbf{A}(\hat{\rho}_n - \rho_{\dagger})$ with $\mathbf{A} := [\mathbf{0} : \mathbf{I}_{kq+p}]$. Since $\mathbf{A} \mathbf{e}_1 = \mathbf{0}$ the smoothing bias drops out, and the limit covariance is $\mathbf{A} \Psi^+ \mathbf{A}'$, which is Ψ by (17). This completes the proof. ■

Proof of Lemma A.7: [Part (i) is proved below. The earlier draft claimed that the weighted projection agrees with the unweighted one, which is false for the centered stationary block $\mathbf{s}_t = (\bar{\mathbf{w}}_t', \mathbf{h}_t')'$: there the martingale part dominates for both components, and the coefficient is $O_{\mathbb{P}}(h^{-5/2})$ rather than $O_{\mathbb{P}}(1)$. The coefficients are now obtained by a Schur complement, and the divergence is harmless because $h^{-5/2} = o(\sqrt{n})$ under $nh^5 \rightarrow \infty$.]

Throughout, we decompose the second-order kernel derivative into its systematic and martingale parts $\phi_t^{(2)} = m_h^{(2)} + \bar{\phi}_t^{(2)}$, where $m_h^{(2)} := \mathbb{E}[\phi^{(2)}(u_{th})] = h^3 f_u^{(2)}(0) \{1 + o(h)\}$ and $\bar{\phi}_t^{(2)} := \phi^{(2)}(u_{th}) - m_h^{(2)}$ by (A.5). By (A.6), $\text{var}(\bar{\phi}_t^{(2)}) = O(h)$, and $\{\bar{\phi}_t^{(2)}, \mathcal{F}_t\}$ is a martingale difference array by the same argument as in Remark 4 (iv), since $\{u_t\}$ is IID and independent of $\{\mathbf{w}_t\}$. We also note that $\mathbf{x}_t$, $\bar{\mathbf{w}}_t$ and $\mathbf{k}_t$ are all $\mathcal{F}_{t-1}$ -measurable.

(i) Since $\tilde{\mathbf{x}}_t$ already removes $(1, \bar{\mathbf{w}}_t)'$, the residual difference involves the *difference* of the weighted

and unweighted coefficients on 1 and $\bar{w}_t$, and the weighted coefficient itself on $\mathbf{h}_t$; this is what d_{0n} and $\mathbf{D}_n$ record. We show that all three are $O_{\mathbb{P}}(h^{-5/2})$.

Because this is a multivariate weighted regression, the intercept and the slopes are not simple weighted ratios: the weighted cross moments among the regressors do not vanish, so the coefficients must be obtained by a block inverse. Let $\mathbf{r}_t := (1, \mathbf{s}_t')'$,

$$\mathbf{T}_n := \begin{bmatrix} t_{00} & \mathbf{t}_{0s} \\ \mathbf{t}_{s0} & \mathbf{T}_{ss} \end{bmatrix} := \sum_{t=1}^n \phi_t^{(2)} \mathbf{r}_t \mathbf{r}_t', \quad \text{and} \quad \mathbf{N}_n := [\mathbf{n}_0 \quad \mathbf{N}_s] := \sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t \mathbf{r}_t'.$$

The blocks have the following orders. First, $t_{00} := \sum_{t=1}^n \phi_t^{(2)} = n m_h^{(2)} + O_{\mathbb{P}}(\sqrt{nh}) = O_{\mathbb{P}}(nh^3)$, since $\sqrt{nh}/(nh^3) = (nh^5)^{-1/2} = o(1)$. Second, $\mathbf{T}_{ss} = O_{\mathbb{P}}(nh^3)$, systematic-dominated by the first part of the proof of Lemma A.6 (i). Third, because $\mathbf{s}_t$ is centered, $\sum_{t=1}^n \mathbf{s}_t = O_{\mathbb{P}}(\sqrt{n})$, so that the systematic part of $\mathbf{t}_{0s}$ is $O_{\mathbb{P}}(h^3 \sqrt{n})$ while its martingale part is $O_{\mathbb{P}}(\sqrt{nh})$; as $\sqrt{nh}/(h^3 \sqrt{n}) = h^{-5/2} \rightarrow \infty$, the martingale part dominates and $\mathbf{t}_{0s} = O_{\mathbb{P}}(\sqrt{nh})$. Fourth, $\mathbf{n}_0 = m_h^{(2)} \sum_{t=1}^n \mathbf{x}_t + O_{\mathbb{P}}(n\sqrt{h}) = O_{\mathbb{P}}(h^3 n^{3/2})$ by (ii), the systematic part dominating because $\sum_{t=1}^n \mathbf{x}_t = O_{\mathbb{P}}(n^{3/2})$. Fifth, $\mathbf{N}_s = O_{\mathbb{P}}(n\sqrt{h})$ by (iii) and (iv), the martingale part dominating for both the $\bar{w}_t$ and the $\mathbf{h}_t$ components, again because $\mathbf{s}_t$ is centered.

By the block inverse, the weighted coefficient on $\mathbf{s}_t$ is the Schur-complement expression

$$\hat{\mathbf{C}}_n = (\mathbf{N}_s - \mathbf{n}_0 t_{00}^{-1} \mathbf{t}_{0s}) (\mathbf{T}_{ss} - \mathbf{t}_{s0} t_{00}^{-1} \mathbf{t}_{0s})^{-1}.$$

Here $\mathbf{n}_0 t_{00}^{-1} \mathbf{t}_{0s} = O_{\mathbb{P}}(h^3 n^{3/2}) O_{\mathbb{P}}((nh^3)^{-1}) O_{\mathbb{P}}(\sqrt{nh}) = O_{\mathbb{P}}(n\sqrt{h})$, which is of the same order as $\mathbf{N}_s$, so that the numerator is $O_{\mathbb{P}}(n\sqrt{h})$; and $\mathbf{t}_{s0} t_{00}^{-1} \mathbf{t}_{0s} = O_{\mathbb{P}}(nh) O_{\mathbb{P}}((nh^3)^{-1}) = O_{\mathbb{P}}(h^{-2})$, which is negligible relative to $\mathbf{T}_{ss}$ because $h^{-2}/(nh^3) = (nh^5)^{-1} = o(1)$, so that the denominator is $O_{\mathbb{P}}(nh^3)$. Therefore

$$\hat{\mathbf{C}}_n = O_{\mathbb{P}}(n\sqrt{h}) O_{\mathbb{P}}(nh^3)^{-1} = O_{\mathbb{P}}(h^{-5/2}).$$

The corresponding intercept is $\hat{\mathbf{a}}_n = (\mathbf{n}_0 - \hat{\mathbf{C}}_n \mathbf{t}_{0s}) t_{00}^{-1}$. Its leading part is $\mathbf{n}_0 t_{00}^{-1}$, and the correction satisfies $\hat{\mathbf{C}}_n \mathbf{t}_{0s} t_{00}^{-1} = O_{\mathbb{P}}(h^{-5/2}) O_{\mathbb{P}}(\sqrt{nh}) O_{\mathbb{P}}((nh^3)^{-1}) = O_{\mathbb{P}}(n^{-1/2} h^{-5}) = o_{\mathbb{P}}(h^{-5/2})$, the last step because $n^{-1/2} h^{-5/2} = (nh^5)^{-1/2} = o(1)$. For the leading part, writing it as a ratio,

$$\mathbf{n}_0 t_{00}^{-1} = \frac{m_h^{(2)} \sum_{t=1}^n \mathbf{x}_t + O_{\mathbb{P}}(n\sqrt{h})}{n m_h^{(2)} \{1 + O_{\mathbb{P}}((nh^5)^{-1/2})\}},$$

the numerator perturbation contributes $O_{\mathbb{P}}(n\sqrt{h})/(n m_h^{(2)}) = O_{\mathbb{P}}(h^{-5/2})$ and the denominator perturbation

contributes $n^{-1} \sum_{t=1}^n \mathbf{x}_t \cdot O_{\mathbb{P}}((nh^5)^{-1/2}) = O_{\mathbb{P}}(\sqrt{n}) O_{\mathbb{P}}((nh^5)^{-1/2}) = O_{\mathbb{P}}(h^{-5/2})$. For the unweighted projection the corresponding sample identity is $\bar{\mathbf{a}}_n = \bar{\mathbf{x}}_n - \bar{\mathbf{B}}_n \bar{\mathbf{w}}_n$ with $\bar{\mathbf{x}}_n := n^{-1} \sum_{t=1}^n \mathbf{x}_t$ and $\bar{\mathbf{w}}_n := n^{-1} \sum_{t=1}^n \bar{\mathbf{w}}_t$, because $\bar{\mathbf{w}}_t$ is centered in population but not in sample. Since $\bar{\mathbf{B}}_n = O_{\mathbb{P}}(1)$ and $\bar{\mathbf{w}}_n = O_{\mathbb{P}}(n^{-1/2})$, we have $\bar{\mathbf{a}}_n = \bar{\mathbf{x}}_n + O_{\mathbb{P}}(n^{-1/2})$, and this extra term is negligible relative to $O_{\mathbb{P}}(h^{-5/2})$. Collecting the three comparisons therefore gives $|d_{0n}| + \|\mathbf{D}_n\| = O_{\mathbb{P}}(h^{-5/2})$, where the $\mathbf{h}_t$ component of $\mathbf{D}_n$ is $\hat{\mathbf{D}}_n$ itself and the $\bar{\mathbf{w}}_t$ component is a difference of coefficients of the same order. This quantity diverges but satisfies $h^{-5/2} = o(\sqrt{n})$, an equivalent form of $nh^5 \rightarrow \infty$.

We now verify the two displays. Expanding the weighted quadratic form with $\ddot{\mathbf{x}}_t - \tilde{\mathbf{x}}_t = -\mathbf{d}_{0n} - \mathbf{D}_n \mathbf{s}_t$ produces cross and quadratic terms. Using $\sum_{t=1}^n \phi_t^{(2)} \mathbf{s}_t \tilde{\mathbf{x}}'_t = O_{\mathbb{P}}(n\sqrt{h})$ by (iii) and (iv), $\sum_{t=1}^n \phi_t^{(2)} \mathbf{s}_t \mathbf{s}'_t = O_{\mathbb{P}}(h^3 n)$, $\sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}'_t = O_{\mathbb{P}}(n\sqrt{h})$ and $\sum_{t=1}^n \phi_t^{(2)} = O_{\mathbb{P}}(h^3 n)$,

$$\frac{\mathbf{D}_n}{n^2 h^3} \sum_{t=1}^n \phi_t^{(2)} \mathbf{s}_t \tilde{\mathbf{x}}'_t = O_{\mathbb{P}}\left(\frac{h^{-5/2} n \sqrt{h}}{n^2 h^3}\right) = O_{\mathbb{P}}((nh^5)^{-1}),$$

$$\frac{\mathbf{D}_n}{n^2 h^3} \left(\sum_{t=1}^n \phi_t^{(2)} \mathbf{s}_t \mathbf{s}'_t \right) \mathbf{D}'_n = O_{\mathbb{P}}\left(\frac{h^{-5} h^3 n}{n^2 h^3}\right) = O_{\mathbb{P}}((nh^5)^{-1}),$$

and the intercept terms $\mathbf{d}_{0n}(n^2 h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \tilde{\mathbf{x}}'_t$ and $d_{0n}^2 (n^2 h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)}$ are of the same two orders. All are $o_{\mathbb{P}}(1)$ under $nh^5 \rightarrow \infty$, which gives the first display. For the score, $\sum_{t=1}^n \phi_t^{(1)} \mathbf{s}_t = O_{\mathbb{P}}(\sqrt{n}h)$ by Lemma A.4 (ii), so that $\mathbf{D}_n(n\sqrt{h})^{-1} \sum_{t=1}^n \phi_t^{(1)} \mathbf{s}_t = O_{\mathbb{P}}(n^{-1/2} h^{-5/2}) = O_{\mathbb{P}}((nh^5)^{-1/2})$. The intercept contribution splits into a centered part of the same order and a systematic part

$$\frac{\mathbf{d}_{0n}}{n\sqrt{h}} n m_h = O_{\mathbb{P}}(h^{-5/2}) O(h^{7/2}) = O_{\mathbb{P}}(h) = o_{\mathbb{P}}(1),$$

using $m_h = O(h^4)$ from (A.2). This gives the second display and completes the proof of (i). [JSCHO: This proof is too lengthy. Please focus on the proof of (i.a) and (i.b) using the definition of $\ddot{\mathbf{x}}_t$, here.]

(ii) We split the sum as follows:

$$\sum_{t=1}^n \phi_t^{(2)} \mathbf{x}'_t = m_h^{(2)} \sum_{t=1}^n \mathbf{x}'_t + \sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{x}'_t.$$

For the first term, $n^{-3/2} \sum_{t=1}^n \mathbf{x}_t \rightsquigarrow \int \bar{\mathbf{B}}_w(r) dr$ by Lemma A.3 (i), so that it equals $h^3 f_u^{(2)}(0) n^{3/2} \{ \int \bar{\mathbf{B}}'_w + o_{\mathbb{P}}(1) \}$. For the second term, the summands form an MDA because $\mathbf{x}_t$ is $\mathcal{F}_{t-1}$ -measurable, so that

$$\text{var} \left(\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{x}'_t \right) = \sum_{t=1}^n \mathbb{E}[(\bar{\phi}_t^{(2)})^2 \mathbf{x}_t \mathbf{x}'_t] = O(h) \sum_{t=1}^n \mathbb{E}[\mathbf{x}_t \mathbf{x}'_t] = O(h n^2),$$

because $\mathbb{E}\|\mathbf{x}_t\|^2 = O(t)$, so that $\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{x}_t' = O_{\mathbb{P}}(n\sqrt{h})$. Comparing the two terms, the ratio of the second to the first is $O(n\sqrt{h}/(h^3 n^{3/2})) = O((nh^5)^{-1/2}) = o(1)$ under $nh^5 \rightarrow \infty$, so that the first term dominates and (ii) follows after dividing $\sum_{t=1}^n \phi_t^{(2)} \mathbf{x}_t'$ by $n^{3/2}h^3$.

(iii) and (iv) Both statistics are handled in the same way as (ii). Letting $\mathbf{v}_t$ be either $\bar{\mathbf{w}}_t$ or $\mathbf{k}_t$, $\mathbb{E}\|\mathbf{v}_t\|^2 < \infty$ by Assumption 1 (iv.a) and (v). Then, the systematic part is $m_h^{(2)} \sum_{t=1}^n \mathbf{v}_t \mathbf{x}_t' = O_{\mathbb{P}}(h^3 n)$ by Lemma A.2 (ii) and (v), and the MDA part satisfies

$$\text{var}\left(\sum_{t=1}^n \bar{\phi}_t^{(2)} \mathbf{v}_t \mathbf{x}_t'\right) = O(h) \sum_{t=1}^n \mathbb{E}[\|\mathbf{v}_t\|^2 \|\mathbf{x}_t\|^2] = O(hn^2),$$

where the last equality holds by Assumption 1 (iv.d), which bounds $\mathbb{E}\|(\bar{\mathbf{w}}_t', \mathbf{k}_t')'\|^4$ by using the Cauchy-Schwarz inequality and $\mathbb{E}\|\mathbf{x}_t\|^4 = O(t^2)$. Therefore, the MDA part is $O_{\mathbb{P}}(n\sqrt{h})$, so that $\sum_{t=1}^n \phi_t^{(2)} \mathbf{v}_t \mathbf{x}_t' = O_{\mathbb{P}}(n\sqrt{h})$, because $h^3 n = o(n\sqrt{h})$.

(v) We first note that $\hat{\alpha}_n - \alpha_* = (\hat{\nu}_n - \alpha_*) - \mu_*' \mathbf{l}_p' (\hat{\phi}_n - \phi_*) = \mathbf{s}'_{\alpha} (\hat{\rho}_n - \rho_*)$ with $\mathbf{s}_{\alpha} = (1, \mathbf{0}'_{kq}, -\mu_*' \mathbf{l}_p')'$. Therefore, its asymptotic distribution trivially follows from Lemma A.6 (iii) such that its asymptotic variance is $\mathbf{s}'_{\alpha} \Psi^+ \mathbf{s}_{\alpha}$ and its bias is $\mathbf{s}'_{\alpha} (b_h \mathbf{e}_1) = b_h$.

(vi) Because the rows of $\mathbf{P}_{\widehat{\mathbf{W}}} \mathbf{X}$ are orthogonal to $\hat{\mathbf{w}}_t = (1, \bar{\mathbf{w}}_t')'$, $\sum_{t=1}^n \tilde{\mathbf{x}}_t = \mathbf{0}$ and $\sum_{t=1}^n \bar{\mathbf{w}}_t \tilde{\mathbf{x}}_t' = \mathbf{0}$. Therefore,

$$\sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}_t' = \sum_{t=1}^n \begin{bmatrix} \phi_t^{(2)} \tilde{\mathbf{x}}_t' \\ \phi_t^{(2)} \bar{\mathbf{w}}_t \tilde{\mathbf{x}}_t' \\ \phi_t^{(2)} \mathbf{k}_t \tilde{\mathbf{x}}_t' \end{bmatrix} = \sum_{t=1}^n \begin{bmatrix} \bar{\phi}_t^{(2)} \tilde{\mathbf{x}}_t' \\ \bar{\phi}_t^{(2)} \bar{\mathbf{w}}_t \tilde{\mathbf{x}}_t' \\ m_h^{(2)} \mathbf{k}_t \tilde{\mathbf{x}}_t' + \bar{\phi}_t^{(2)} \mathbf{k}_t \tilde{\mathbf{x}}_t' \end{bmatrix} = \begin{bmatrix} O_{\mathbb{P}}(n\sqrt{h}) \\ O_{\mathbb{P}}(n\sqrt{h}) \\ O_{\mathbb{P}}(nh^3) + O_{\mathbb{P}}(n\sqrt{h}) \end{bmatrix}$$

by applying the proofs of (ii, iii, and iv). Therefore, $(n^{3/2}h^3)^{-1} \sum_{t=1}^n \phi_t^{(2)} \mathbf{j}_t \tilde{\mathbf{x}}_t' = o_{\mathbb{P}}(1)$. This completes the proof. ■

A.3 Proofs

In this section, we prove the main claims of this study.

Proof of Theorem 1: The autoregressive parameter ϕ_* is identical to the coefficient of $\mathbf{k}_t$ in the reparameterized model, viz., ϕ_* , and the estimators for them is identical, which is given as $\hat{\phi}_n$ in $\hat{\rho}_n$. Lemma A.6 (iii) already provides the asymptotic distribution of $\hat{\phi}_n$, viz., $\sqrt{nh^3}(\hat{\phi}_n - \phi_*) \overset{A}{\rightsquigarrow} N(\mathbf{0}, \Psi_{\phi\phi}^+)$, where $\Psi_{\phi\phi}^+$ denotes the covariance matrix corresponding to ϕ_* in Ψ^+ , which is formed by the last p rows and p columns of $\Psi^+ := \omega_* \mathbb{E}[\mathbf{j}_t \mathbf{j}_t']^{-1}$. We further note that $\Psi_{\phi\phi}^+ = \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1}$ by the definition of $\mathbf{h}_t$, so that

$\sqrt{nh^3}(\hat{\phi}_n - \phi_*) \stackrel{A}{\sim} N(\mathbf{0}, \mathbf{\Pi})$ using the definition of $\mathbf{\Pi} := \omega_* \mathbb{E}[\mathbf{h}_t \mathbf{h}_t']^{-1}$. This completes the proof. $\blacksquare$

Proof of Theorem 2: (i) By the reparameterization given in (14), we note that $(\hat{\gamma}_n - \gamma_*) + \beta_* \iota'_p(\hat{\phi}_n - \phi_*) = \hat{\kappa}_n - \beta_*$, and Lemma A.6 (ii) already provides the desired weak limit.

(ii) If we rearrange $(\hat{\gamma}_n - \gamma_*) + \beta_* \iota'_p(\hat{\phi}_n - \phi_*) = \hat{\kappa}_n - \beta_*$,

$$\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*) = \sqrt{nh^3}(\hat{\kappa}_n - \beta_*) - \beta_* \iota'_p \sqrt{nh^3}(\hat{\phi}_n - \phi_*) = -\beta_* \iota'_p \sqrt{nh^3}(\hat{\phi}_n - \phi_*) + o_{\mathbb{P}}(1),$$

where the second equality follows from the fact that $(\hat{\kappa}_n - \beta_*) = O_{\mathbb{P}}((n^2 h^3)^{-1/2})$. Furthermore, Theorem 1 already shows that $\sqrt{nh^3}(\hat{\phi}_n - \phi_*) \stackrel{A}{\sim} N(\mathbf{0}, \mathbf{\Pi})$, so that $\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*) \stackrel{A}{\sim} N(\mathbf{0}, \sigma_{\phi}^2 \beta_* \beta_*')$ with $\sigma_{\phi}^2 = \iota'_p \mathbf{\Pi} \iota_p$ as desired. $\blacksquare$

Proof of Theorem 3: From (22) and $\sum_j (\hat{\phi}_{j,n} - \phi_{j*}) = \iota'_p(\hat{\phi}_n - \phi_*)$, it follows that

$$\hat{\beta}_n - \beta_* = \left[1 - \sum_j \hat{\phi}_{j,n}\right]^{-1} \left[(\hat{\gamma}_n - \gamma_*) + \beta_* \iota'_p(\hat{\phi}_n - \phi_*)\right] = \left[1 - \sum_j \hat{\phi}_{j,n}\right]^{-1} (\hat{\kappa}_n - \beta_*).$$

Given this, $\sum_j \hat{\phi}_{j,n} \xrightarrow{\mathbb{P}} \sum_j \phi_{j*}$ and the continuous mapping theorem applied to Theorem 2 (i) lead to the desired limit distribution. $\blacksquare$

Proof of Theorem 4: Since $\tilde{\varphi}_n := -\mathbf{R}_1 \hat{\phi}_n$, $\sqrt{nh^3}(\tilde{\varphi}_n - \varphi_*) = -\mathbf{R}_1 \sqrt{nh^3}(\hat{\phi}_n - \phi_*) \stackrel{A}{\sim} N(\mathbf{0}, \mathbf{R}_1 \mathbf{\Pi} \mathbf{R}_1')$ by Theorem 1. This completes the proof. $\blacksquare$

Proof of Theorem 5: (i) By the reparameterization, we obtain that

$$\hat{\delta}_n - \delta_* = (\hat{v}_n - v_{\dagger}) - \Xi_*(\hat{\phi}_n - \phi_*) = [\mathbf{I}_{kq} \vdots -\Xi_*]((\hat{v}_n - v_{\dagger})', (\hat{\phi}_n - \phi_*)')'.$$

By Lemma A.6 (v), $\sqrt{nh^3}((\hat{v}_n - v_{\dagger})', (\hat{\phi}_n - \phi_*)')' \stackrel{A}{\sim} N(\mathbf{0}, \mathbf{\Psi})$. Therefore, it follows that $\sqrt{nh^3}(\hat{\delta}_n - \delta_*) \stackrel{A}{\sim} N(\mathbf{0}, \mathbf{\Omega}_{\delta})$ with $\mathbf{\Omega}_{\delta} = [\mathbf{I}_{kq} \vdots -\Xi_*] \mathbf{\Psi} [\mathbf{I}_{kq} \vdots -\Xi_*]'$.

(ii) Using the fact that $\tilde{\psi}_n = \hat{\delta}_n + \mathbf{S} \hat{\gamma}_n$, $\psi_* = \delta_* + \mathbf{S} \gamma_*$, and (21), we obtain that

$$\sqrt{nh^3}(\tilde{\psi}_n - \psi_*) = [\mathbf{I}_{kq} \vdots -\mathbf{C}] \sqrt{nh^3} \begin{pmatrix} \hat{v}_n - v_{\dagger} \\ \hat{\phi}_n - \phi_* \end{pmatrix} + O_{\mathbb{P}}(n^{-1/2}).$$

Here, the remainder terms is $O_{\mathbb{P}}(n^{-1/2})$, because $\mathbf{S}(\hat{\kappa}_n - \beta_*) = O_{\mathbb{P}}((n \sqrt{h^3})^{-1})$. Therefore, $\sqrt{nh^3}(\tilde{\psi}_n - \psi_*) \stackrel{A}{\sim} N(\mathbf{0}, \mathbf{\Omega}_{\theta})$ by Lemma A.6 (v). $\blacksquare$

Proof of Corollary 1. (i) By Theorem 3 (ii), we obtain that

$$\left(1 - \sum_{j=1}^p \phi_{j*}\right) n\sqrt{h^3} (\hat{\beta}_n - \beta_*) \stackrel{A}{\sim} N\left(\mathbf{0}, \frac{v^2 f_u(0)}{[f_u^{(2)}(0)]^2} \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1}\right)$$

conditional on $\bar{\mathbf{B}}_w(\cdot)$. The weight matrix of $\mathcal{W}_n(\beta)$ weakly converges to the conditional covariance by Lemma A.4 (iv):

$$\hat{\omega}_n \left(\frac{1}{n^2} \sum_{t=1}^n \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t' \right)^{-1} \rightsquigarrow \omega_* \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1},$$

because $\hat{f}_u(0) \xrightarrow{\mathbb{P}} f_u(0)$ and $\hat{f}_u^{(2)}(0) \xrightarrow{\mathbb{P}} f_u^{(2)}(0)$. Furthermore, $\sum_j \hat{\phi}_{j,n} \xrightarrow{\mathbb{P}} \sum_j \phi_{j*}$ by Theorem 1 (i). Therefore, $\mathcal{W}_n(\beta)$ asymptotically follows $\mathcal{X}_r^2$ conditional on $\mathbf{B}_w(\cdot)$. Furthermore, the conditional weak limit does not depend on $\bar{\mathbf{B}}_w(\cdot)$, so that it asymptotically follows $\mathcal{X}_r^2$ unconditionally.

(ii) Under $\mathbb{H}_1^{(1)}$, $\mathbf{Q}\beta_* - \mathbf{q} \neq \mathbf{0}$ and $\mathcal{W}_n(\beta)$ diverges at the rate $n^2 h^3$, so that $\mathbb{P}(\mathcal{W}_n(\beta) \geq s_n^{(1)}) \xrightarrow{\mathbb{P}} 1$ for any $s_n^{(1)} = o(n^2 h^3)$. ■

Proof of Corollary 3. (i) We prove both (i.a) and (i.b). If $\mathbf{Q}\beta_* = \mathbf{0}$, then (25) reduces to $\mathbf{Q}(\hat{\gamma}_n - \gamma_*) = \mathbf{Q}(\hat{\kappa}_n - \beta_*)$, so that we obtain the following by Theorem 2(i),

$$n\sqrt{h^3} \mathbf{Q}(\hat{\gamma}_n - \gamma_*) \stackrel{A}{\sim} N\left(\mathbf{0}, \mathbf{Q} \frac{v^2 f_u(0)}{[f_u^{(2)}(0)]^2} \left[\int \tilde{\mathbf{B}}_w \tilde{\mathbf{B}}_w' \right]^{-1} \mathbf{Q}'\right)$$

conditional on $\bar{\mathbf{B}}_w(\cdot)$. The weight matrix of $\mathcal{W}_n^{(0)}(\gamma)$ converges to the conditional covariance by Lemma A.4 (iv) together with the consistency of $\hat{\omega}_n$ for ω_* . Therefore, $\mathcal{W}_n^{(0)}(\gamma) \stackrel{A}{\sim} \mathcal{X}_r^2$ under $\mathbb{H}_1^{(3)}$. Under $\mathbb{H}_1^{(3)}$, $\mathbf{Q}\gamma_* - \mathbf{q} \neq \mathbf{0}$ and the Wald test diverges at the rate of $n^2 h^3$, so that $\mathbb{P}(\mathcal{W}_n^{(0)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ for any $s_n = o(n^2 h^3)$.

(ii) Without loss of generality, we let $\mathbf{Q} = \mathbf{q}'_1$. If $\mathbf{Q} = \mathbf{q}'_1$ such that $\mathbf{q}'_1 \beta_* \neq 0$, then

$$\sqrt{nh^3} \mathbf{q}'_1 (\hat{\gamma}_n - \gamma_*) = \sqrt{nh^3} \mathbf{q}'_1 (\hat{\kappa}_n - \beta_*) - (\mathbf{q}'_1 \beta_*) \boldsymbol{\iota}'_p \sqrt{nh^3} (\hat{\phi}_n - \phi_*)$$

by (25). The first term on the right side is $n^{-1/2} \cdot n\sqrt{h^3} \mathbf{q}'_1 (\hat{\kappa}_n - \beta_*) = O_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1)$ by Theorem 2(i) and $\boldsymbol{\iota}'_p \sqrt{nh^3} (\hat{\phi}_n - \phi_*) \stackrel{A}{\sim} N(0, \sigma_\phi^2)$ by Theorem 1. Therefore, it follows that

$$\sqrt{nh^3} \mathbf{q}'_1 (\hat{\gamma}_n - \gamma_*) \stackrel{A}{\sim} N(0, \sigma_\phi^2 (\mathbf{q}'_1 \beta_*)^2).$$

Since $\hat{\sigma}_\phi^2 \xrightarrow{\mathbb{P}} \sigma_\phi^2$ and $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_*$, Slutsky's theorem delivers $\mathcal{W}_n^{(1)}(\gamma) \stackrel{A}{\sim} \mathcal{X}_1^2$ under $\mathbb{H}_0^{(3)} : \mathbf{q}'_1 \gamma_* = q_1$. Under

$\mathbb{H}_1^{(3)}$, $\mathbf{q}_1' \gamma_* - q_1 \neq 0$ and the Wald test diverges at the rate of nh^3 , so that $\mathbb{P}(\mathcal{W}_n^{(1)}(\gamma) \geq s_n) \xrightarrow{\mathbb{P}} 1$ for any $s_n = o(nh^3)$.

A.4 Supplementary Simulations

In this section we discuss supplementary simulation results.

A.4.1 Simulation Using MARDL Model

We assume the same DGPs and model condition as given in Section 3.1 and report the simulation results on the estimators using the MARDL model. First, Figure A.1 shows the empirical densities of the estimators obtained by simulation under DGPs 1, 2, and 3. Each estimator is standardized so that it follows the standard normal distribution:

$$\frac{\sqrt{nh^3}(\hat{\phi}_n - \phi_*)}{\hat{\Pi}_n^{1/2}}, \quad \frac{\sqrt{nh^3}(\hat{\gamma}_n - \gamma_*)}{\hat{\sigma}_{\phi_n} |\hat{\beta}_n|}, \quad \text{and} \quad \frac{n\sqrt{h^3}(\hat{\beta}_n - \beta_*)}{|1 - \hat{\phi}_n|^{-1} \hat{\omega}_n^{1/2} [n^{-2} \sum_{t=1}^n \tilde{x}_t^2]^{-1/2}}.$$

Figure A.1 shows the empirical densities of these three statistics for $n = 200, 400, 600$, and 800 , and we observe that they approach the standard normal density function as the sample size increases, corroborating Theorems 1, 2 (ii) and 3.

Second, Figure A.2 presents the QQ plots between the three statistics and standard normal random variables for $n = 200, 1000$, and 3000 . We also observe that the QQ plot approaches the 45-degree line as the sample size grows. This further corroborates the asymptotic theory on the parameter estimators given in Theorems 1, 2 (ii) and 3.

Finally, we conduct 1,000 independent experiments for different sample sizes $n = 200, 400, 600, 800, 1000, 2000$ and $3,000$ and report the sample average of the estimated coefficients and their MSEs in Table A.1. For all the three DGPs, all MSEs converge to zero, implying that the estimated parameters are consistent for the DGP parameters. Specifically, regardless of the error distribution, the modal estimators exhibit slight bias for small n , but significant improvements are attained as the sample size increases. This corroborates the consistency of the estimators. In particular, the MSEs of $\hat{\beta}_n$ is smaller than the other estimators, corroborating that the long-run parameter estimator is super-consistent.

A.4.2 Simulation Using MARDL-EC

We assume the same DGPs and model condition as given in Section 3.2 and report the simulation results on the estimators using the MARDL-EC representation. First, Figure A.3 shows the empirical density functions

of the standardized EC parameter estimators under DGPs 1, 2, and 3. The standardized EC parameter estimators are obtained by using the sample analogs of the asymptotic covariance matrices given in Theorems 4 and 5. As the sample size increases, the empirical density functions approach the standard normal density function for both $\tilde{\varphi}_n$ and $\tilde{\psi}_n$. This feature corroborates Theorems 4 and 5.

Second, Figure A.4 draws the QQ plots between the standardized statistics and standard normal random variable under DGPs 1, 2, and 3. As the sample size increases, the QQ plot approaches the 45-degree line. This also corroborates Theorems 4 and 5.

Finally, we examine the finite sample bias and MSE of the MARDL-EC parameter estimators in parallel to Table A.1. We report the simulation results in Table A.2. For all the cases of consideration, we observe consistent decreases in MSE as the sample size grows. This indicates that the estimators are consistent for the MARDL-EC parameters.

A.5 Supplementary Empirics

In this section we provide supplementary empirical information.

A.5.1 Empirical Data Source

We obtain global COVID-19 data for various countries from Our World in Data, primarily sourced from the World Health Organization (WHO).¹ This dataset provides confirmed cases, deaths, and recoveries of COVID-19 for countries (sometimes including states) since January 3, 2020. The confirmed cases and deaths data are collected from the World Health Organization’s Coronavirus Dashboard, and the dataset is updated daily.

We also obtain other variables related to COVID-19 from Our World in Data², such as location, population, GDP per capita (Gross domestic product at purchasing power parity, constant 2011 international dollars, most recent year available), population density (Number of people divided by land area, measured in square kilometers, most recent year available), median age (Median age of the population, UN projection for 2020), aged 65 and older (Share of the population that is 65 years and older, most recent year available), aged 70 and older (Share of the population that is 70 years and older in 2015), female smokers (Share of women who smoke, most recent year available), male smokers (Share of men who smoke, most recent year available), hospital beds per thousand (Hospital beds per 1,000 people, most recent year available since 2010), life

¹Data are available at <https://ourworldindata.org/covid-vaccinations>

²Data are available at <https://github.com/owid/covid-19-data/tree/master>

expectancy (Life expectancy at birth in 2019), and human development index (A composite index measuring average achievement in three basic dimensions of human development. Values for 2019).

Furthermore, the Oxford COVID-19 Government Response Tracker (OxCGRT) systematically gathers information regarding the timing and nature of pandemic response measures enacted by governments. Throughout the years 2020, 2021, and 2022, OxCGRT meticulously compiles data on diverse policy responses reflecting the breadth of government actions.³ This data are aggregated to form a series of policy indices with the Government Response Index assessing the government's level of intervention across several policy domains: C - Containment and closure policies, E - Economic policies, H - Health system policies, V - Vaccination policies. These indices gauge the extent to which governments have adopted various policy approaches and the degree to which they have been implemented. Higher values indicate stricter measures reflecting the comprehensiveness and stringency of government interventions across the specified policy domains. Notably, it is important to clarify that this index does not provide an evaluation of the effectiveness of the government's policies.

A.5.2 Descriptive Statistics

We provide the descriptive statistics related to COVID-19 of Denmark, Norway, Belgium, and Germany in Table A.3.

Denmark and Norway are two European countries that are comparable in terms of population size and economic strength. Denmark has a population of approximately 5,882,259, while Norway's population is 5,434,324. The per capita GDP of Denmark is around \$46,683, while Norway's per capita GDP is \$64,800. Both countries belong to the Scandinavian region and are situated adjacent to the Scandinavian Peninsula. They share relatively low population densities, with Denmark at 137/km² and Norway at just 14/km². Furthermore, these two nations also exhibit significant similarities in the variables related to COVID-19, such as the proportion of elderly population, smoking rates, and hospital bed capacity.

On the other hand, Belgium and Germany belong to a group of European countries with more concentrated populations and are geographically adjacent to each other. Belgium has a population density of 376/km² while Germany's population density is 237/km². Belgium's population is 11,655,923, while Germany's is 83,369,840. The economic development levels of these two countries also show similarities, with Belgium's per capita GDP at \$42,659 and Germany's at \$45,229. Similarly, Belgium and Germany demonstrate noticeable similarities in the variables related to COVID-19.

³Data are available at <https://www.bsg.ox.ac.uk/research/covid-19-government-response-tracker>

DGP	Para.	Stat. \ n	200	400	600	800	1,000	2,000	3,000
1	α_*	Est.	-0.0087	-0.0018	-0.0058	0.0062	0.0038	0.0009	-0.0012
		Std. Err.	(0.1305)	(0.0983)	(0.0935)	(0.0794)	(0.0744)	(0.0590)	(0.0548)
		MSE	0.0171	0.0097	0.0088	0.0063	0.0055	0.0035	0.0030
	β_*	Est.	2.0007	2.0034	2.0022	1.9987	2.0002	1.9999	1.9999
		Std. Err.	(0.0969)	(0.0496)	(0.0345)	(0.0296)	(0.0251)	(0.0137)	(0.0093)
		MSE	0.0094	0.0025	0.0012	0.0009	0.0006	0.0002	0.0001
	γ_*	Est.	1.0097	1.0026	1.0122	1.0110	1.0043	1.0045	0.9999
		Std. Err.	(0.2015)	(0.1480)	(0.1267)	(0.1204)	(0.1069)	(0.0855)	(0.0788)
		MSE	0.0406	0.0219	0.0162	0.0146	0.0114	0.0073	0.0062
	ϕ_*	Est.	0.4949	0.4993	0.4943	0.4941	0.4979	0.4977	0.5000
		Std. Err.	(0.1004)	(0.0744)	(0.0635)	(0.0604)	(0.0536)	(0.0429)	(0.0393)
		MSE	0.0101	0.0055	0.0041	0.0037	0.0029	0.0018	0.0015
2	α_*	Est.	0.0323	0.0275	0.0224	0.0206	0.0186	0.0155	0.0141
		Std. Err.	(0.0378)	(0.0301)	(0.0263)	(0.0229)	(0.0226)	(0.0170)	(0.0150)
		MSE	0.0025	0.0017	0.0012	0.0009	0.0009	0.0005	0.0004
	β_*	Est.	2.0000	2.0009	2.0006	1.9996	2.0002	2.0000	2.0000
		Std. Err.	(0.0252)	(0.0138)	(0.0103)	(0.0075)	(0.0065)	(0.0036)	(0.0024)
		MSE	0.0006	0.0002	0.0001	0.0001	0.0000	0.0000	0.0000
	γ_*	Est.	1.0022	1.0020	0.9998	1.0015	1.0003	1.0013	1.0010
		Std. Err.	(0.0706)	(0.0564)	(0.0464)	(0.0438)	(0.0400)	(0.0322)	(0.0290)
		MSE	0.0050	0.0032	0.0021	0.0019	0.0016	0.0010	0.0008
	ϕ_*	Est.	0.4988	0.4992	0.5002	0.4991	0.4999	0.4993	0.4995
		Std. Err.	(0.0358)	(0.0285)	(0.0232)	(0.0221)	(0.0201)	(0.0161)	(0.0145)
		MSE	0.0013	0.0008	0.0005	0.0005	0.0004	0.0003	0.0002
3	α_*	Est.	0.3487	0.2586	0.2259	0.2070	0.1979	0.1706	0.1412
		Std. Err.	(0.4358)	(0.2953)	(0.2626)	(0.2313)	(0.2109)	(0.1641)	(0.1482)
		MSE	0.3113	0.1540	0.1199	0.0963	0.0836	0.0560	0.0419
	β_*	Est.	1.9995	2.0008	1.9994	2.0004	1.9987	2.0004	2.0007
		Std. Err.	(0.2527)	(0.1254)	(0.0979)	(0.0716)	(0.0558)	(0.0308)	(0.0227)
		MSE	0.0638	0.0157	0.0096	0.0051	0.0031	0.0009	0.0005
	γ_*	Est.	1.0146	1.0046	1.0013	0.9984	1.0033	1.0022	0.9966
		Std. Err.	(0.2264)	(0.1563)	(0.1258)	(0.1230)	(0.1064)	(0.0825)	(0.0781)
		MSE	0.0514	0.0244	0.0158	0.0151	0.0113	0.0068	0.0061
	ϕ_*	Est.	0.4928	0.4979	0.4989	0.5009	0.4980	0.4990	0.5019
		Std. Err.	(0.0945)	(0.0724)	(0.0609)	(0.0590)	(0.0520)	(0.0409)	(0.0386)
		MSE	0.0090	0.0052	0.0037	0.0035	0.0027	0.0017	0.0015

Table A.1: SIMULATION RESULTS FOR THE MARDL ESTIMATOR. Notes: (i) Number of replications 1,000. Empirical standard deviations are in parentheses and MSE denotes the Monte Carlo mean squared error. (ii) The DGP and true parameter values are as in Table 1. (iii) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2 $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

DGP	Para.	Stat. \ n	200	400	600	800	1,000	2,000	3,000
1	α_*	Est.	-0.0040	0.0005	-0.0015	-0.0015	0.0001	0.0015	-0.0035
		Std. Err.	(0.1308)	(0.0997)	(0.0948)	(0.0833)	(0.0776)	(0.0609)	(0.0548)
		MSE	0.0171	0.0099	0.0090	0.0069	0.0060	0.0037	0.0030
	β_*	Est.	2.0014	1.9994	1.9991	2.0000	2.0008	1.9997	2.0001
		Std. Err.	(0.0946)	(0.0493)	(0.0370)	(0.0286)	(0.0235)	(0.0138)	(0.0097)
		MSE	0.0089	0.0024	0.0014	0.0008	0.0006	0.0002	0.0001
	ζ_*	Est.	-0.5123	-0.5028	-0.5019	-0.5036	-0.5017	-0.4981	-0.5017
		Std. Err.	(0.0921)	(0.0772)	(0.0662)	(0.0563)	(0.0570)	(0.0425)	(0.0383)
		MSE	0.0086	0.0060	0.0044	0.0032	0.0033	0.0018	0.0015
	φ_*	Est.	0.0026	-0.0001	-0.0041	-0.0014	-0.0037	-0.0010	-0.0000
		Std. Err.	(0.1087)	(0.0868)	(0.0747)	(0.0671)	(0.0660)	(0.0504)	(0.0476)
		MSE	0.0118	0.0075	0.0056	0.0045	0.0044	0.0025	0.0023
	ψ_*	Est.	0.4959	0.4972	0.5035	0.4968	0.5011	0.5005	0.5000
		Std. Err.	(0.2783)	(0.2094)	(0.1853)	(0.1653)	(0.1615)	(0.1227)	(0.1098)
		MSE	0.0774	0.0438	0.0343	0.0273	0.0260	0.0150	0.0120
2	α_*	Est.	0.0328	0.0269	0.0235	0.0209	0.0189	0.0152	0.0146
		Std. Err.	(0.0382)	(0.0316)	(0.0267)	(0.0236)	(0.0223)	(0.0175)	(0.0150)
		MSE	0.0025	0.0017	0.0013	0.0010	0.0009	0.0005	0.0004
	β_*	Est.	1.9986	2.0001	1.9993	2.0001	2.0003	2.0000	2.0000
		Std. Err.	(0.0264)	(0.0146)	(0.0106)	(0.0080)	(0.0067)	(0.0039)	(0.0026)
		MSE	0.0007	0.0002	0.0001	0.0001	0.0000	0.0000	0.0000
	ζ_*	Est.	-0.5036	-0.4999	-0.4985	-0.4999	-0.5007	-0.5000	-0.5000
		Std. Err.	(0.0551)	(0.0436)	(0.0369)	(0.0326)	(0.0308)	(0.0245)	(0.0221)
		MSE	0.0030	0.0019	0.0014	0.0011	0.0009	0.0006	0.0005
	φ_*	Est.	-0.0023	0.0017	0.0012	0.0003	-0.0022	-0.0004	0.0009
		Std. Err.	(0.0820)	(0.0641)	(0.0552)	(0.0479)	(0.0458)	(0.0362)	(0.0326)
		MSE	0.0067	0.0041	0.0030	0.0023	0.0021	0.0013	0.0011
	ψ_*	Est.	0.4983	0.5012	0.5001	0.4986	0.5015	0.5012	0.4994
		Std. Err.	(0.0797)	(0.0591)	(0.0523)	(0.0475)	(0.0444)	(0.0344)	(0.0304)
		MSE	0.0063	0.0035	0.0027	0.0023	0.0020	0.0012	0.0009
3	α_*	Est.	0.3342	0.2672	0.2295	0.2278	0.1998	0.1643	0.1494
		Std. Err.	(0.4216)	(0.3350)	(0.2610)	(0.2444)	(0.2209)	(0.1775)	(0.1552)
		MSE	0.2893	0.1835	0.1207	0.1116	0.0886	0.0585	0.0464
	β_*	Est.	2.0028	2.0040	2.0003	1.9983	2.0006	1.9984	1.9994
		Std. Err.	(0.2517)	(0.1353)	(0.0908)	(0.0686)	(0.0592)	(0.0308)	(0.0238)
		MSE	0.0633	0.0183	0.0082	0.0047	0.0035	0.0009	0.0006
	ζ_*	Est.	-0.5039	-0.4994	-0.5040	-0.5022	-0.5013	-0.5017	-0.4981
		Std. Err.	(0.1103)	(0.0850)	(0.0726)	(0.0634)	(0.0588)	(0.0468)	(0.0417)
		MSE	0.0122	0.0072	0.0053	0.0040	0.0035	0.0022	0.0017
	φ_*	Est.	-0.0001	-0.0018	0.0032	0.0018	0.0020	0.0014	-0.0004
		Std. Err.	(0.1075)	(0.0854)	(0.0697)	(0.0660)	(0.0599)	(0.0465)	(0.0400)
		MSE	0.0115	0.0073	0.0049	0.0044	0.0036	0.0022	0.0016
	ψ_*	Est.	0.5057	0.5167	0.4729	0.5096	0.4938	0.4993	0.5009
		Std. Err.	(0.6970)	(0.5458)	(0.4642)	(0.4276)	(0.3648)	(0.3005)	(0.2618)
		MSE	0.4854	0.2979	0.2160	0.1828	0.1330	0.0902	0.0685

Table A.2: SIMULATION RESULTS FOR THE MARDL–EC ESTIMATOR. Notes: (i) Number of replications 1,000. Empirical standard deviations are in parentheses and MSE denotes the Monte Carlo mean squared error. (ii) The DGP and true values are $(\alpha_{3*}, \beta_{2*}, \zeta_*, \varphi_*, \psi_*) = (0, 2, -0.5, 0, 0.5)$, as in Table 3. (iii) The three error distributions are as in Table 3.

	Denmark	Norway	Belgium	Germany
Population	5,882,259	5,434,324	11,655,923	83,369,840
GDP per capita	46,683	64,800	42,659	45,229
Population density	137	14	376	237
Median age	42	40	42	47
Aged 65 older	20	17	19	21
Aged 70 older	12	11	13	16
Female smokers	19	20	25	28
Male smokers	19	21	31	33
Hospital beds per thousand	3	4	6	8
Life expectancy	81	82	82	81
Human development index	0.94	0.96	0.93	0.95

Table A.3: MEDICAL STATISTICS ON COVID-19 FOR TWO SETS OF EUROPEAN COUNTRIES. THE DATA IS SOURCED FROM OUR WORLD IN DATA. (<https://github.com/owid/covid-19-data/tree/master>).

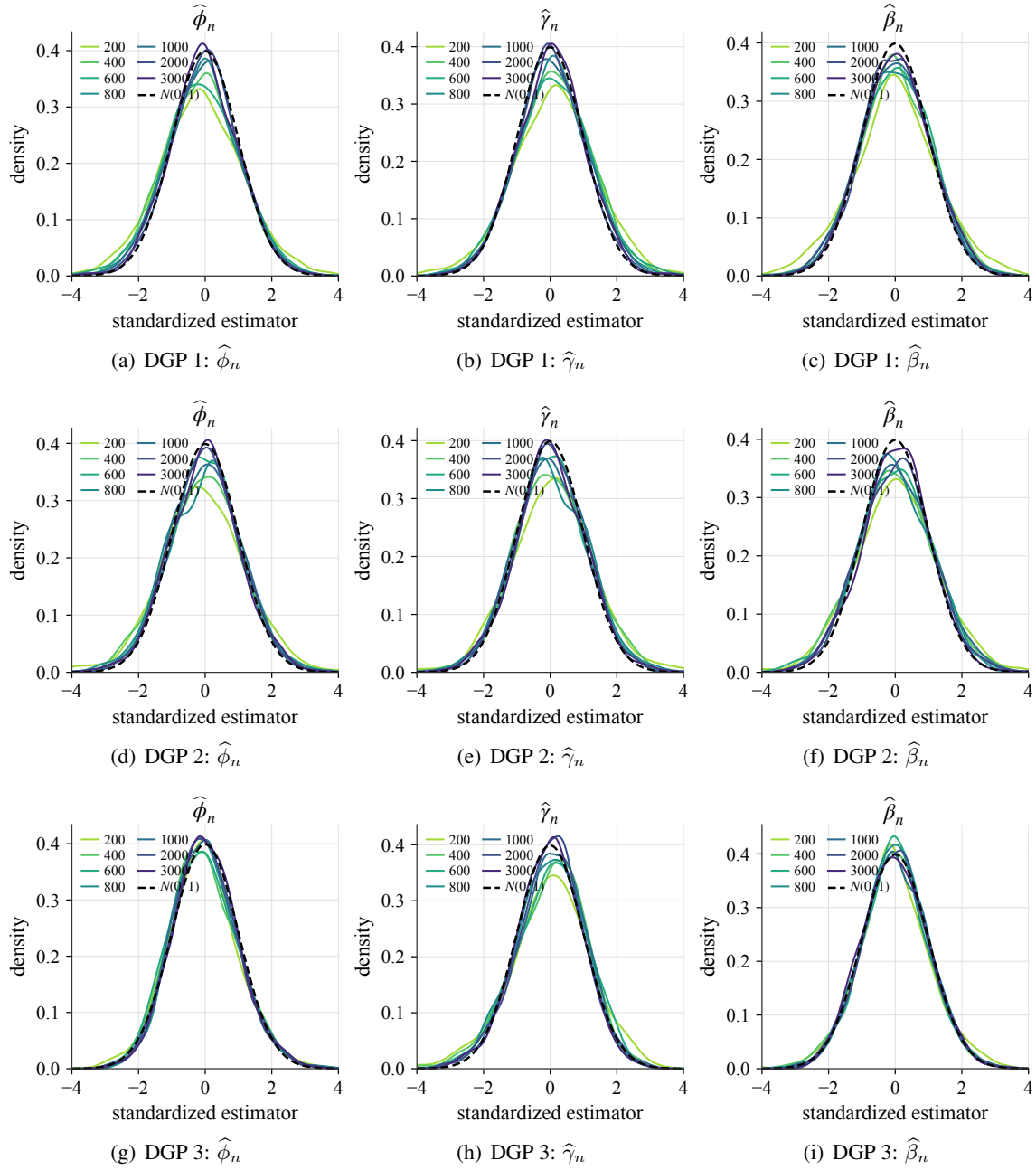


Figure A.1: SAMPLING DENSITIES OF THE MARDL ESTIMATORS UNDER DGPs 1–3. The figures display the probability densities of $\hat{\phi}_n$, $\hat{\gamma}_n$ and $\hat{\beta}_n$. The seven sample sizes range from $n = 200$ to $n = 3000$, with $R = 1500$ iterations at each sample size. DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

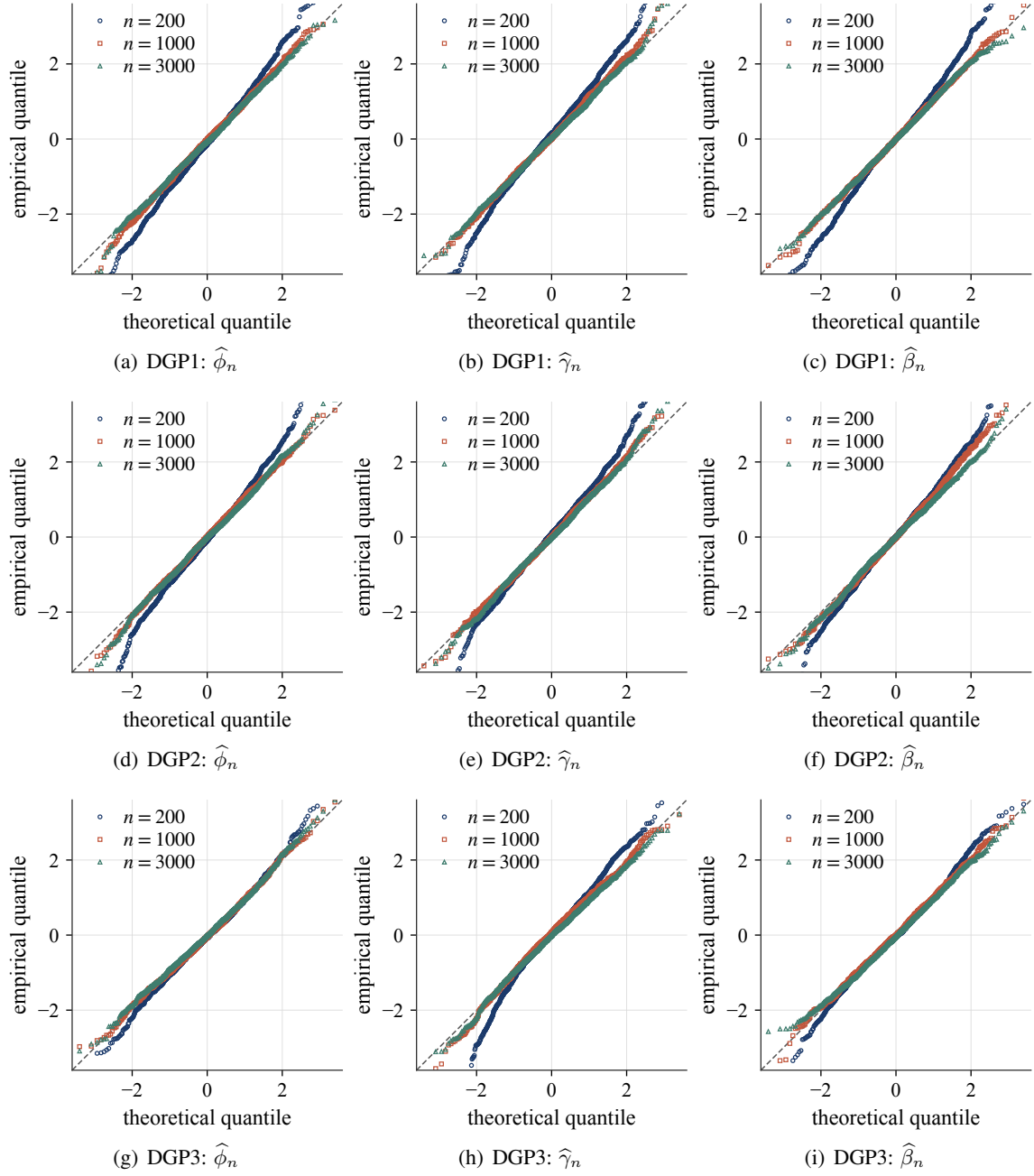


Figure A.2: QQ PLOTS OF THE STUDENTIZED MARDL ESTIMATORS USING DGP 1. Notes: (i) $R = 1500$ iterations at each sample size. (ii) Each estimator is centred at its probability limit and divided by its estimated asymptotic standard error. (iii) The horizontal axis gives standard-normal quantiles and the vertical axis the quantiles of the simulated studentized estimator; the dashed line is the 45-degree line. (iv) The plots approach the 45-degree line as the sample size increases, with the remaining discrepancies concentrated mainly in the tails. (v) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

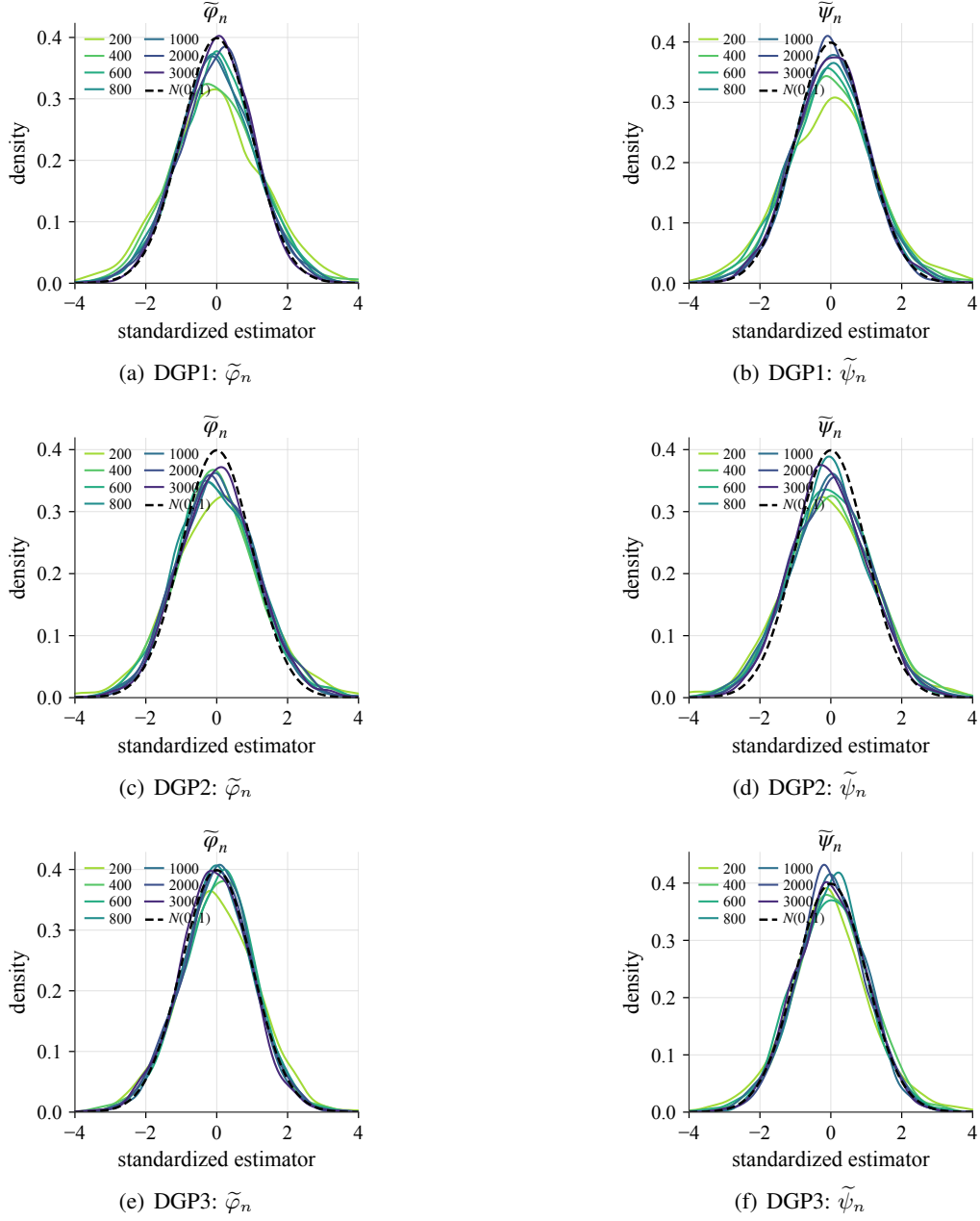


Figure A.3: SAMPLING DENSITIES OF THE MARDL-EC ESTIMATORS UNDER DGP 1. Notes: (i) seven sample sizes from $n = 200$ to $n = 3,000$, with $R = 1,500$ iterations at each sample size. (ii) Each estimator is standardized by its asymptotic standard error, and the dashed curve is the standard normal density. (iii) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.

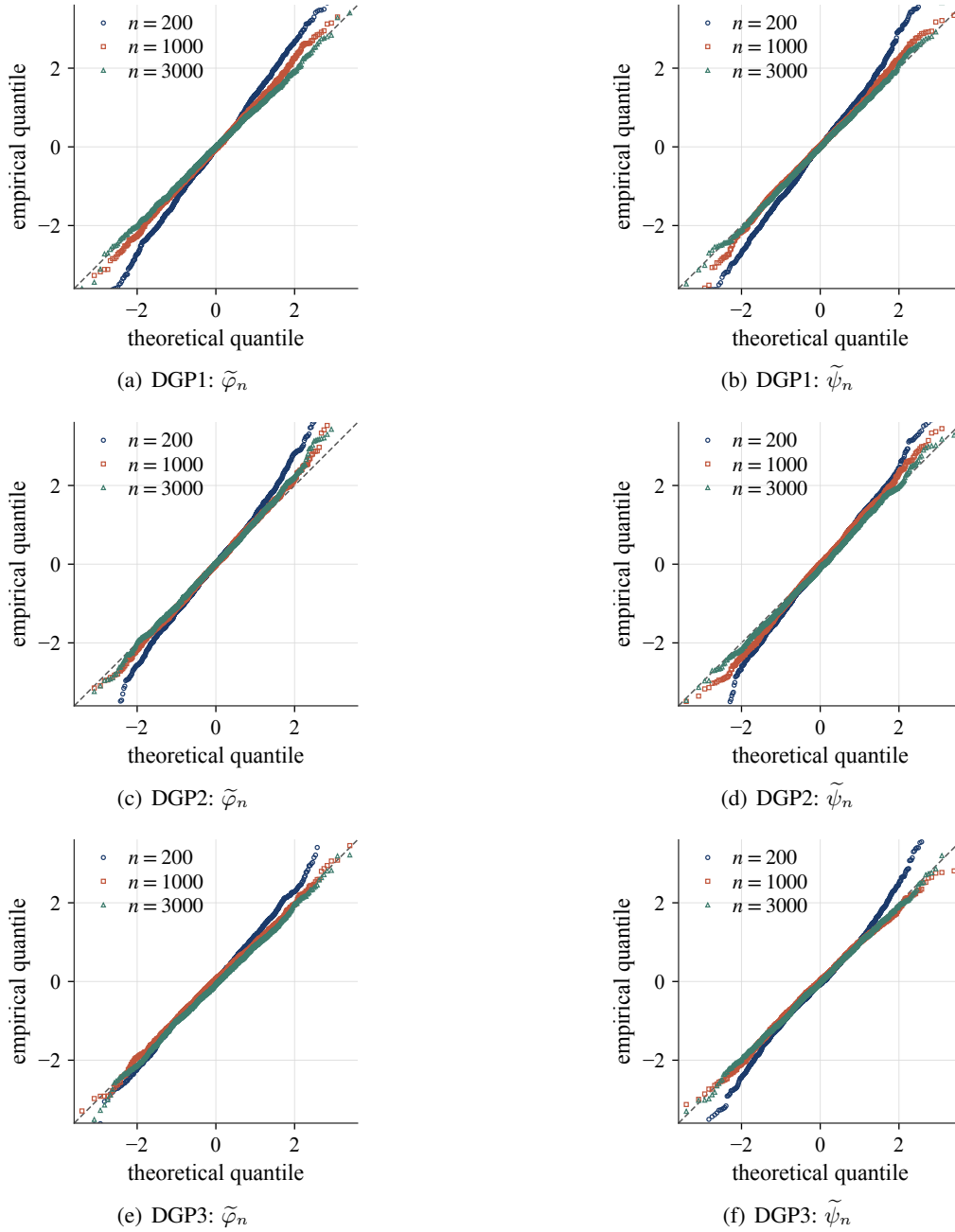


Figure A.4: QQ PLOTS OF THE STUDENTIZED MARDL-EC ESTIMATORS USING DGP 1. Notes: (i) $R = 1,500$ iterations at each sample size. (ii) Each estimator is centred at its probability limit and divided by its estimated asymptotic standard error. (iii) The horizontal axis gives standard-normal quantiles and the vertical axis the quantiles of the simulated studentized estimator; the dashed line is the 45-degree line (iv) The plots approach the 45-degree line as the sample size increases, with the remaining discrepancies concentrated mainly in the tails. (v) DGP 1: $u_t \sim_{\text{iid}} N(0, 0.25)$; DGP 2: $u_t \sim_{\text{iid}} \text{Beta}(2, 6) - 1/6$; DGP 3: $u_t \sim_{\text{iid}} \Gamma(3, 1) - 2$.